

NORMAN FOSTER INSTITUTE
MASTER'S PROGRAMME IN SUSTAINABLE CITIES

GEORGINA KINUTHIA

Urban Design + Social Impact
Architect

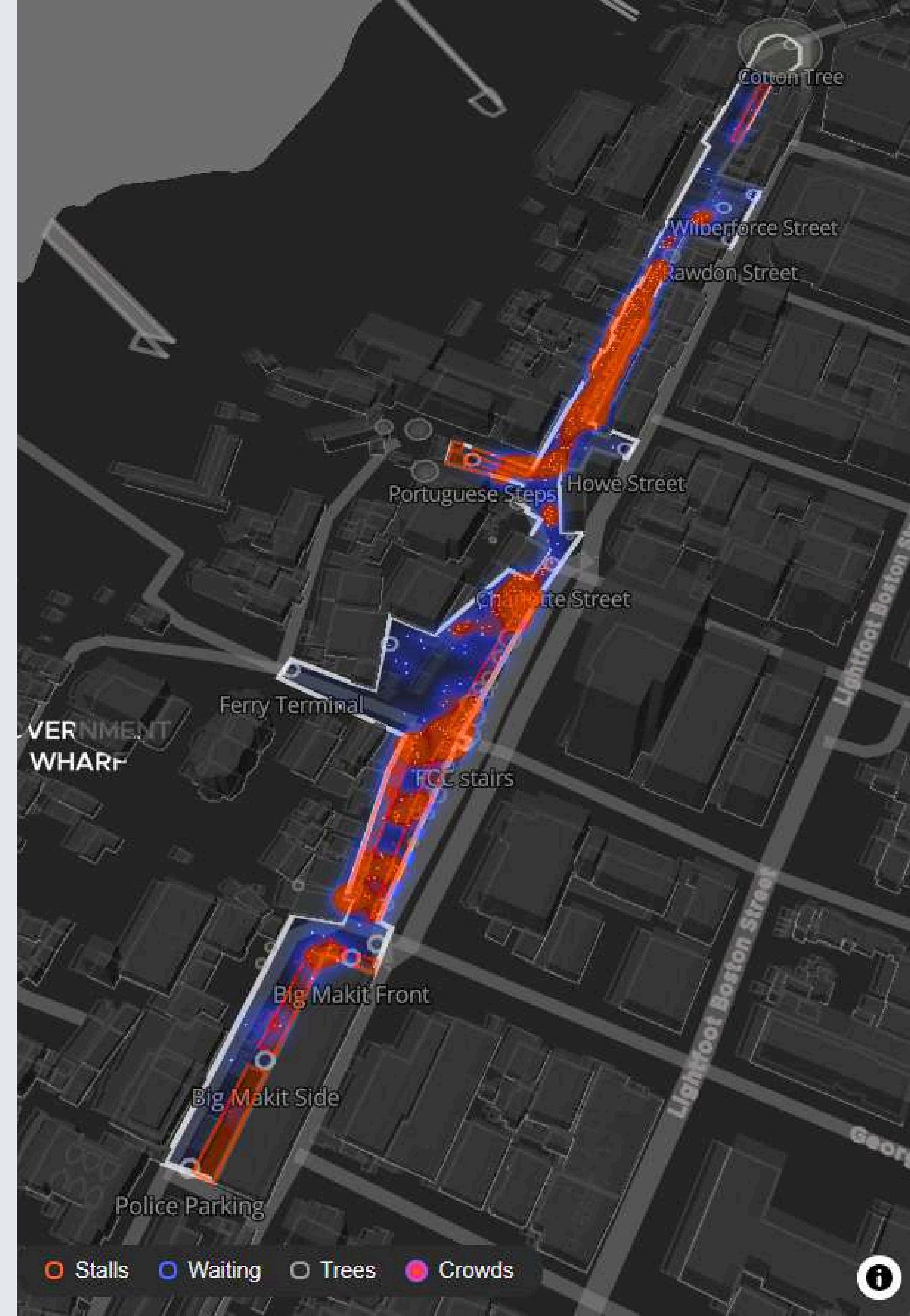
I'm an architect developing expertise in computational urban design and spatial analytics, with a particular interest in using data and simulation to test how design decisions affect people and quality of life.

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SOCIAL IMPACT PROJECTS PORTFOLIO



COMPUTATIONAL & TECHNICAL TOOLKIT

Spatial Analytics & ABM: QGIS(Network Topology, EPSG:6372), Python (GeoPandas, NetworkX), GAMA Platform (GAML Scenario Simulation), LLMs
Architectural & Visualization: Revit, Rhino 3D, Grasshopper, D5 Render, Adobe Creative Suite, Lumion, ArchiCAD, Miro, Figma
Methods & Frameworks: Agent-Based Behavior Modeling, Participatory Planning, Climate-Responsive Passive Strategies, Low-Carbon Sourcing

SELECTED SOCIAL IMPACT PROJECTS (2021 — 2026)

01 / BUEN CAMINO • Tacubaya Active Mobility Study
Location: Mexico City, Mexico | Focus: Computational Design, ABM & Spatial Analytics

02 / HEALING FARM • Trauma-Informed Sanctuary
Location: Kitui, Kenya | Focus: Vernacular Planning, Universal Accessibility & Passive Cooling

03/ MAILI TISA CBC SCHOOL • Eco-Friendly Campus Master Plan
Location: Namanga, Kenya | Focus: CBC Curriculum Design, Passive Shading & Local Materials

04/ IFTIIN PEACE HUB • Resource & Vocational Sanctuary
Location: Abudwak, Somalia | Focus: Climate Comfort, Local Masonry & Circular Design

05 / FDB CIVIC CENTER • Resilient Community Hub
Location: Fond des Blancs, Haiti | Focus: Architecture, Civic Resilience & Adaptability

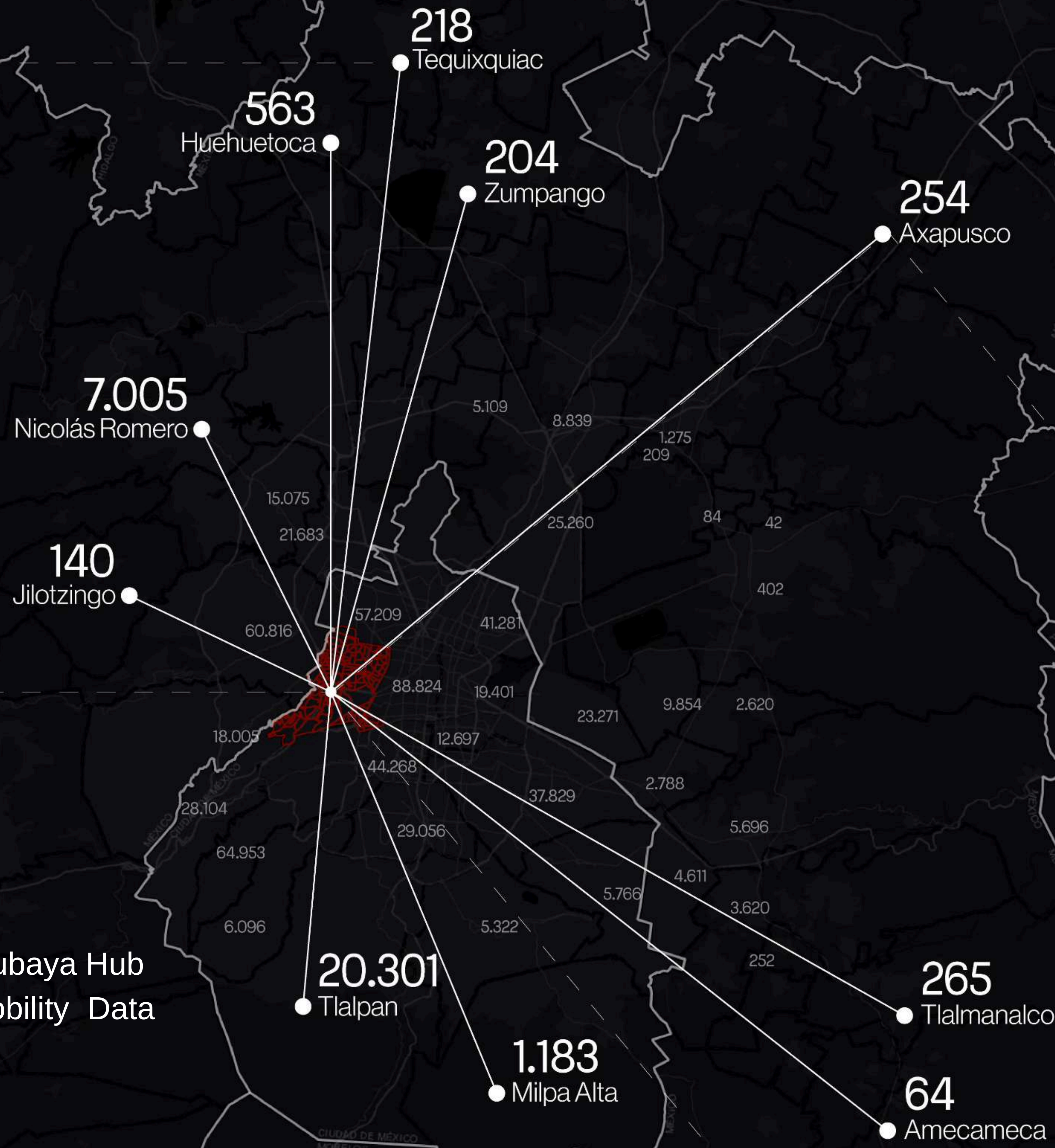
06 / SUMMITS EDUCATION • Modular School Prototype
Location: Mogé & Platanalé, Haiti | Focus: Spatial Equity, Replicable Systems & Material Logic

07 / SAFE HOMES FOR GBV SURVIVORS • Modular Crisis Housing
Location: 47 Counties, Kenya | Focus: Therapeutic Architecture, Security & Privacy

08 / SHOFCO GIRLS' SCHOOL • Informal Settlement Educational Hub
Location: Kibera, Nairobi, Kenya | Focus: High-Density Urban Insertion & Passive Stack Ventilation

BUEN CAMINO: SAFE ACTIVE MOBILITY CORRIDORS

Alcaldía Miguel Hidalgo — Mexico City's Western Transit Gateway



Study Area: Tacubaya Hub
Metropolitan Mobility Data
(INEGI EOD)



THE URBAN DIAGNOSTIC

- 712,235 Daily Commuter Trips across Metro Lines 1, 7, 9 & Metrobús Line 2
- Severe highway severance & narrow 0.60 m sidewalk chokepoints
- High night-time spatial insecurity impacting vulnerable commuters

MY COMPUTATIONAL CONTRIBUTION

1. Network Graph Hygiene: Standardized street centerlines in QGIS
2. Agent Friction Rules: Formulated pedestrian cost functions in Python
3. ABM Scenario Simulation: Modeled baseline vs. intervention flows in GAMA Platform

PRACTICAL DESIGN VALUE

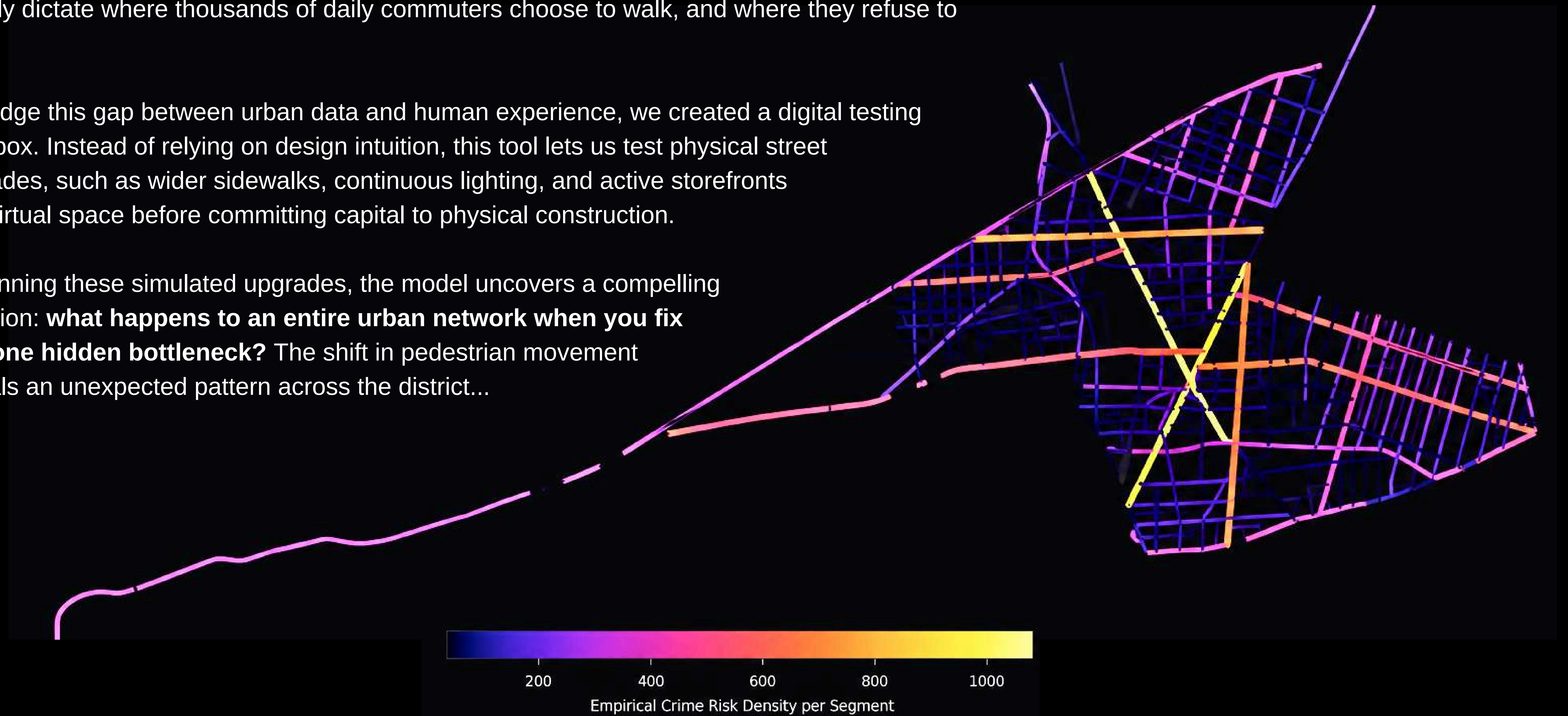
- Converts raw mobility & crime data into streetscape parameters
- Stress-tests street transformations before capital allocation

MAPPING HUMAN MOVEMENT & SCENARIO TESTING

When people walk through a busy transit hub, they don't choose their route based on geographic distance alone, they choose the route that feels safe, clear, and well-lit. In a congested interchange like Tacubaya, dark corners, narrow sidewalks, and isolated streets silently dictate where thousands of daily commuters choose to walk, and where they refuse to go.

To bridge this gap between urban data and human experience, we created a digital testing sandbox. Instead of relying on design intuition, this tool lets us test physical street upgrades, such as wider sidewalks, continuous lighting, and active storefronts in a virtual space before committing capital to physical construction.

By running these simulated upgrades, the model uncovers a compelling question: **what happens to an entire urban network when you fix just one hidden bottleneck?** The shift in pedestrian movement reveals an unexpected pattern across the district...



Workflow: Urban Observation → Data/Criteria → Scenario Simulation → Spatial Design Decision

FEMINIST REPUBLIK HEALING FARM

KITUI, KENYA 2021

Architecture: Precise Architects
Lead Principal: Arthur Mwangi
Architect: Georgina Kinuthia

The Healing Farm in Muselele is a women-led sanctuary where African Women Human Rights Defenders (AWHRDs) can rest, reconnect with nature, and heal. It is both a retreat and a continental hub for collective care, resilience, and feminist solidarity.

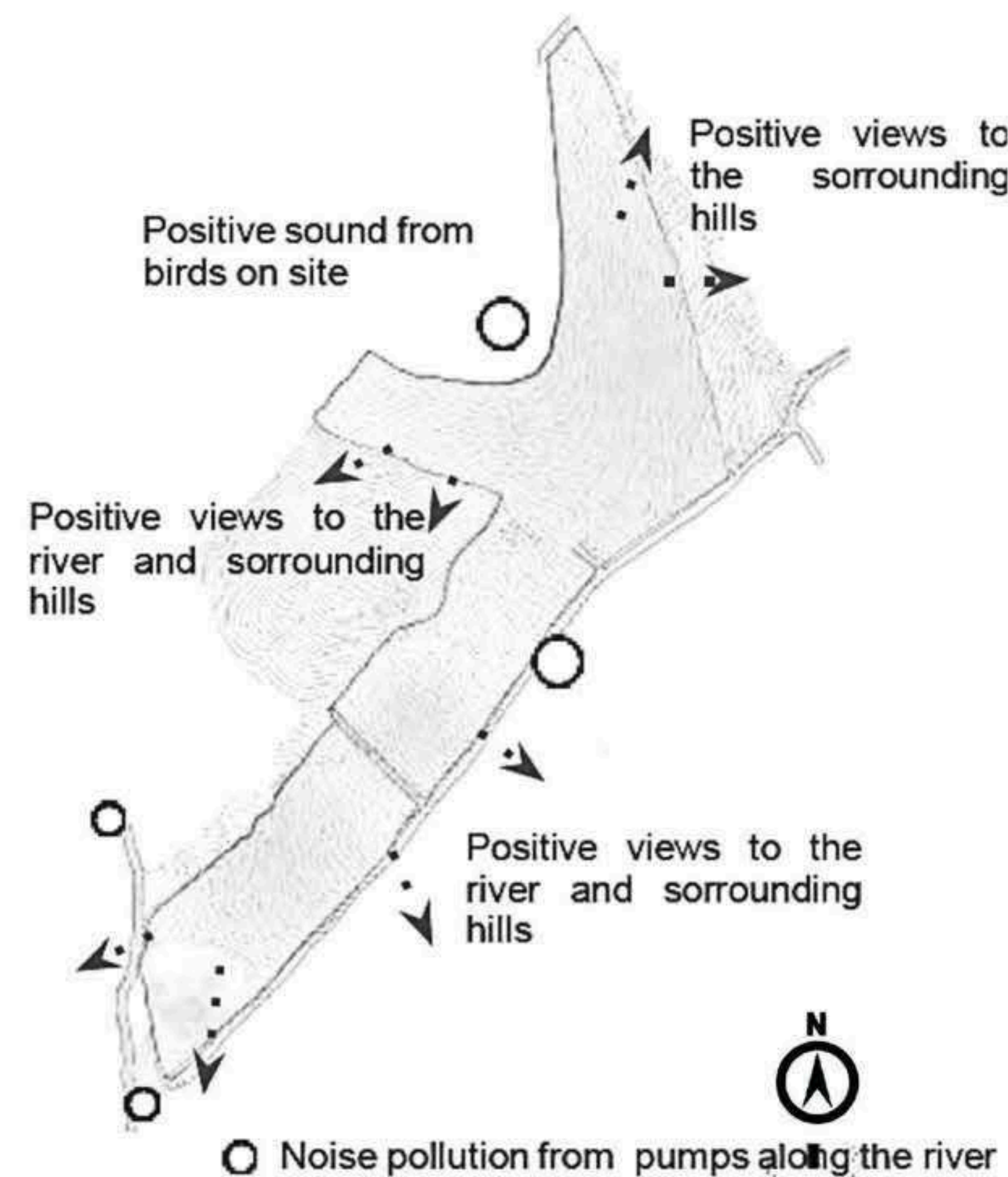
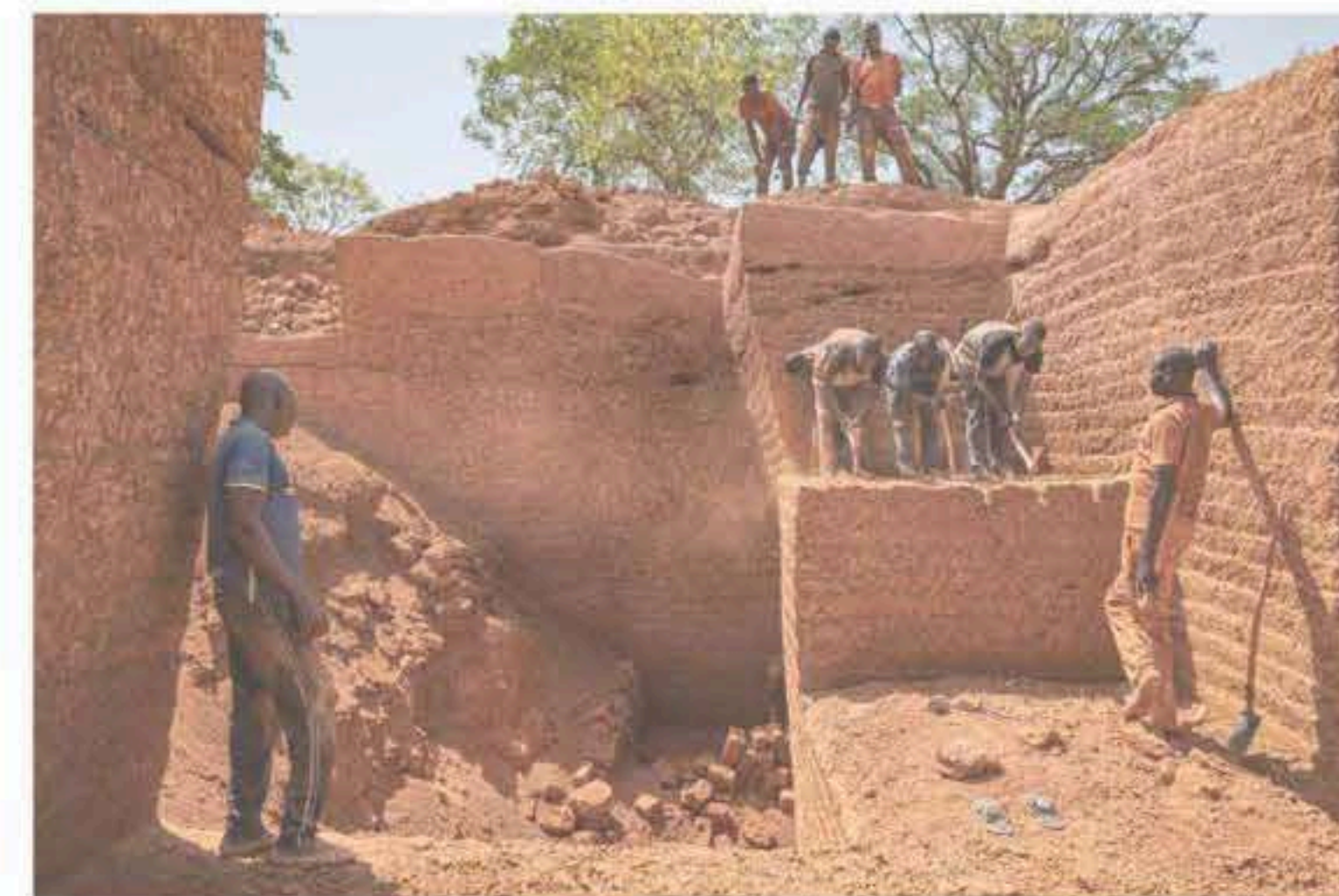
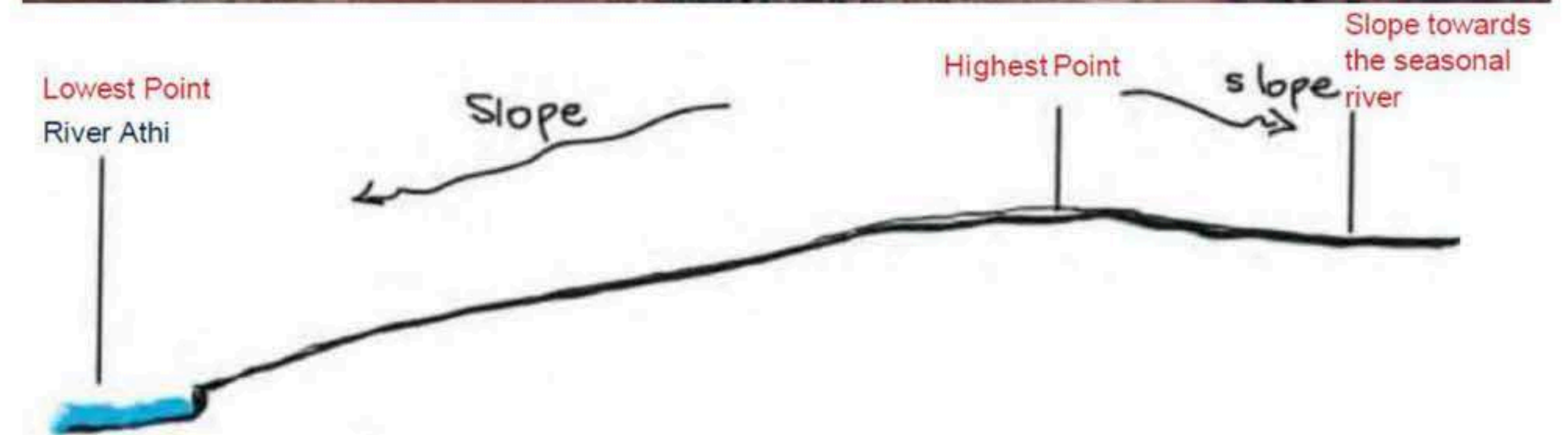
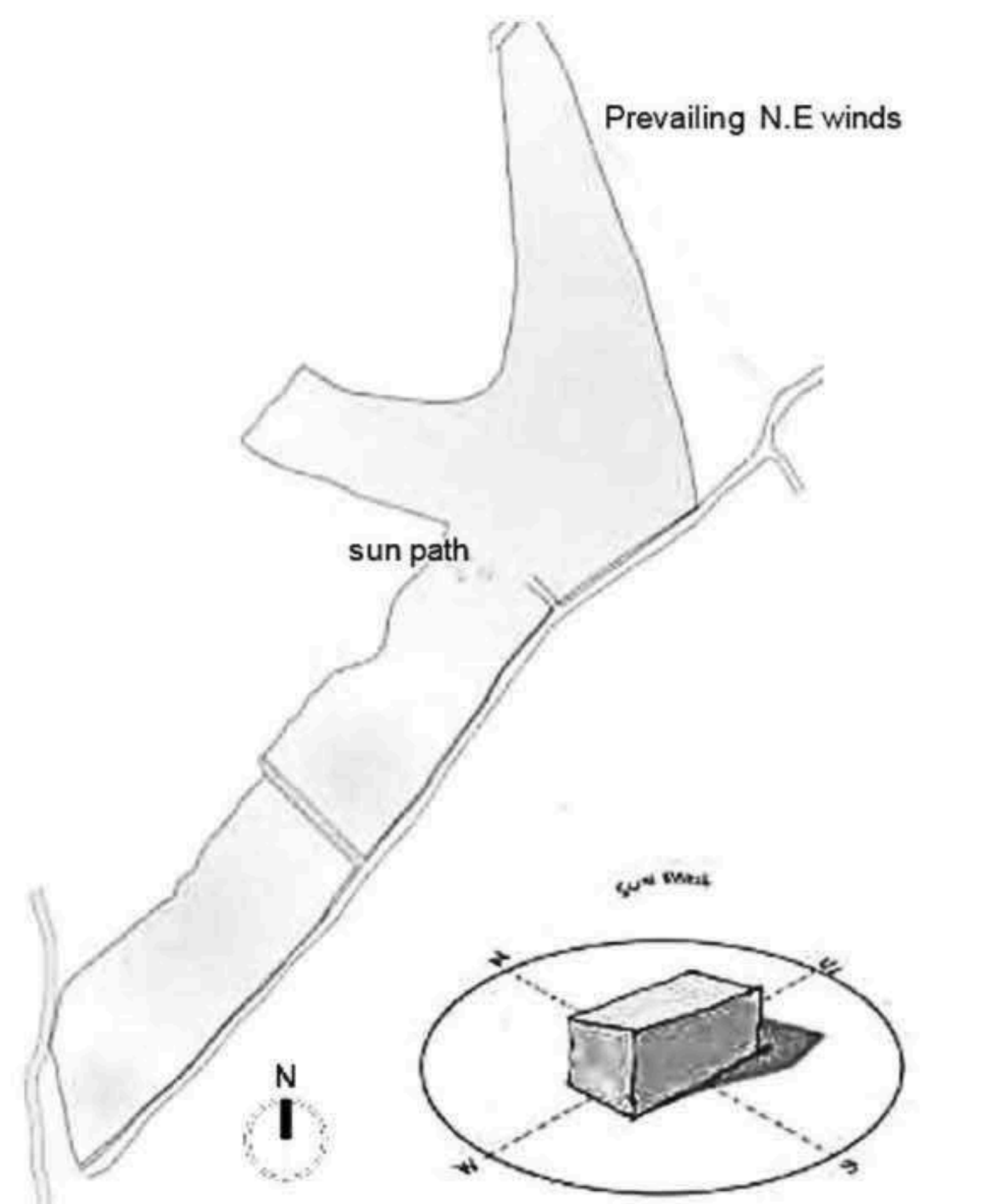
Rooted in vernacular planning and circular forms, the architecture uses locally available materials—bamboo, mud bricks, and stone—reflecting traditional practices while ensuring environmental sustainability. These natural, low-embodied-energy materials create calming, restorative spaces and reduce ecological impact. Bamboo is also used for railings and shaded walkways, chosen for its renewable, fast-growing nature.

Facilities include wellness centers, therapeutic gardens, accommodation, and communal gathering spaces. Accessibility is integral: the design incorporates ramps, tactile and audio assistance, and fully equipped paraplegic rooms with universal design features such as grab bars, wide doors, adjusted countertops, and assistive technology.

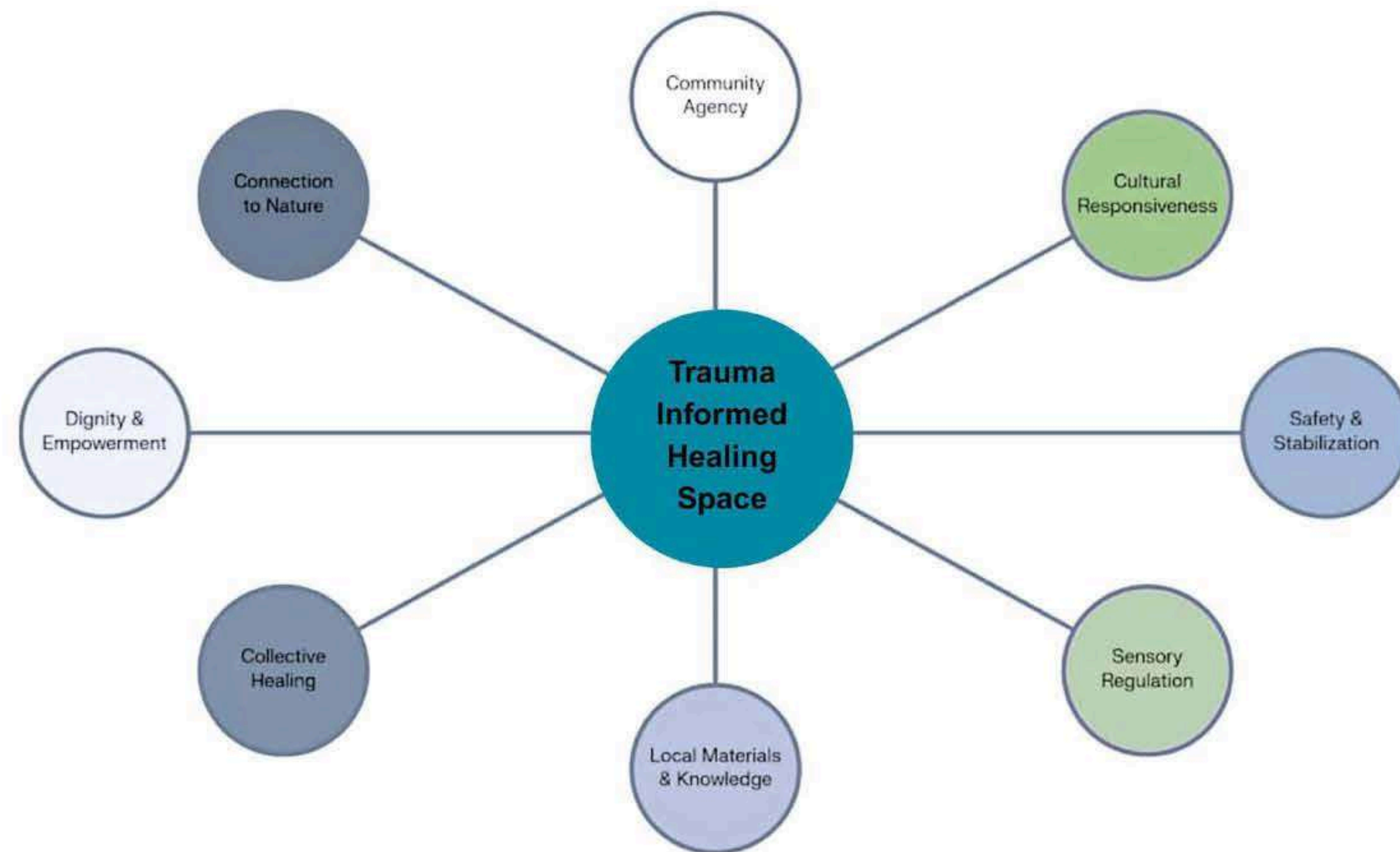
The Healing Farm offers a holistic model for trauma recovery—where dignity, sustainability, and inclusivity shape every experience.



GEORGINA KINUTHIA • PRECISE ARCHITECTS • AFDB



The design responds sensitively to the site's natural topography, following contour lines to reduce extensive earthworks. Public spaces—including the auditorium, reception, wellness center, library, and museum—will be located at the highest point, while accommodation areas will cascade down towards the river. Staff and maintenance facilities will be positioned along the slope toward the seasonal stream on the northern edge. With no connection to the municipal water supply (KITWASCO), the project will rely on rainwater harvesting and a water pan system, supplemented by nearby access to the Athi River. The site experiences a hot, dry tropical savanna climate (Aw), with low rainfall and significant temperature variation, making passive cooling strategies essential. The terrain varies significantly, with a 45-meter elevation difference from the highest to lowest points, informing both water management and climate-responsive planning.



Rooted in traditional African homestead design, the master plan uses soft curves and clustered layouts to create an inclusive, healing flow from public to private spaces, following the natural slope of the land.

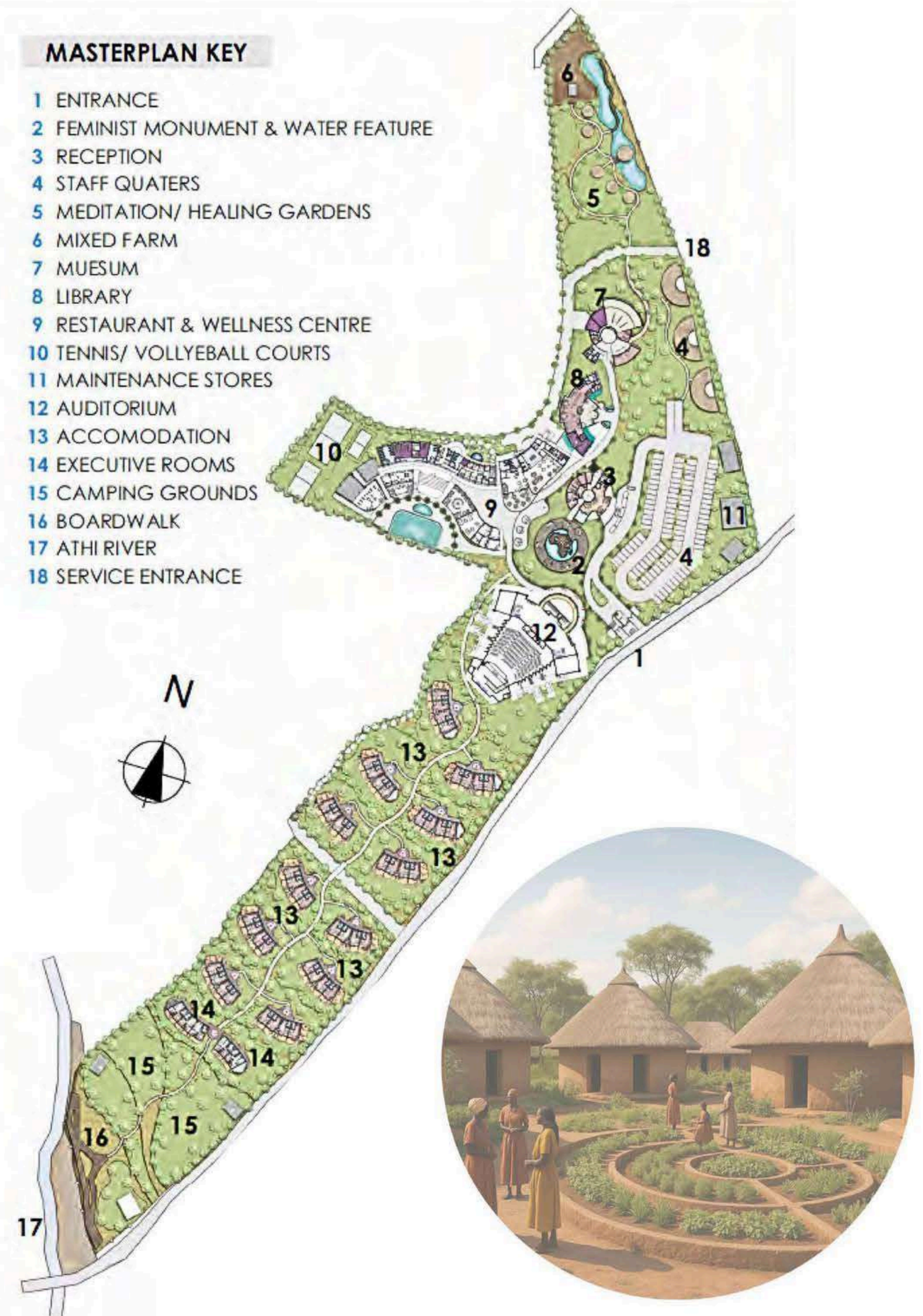
At the top, a Feminist Monument and water feature welcome visitors into communal zones; reception, auditorium, museum, library, wellness center, and restaurant; designed to foster connection and care. As the path descends, healing gardens, sports spaces, and layered accommodation zones offer increasing privacy and comfort, ending at the Athi River with a tranquil boardwalk for reflection.

Local materials such as stone, bamboo, and earth bricks are used not only in the architecture but throughout the landscape; shaping walkways, garden edges, retaining walls, and shaded structures; to ensure cohesion, cultural resonance, and environmental sustainability. This approach weaves together land, tradition, and trauma-informed design to create a space of dignity, restoration, and belonging.



MASTERPLAN KEY

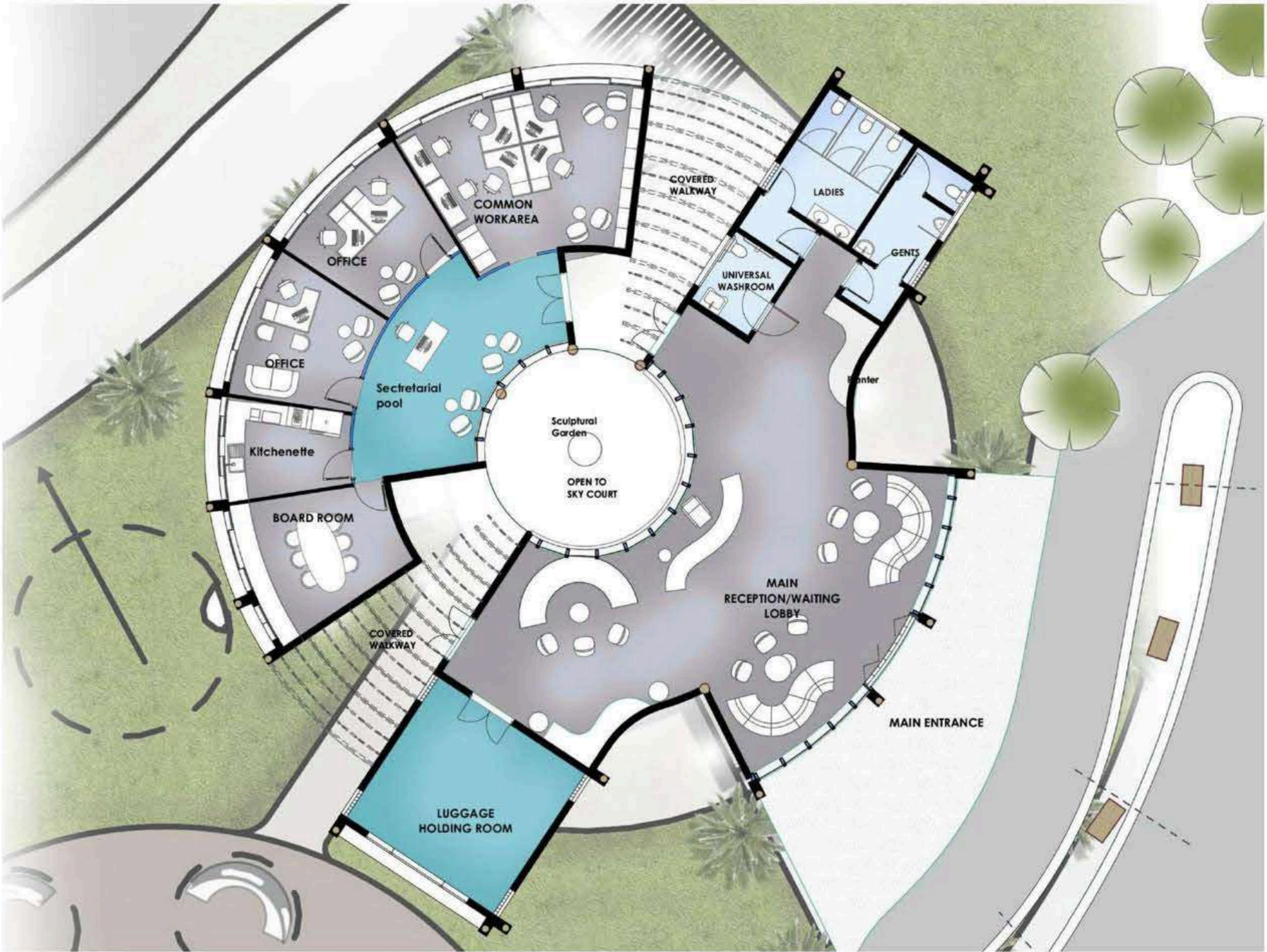
- 1 ENTRANCE
- 2 FEMINIST MONUMENT & WATER FEATURE
- 3 RECEPTION
- 4 STAFF QUARTERS
- 5 MEDITATION/ HEALING GARDENS
- 6 MIXED FARM
- 7 MUESUM
- 8 LIBRARY
- 9 RESTAURANT & WELLNESS CENTRE
- 10 TENNIS/ VOLLYEBALL COURTS
- 11 MAINTENANCE STORES
- 12 AUDITORIUM
- 13 ACCOMODATION
- 14 EXECUTIVE ROOMS
- 15 CAMPING GROUNDS
- 16 BOARDWALK
- 17 ATHI RIVER
- 18 SERVICE ENTRANCE



PASSIVE MOVE	HOW IT ENHANCES COMFORT
Stack ventilation via the raised central roof	Draws warm air upward and expels it, maintaining interior temperatures 4–6 °C cooler than ambient.
Thick adobe walls (high thermal mass)	Absorb daytime heat and release it slowly at night, evening-out diurnal temperature swings.
Deep roof overhangs	Block high-angle midday sun and shelter outdoor walkways from rain, enabling movement in all weather.
Central breezeway corridor	Orchestrates cross-drafts from prevailing winds, so every room enjoys fresh air without fans.
Single-banked rooms	Guarantee at least two exterior exposures per space—maximizing daylight, views, and purge ventilation.
Vent-blocks & clerestory windows	Provide secure, always-open apertures for continuous airflow, even when doors are closed.
Open commons & courtyards	Act as thermal release valves and social living rooms, encouraging informal gathering and cultural rituals.

The Healing Farm embodies social inclusion and environmental stewardship through its thoughtful design. A circular layout fosters communal interaction and intergenerational care, while universal accessibility ensures mobility for all. Spaces for health, prayer, and learning are equitably distributed, and gender-safe design enhances security and dignity for women and girls.

Environmentally, the project champions carbon-conscious construction using local materials, a regenerative water cycle with rain harvesting and greywater reuse, energy positivity through solar and biogas systems, and biodiversity corridors that reconnect the site to its ecosystem—all contributing to a sustainable and inclusive care infrastructure.





The Healing Farm faced several construction challenges, including sloping terrain, uneven ground, water access, and the need for sustainable materials. To stay within budget, the project was phased—starting with the auditorium, reception and library to establish essential functions early.

In response to Kitui's semi-arid climate, the project integrates a hybrid approach of traditional and modern construction techniques, with strong emphasis on community inclusion and environmental performance.

Phased construction

- Enables flexibility and budget control
- Prioritizes essential facilities to activate the site early

Stepped buildings using cut-and-fill techniques

- Reduces grading and excavation, minimizing site disturbance
- Allows structures like the auditorium to nestle naturally into the slope
- Creates shaded outdoor terraces and amphitheater-style gathering areas

Use of local, low-carbon materials

- Mud, stone, and Mwalanite Limesoil reduce emissions and costs
- Supports the local economy and material supply chains

Inspiration from traditional architecture

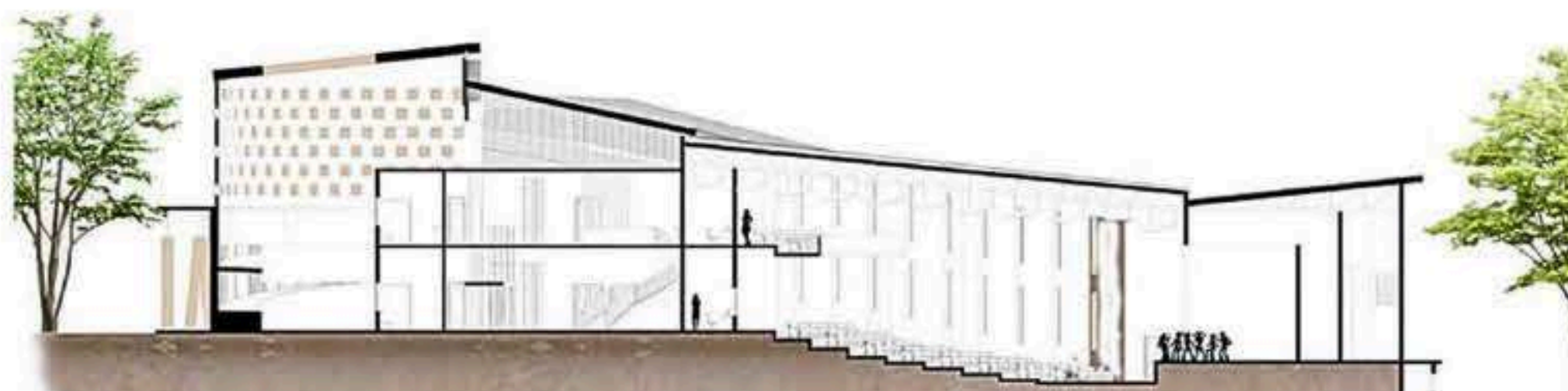
- Easily understood and accepted by local builders

Modern enhancements

- CSEBs (Compressed Stabilized Earth Blocks): Durable, low-energy alternative to concrete
- Rainwater harvesting: Supports irrigation and water resilience

Local capacity-building

- Community training programs promote skills transfer
- Engaging local contractors ensures cultural fit and long-term maintenance



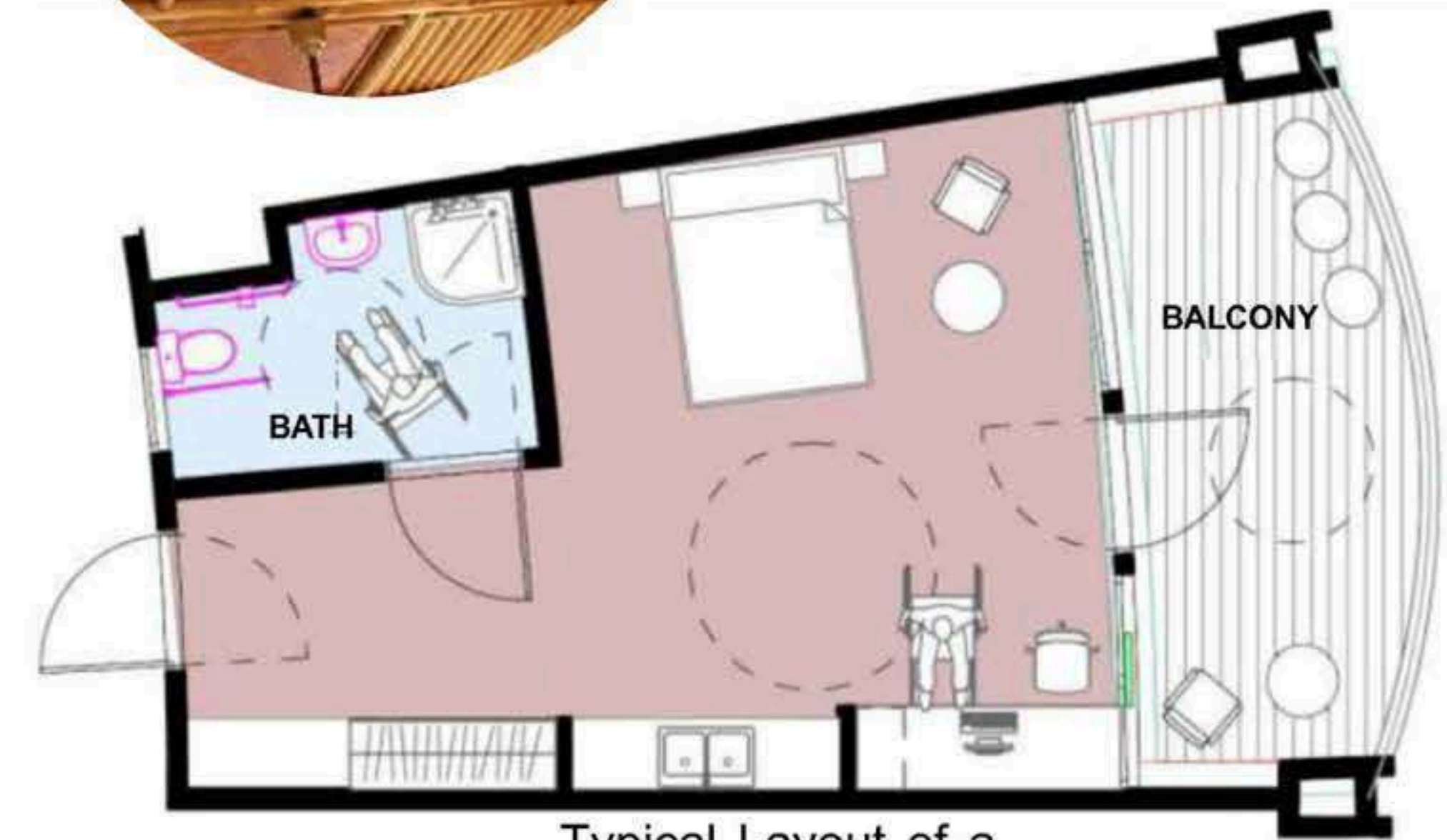
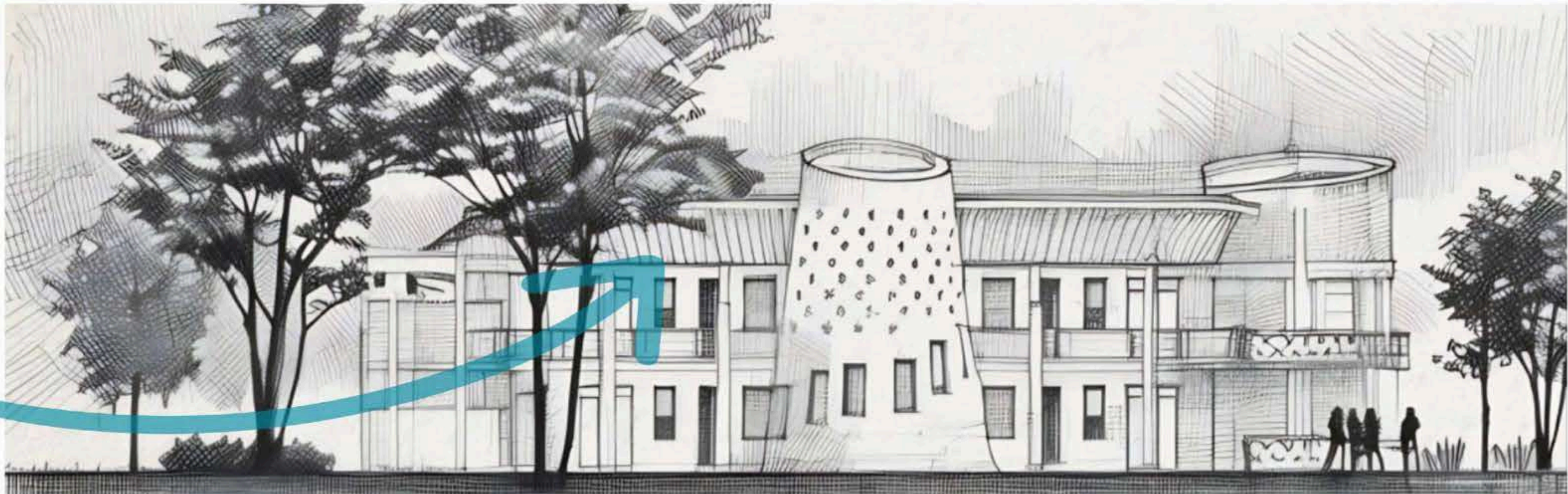
The Healing Farm is designed to be both physically accessible and emotionally supportive, particularly for users recovering from trauma. All circulation paths incorporate ramps, audio and tactile navigation, and assistive technologies to ensure ease of movement for people with disabilities, elders, and children.

Paraplegic-accessible rooms are fully equipped with:

- Grab bars for stability and safety
- Wide, double-acting doors to accommodate wheelchairs
- Adjustable bed heights and countertops
- Tactile aids to assist low-vision users

To create a trauma-sensitive environment, interior finishes emphasize natural materials and sensory comfort. Bamboo ceiling panels were selected for their acoustic performance, warmth, and calming effect. These are fire-proofed for added safety and paired with overhead ventilation slots that allow quiet, uninterrupted airflow.

Earth-toned finishes, non-reflective surfaces, and soft lighting contribute to a peaceful interior that grounds occupants, reduces overstimulation, and encourages healing. The result is an environment that promotes dignity, safety, and wellbeing—fully aligned with the project's restorative mission.



Typical Layout of a Paraplegic Room



MAILI TISA CBC SCHOOL

NAIROBI, KENYA, 2022
Architecture: UDesign
Lead Principal: Paul Mutahi
Architect: Georgina Kinuthia

In a tranquil rural Namanga district, a vibrant school embraced the Maasai community's spirit. Its colorful, eco-friendly architecture houses open-air classrooms with panoramic views connect the children and educators to nature.

Sustainability was key in the design, construction and operation considerations. Efficient energy solutions, water conservation, sustainable site excavation, responsible materials, healthy indoors, innovative design, wildlife harmony, regional priority.

Childhood unfolds, fostering holistic development in this school, a symbol of education and community, wove tradition into modernity, creating a lively haven where the roots of the Maasai community intertwined with the dreams of the future. The school design was also influenced by the new Competency-Based Curriculum (CBC) education framework, which advocates for specialization and emphasizes technical learning.





THE MASTERPLAN

STUDENTS'
ACCOMODATION

SENIOR HIGH-SCHOOL BLOCK

LIBRARY

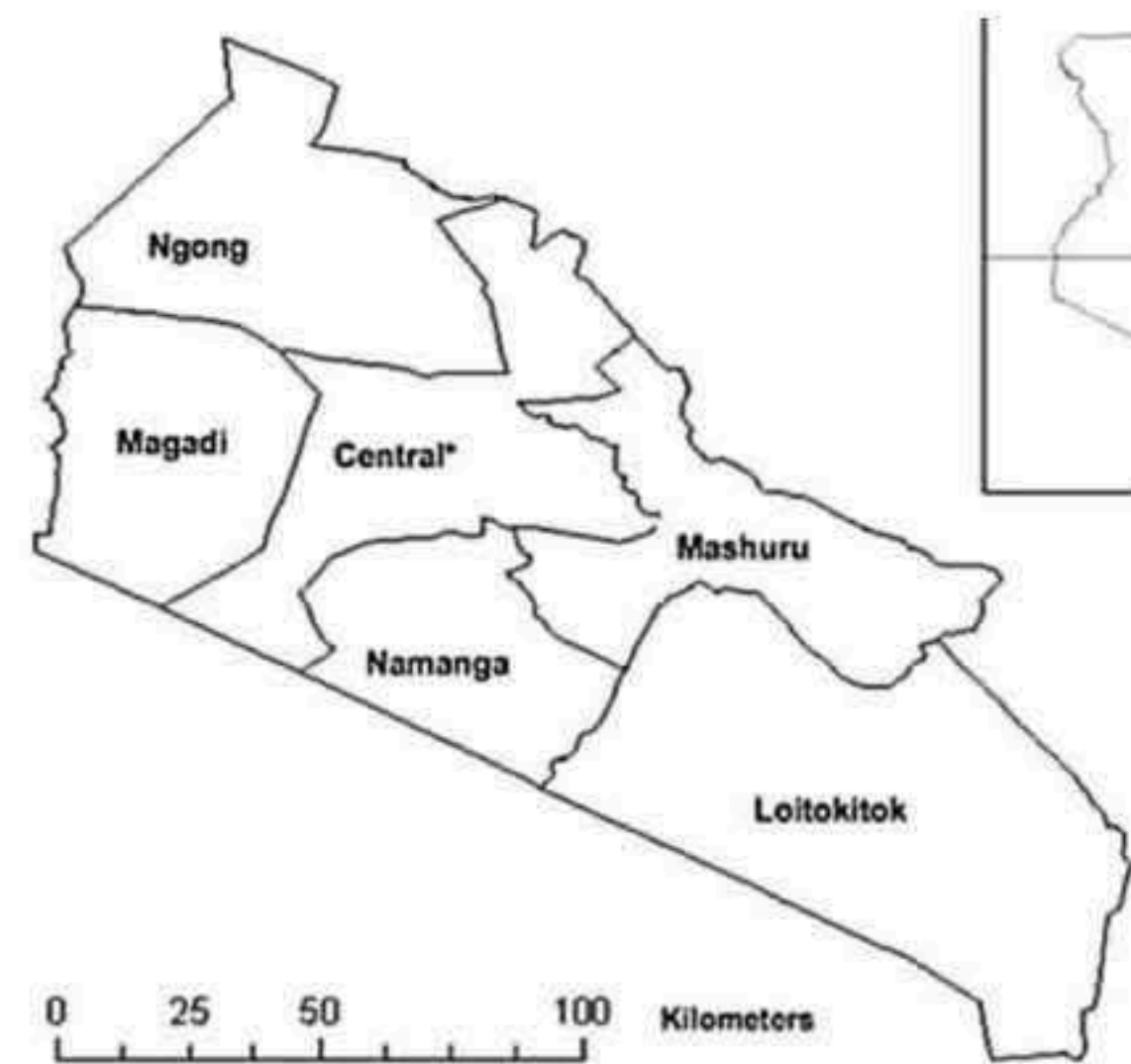
STEM
(TECH)
BLOCK

DINING
HALL

ADMINISTRATION & JUNIOR HIGH-
SCHOOL BLOCK

SWIMMING
POOL





Mt. Ol Donyo Orok



Services Available



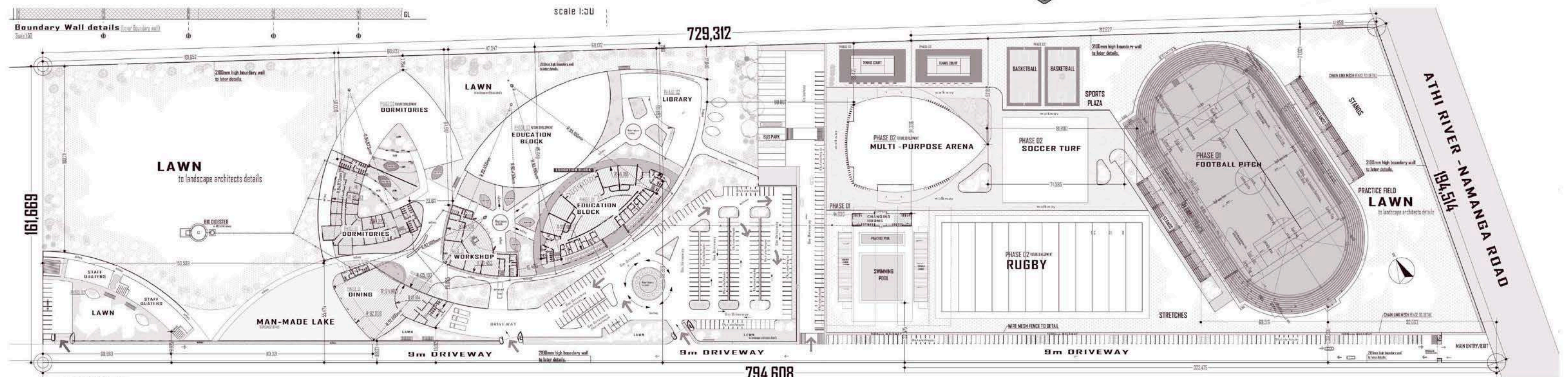
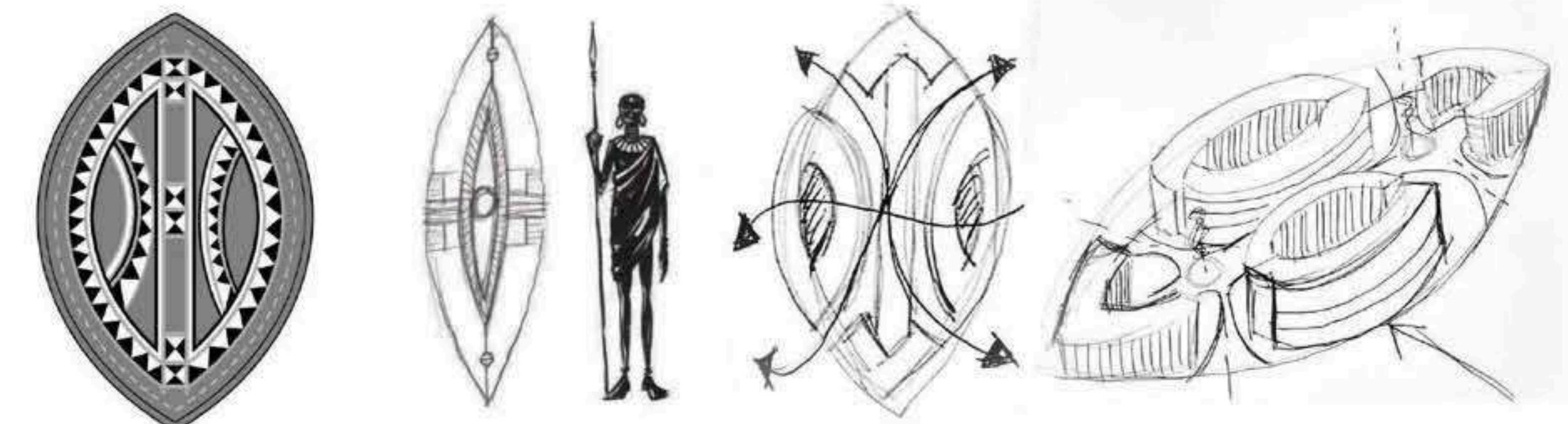
Nairobi-Namanga Rd.

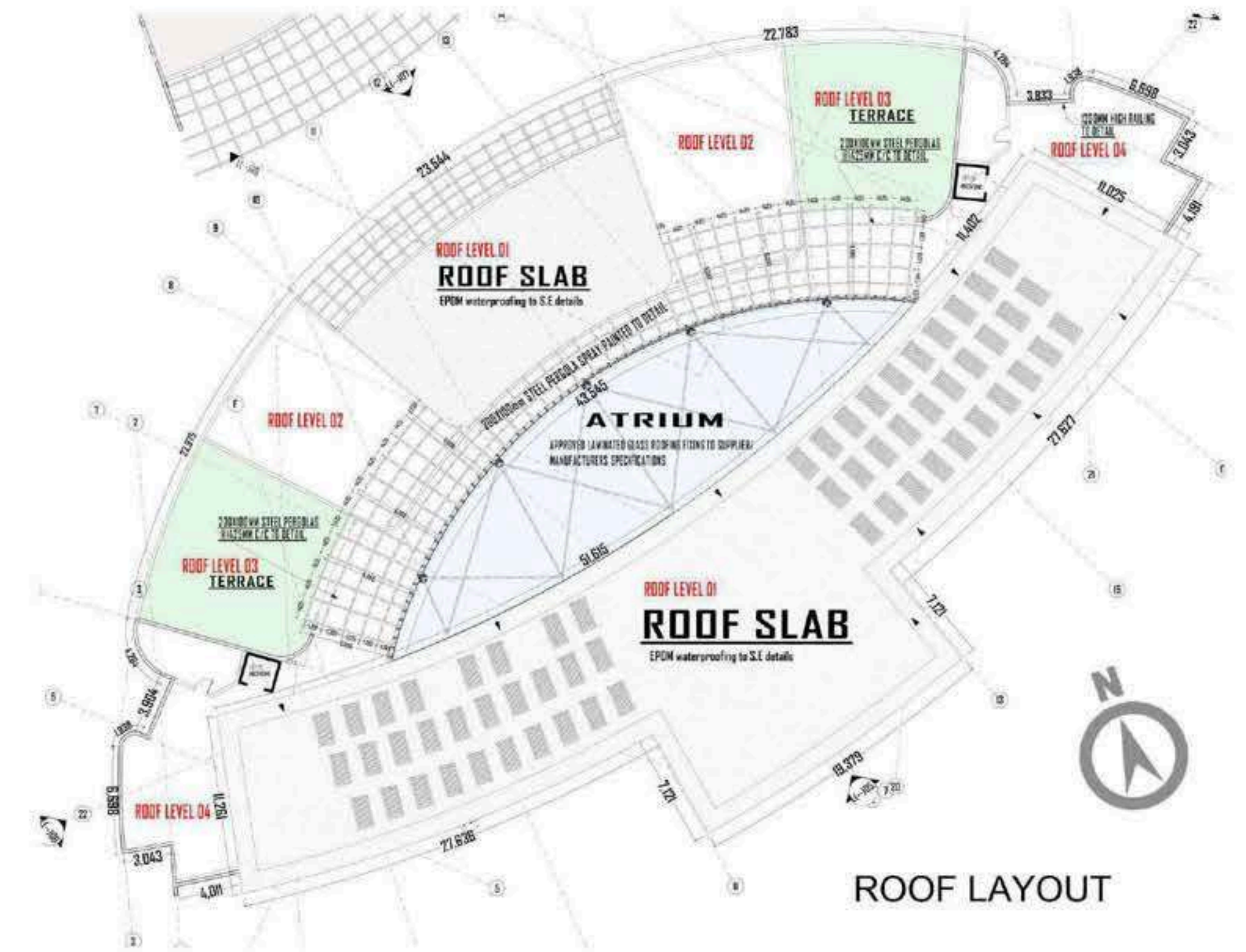
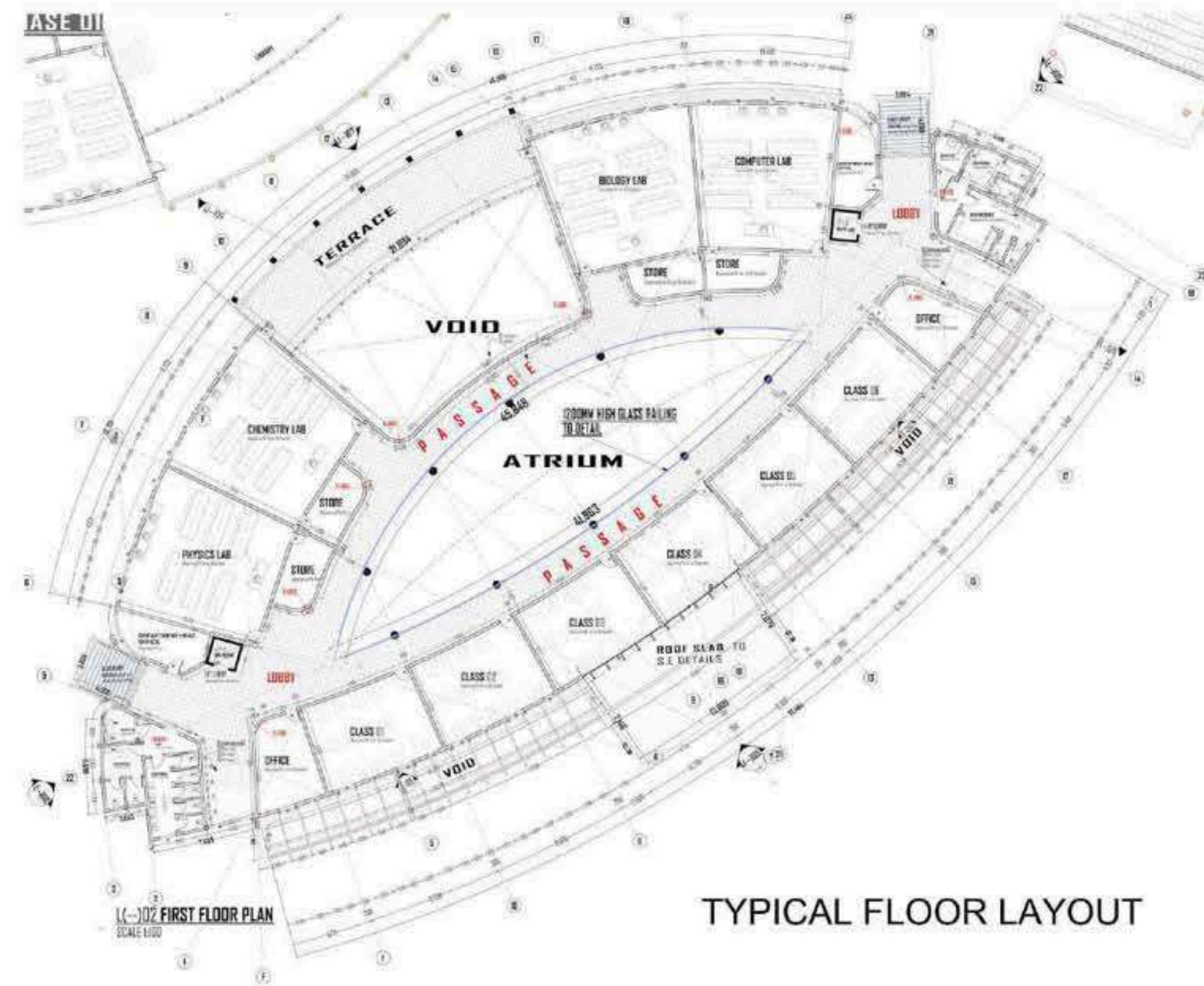
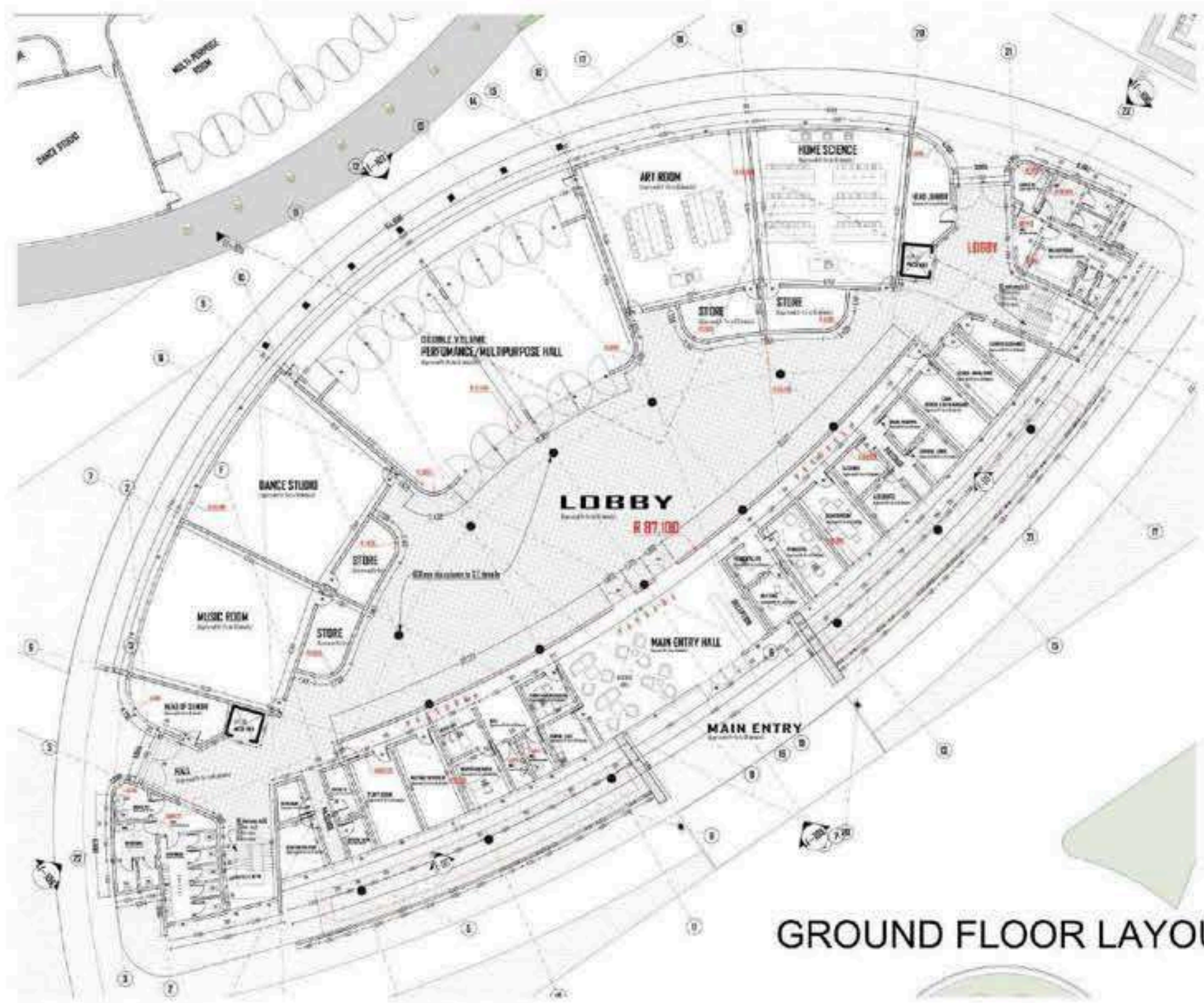
Inspired by the Maasai warrior shield, the master plan translates cultural symbolism into a spatial framework that balances protection, connection, and openness. The shield's form guides a radial, flowing organization, with a central spine anchoring academic and communal zones in a clear, intuitive sequence.

Curved arms extend from this axis, embracing a series of negative spaces—shaded courtyards, outdoor classrooms, and quiet retreats—designed not as filler, but as vital social infrastructure. These in-between spaces foster interaction, reflection, and cultural expression, grounding the school in a rhythm of gathering and openness.

Buildings are sited in harmony with the existing topography and oriented to optimize natural ventilation and daylight. Green buffers and permeable surfaces support passive cooling, while key sightlines connect to Mount Ol Donyo Orok, reinforcing a deep sense of place.

Programmatic zoning places learning clusters at the quiet core, with recreation, vocational areas, and access roads positioned around the perimeter. The result is a climate-conscious, pedestrian-friendly campus that celebrates tradition while embracing innovation—creating a nurturing environment for both learning and community life.

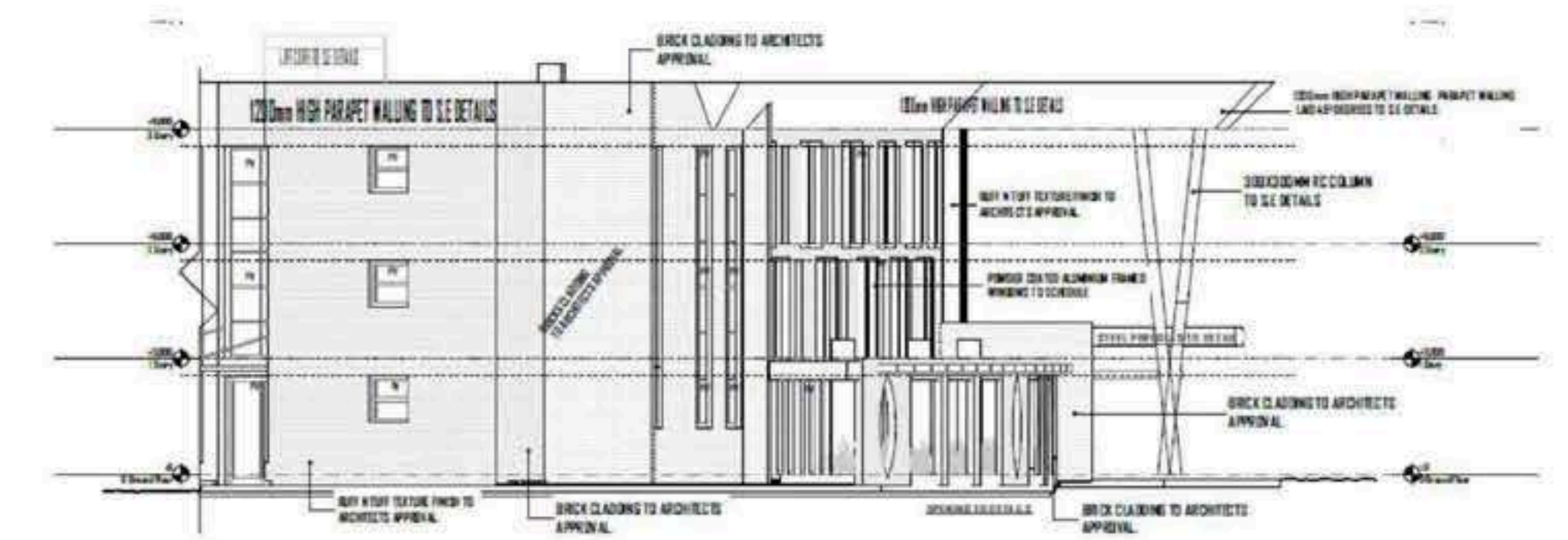




The design of the administration and classroom blocks focuses on optimizing the thermal comfort of students in response to Namanga's hot and dry climate. The layout includes numerous social spaces, versatile areas for presentations and events, abundant natural lighting, and dedicated offices for staff members.

The classrooms feature generous space to facilitate effective air circulation. Large windows, equipped with permanent ventilation, ensure ample natural light and fresh air. The atrium serves to naturally illuminate and ventilate the corridors. Additionally, the brick walls offer substantial thermal mass, effectively reducing daytime heat.

The design for Namanga based school responds to the hot and dry climate by incorporating deep roof overhangs for sun shading, solar panels for sustainable energy, and strategically placed wet and circulation cores along the eastern and western axes to optimize natural ventilation and cooling.



IFTIIN FOUNDATION PEACE HUB

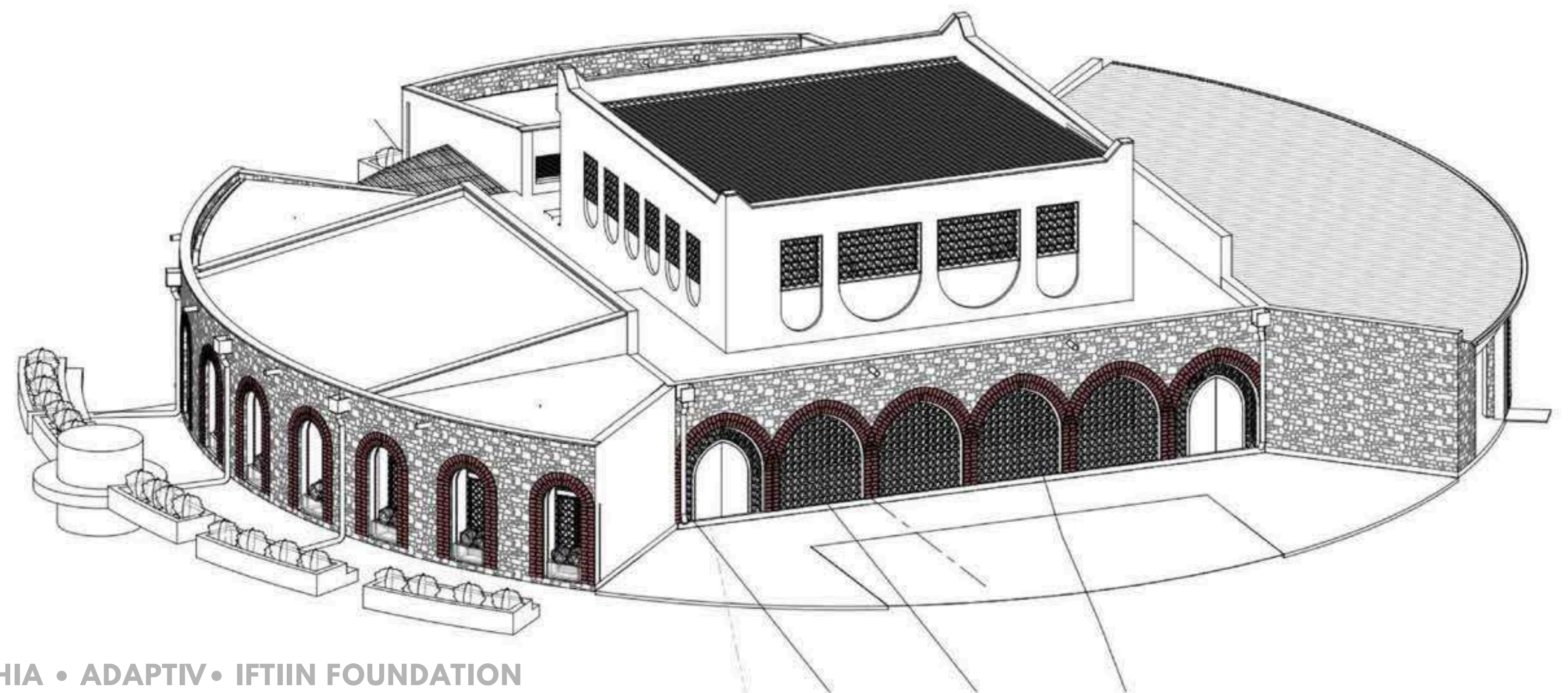
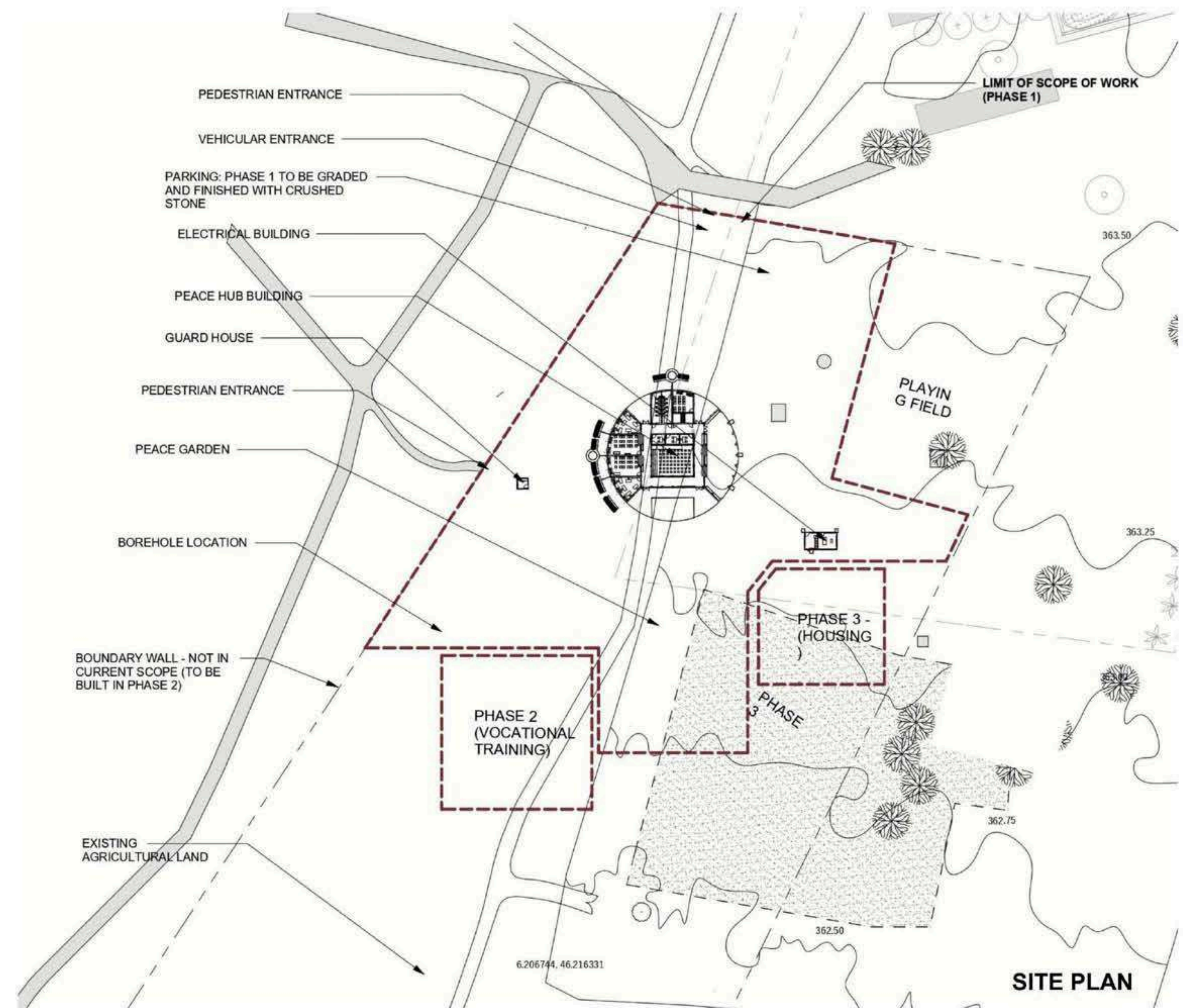
ABUD WAK, SOMALIA 2024
Architecture: Adaptiv
Lead Principal: Rob Freni
Architect: Georgina Kinuthia,

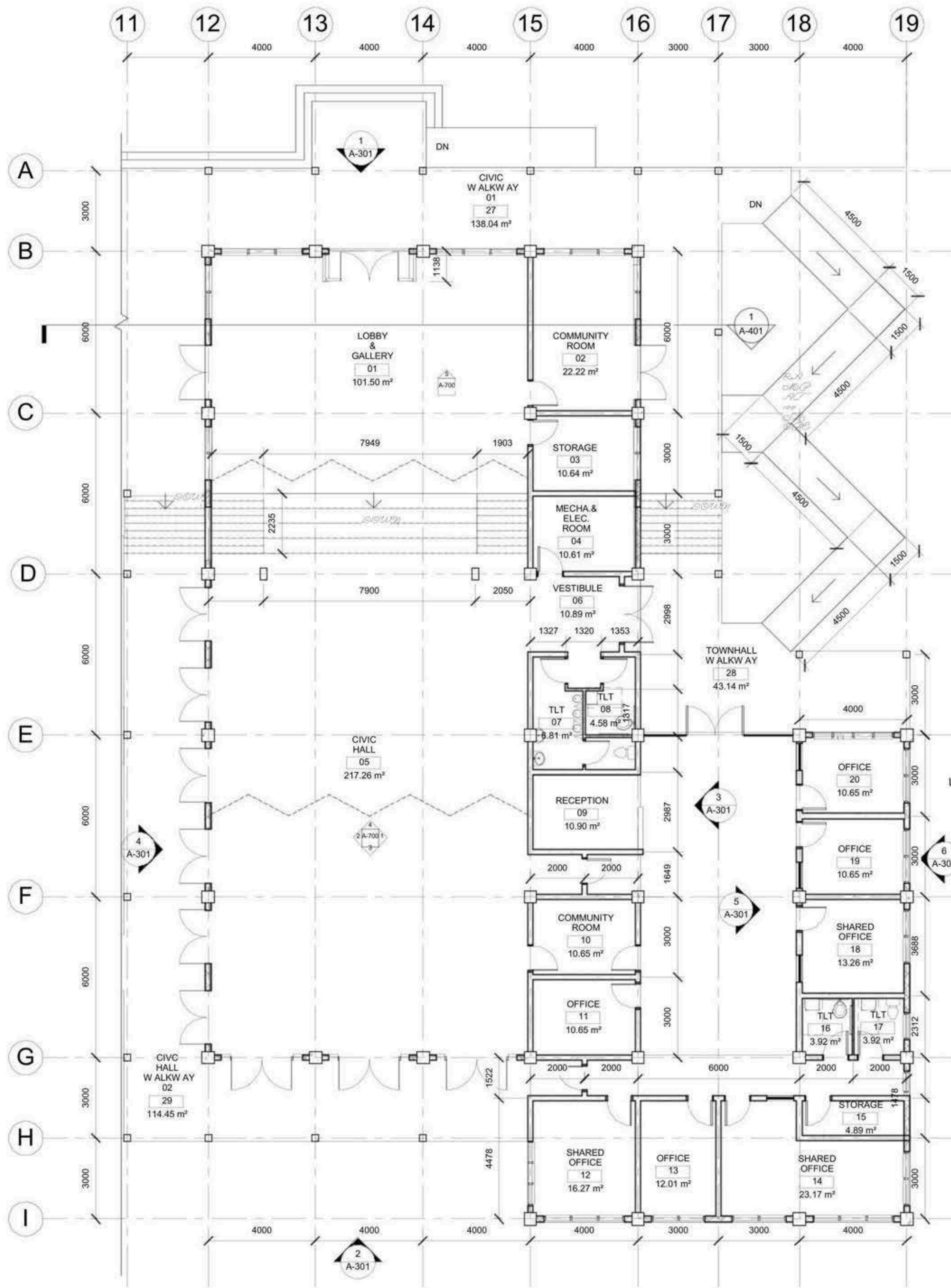
In Abudwak, Somalia—a place shaped by displacement and resource scarcity—the Iftiin Peace Hub reimagines civic space as a platform for opportunity. Designed in partnership with the Iftiin Foundation and supported by USAID, the project blends architectural sensitivity with economic and environmental intelligence.

The circular layout centers around a shared courtyard—a nod to Somali communal culture and a passive strategy for climate comfort. Thick masonry walls of locally quarried limestone ensure thermal mass, reducing energy needs. Shaded colonnades and natural cross-ventilation minimize reliance on mechanical cooling, lowering operational costs in a fuel-constrained context. Socially, the Hub functions as both anchor and launchpad: it houses vocational classrooms, psychosocial services, and enterprise incubators—each space designed for flexibility and long-term relevance. We considered not just form and flow, but future proofing: adaptable floor plans, low-maintenance materials, and user-friendly systems.

Environmentally, the design includes rainwater harvesting, permeable walkways, and native landscaping—reducing strain on the fragile local water table. Economically, we prioritized local labor and materials, injecting value directly into the community supply chain.

This project is a civic node and a proof of concept—showing how architecture can help stabilize lives, circulate resources, and grow possibility from the ground up.

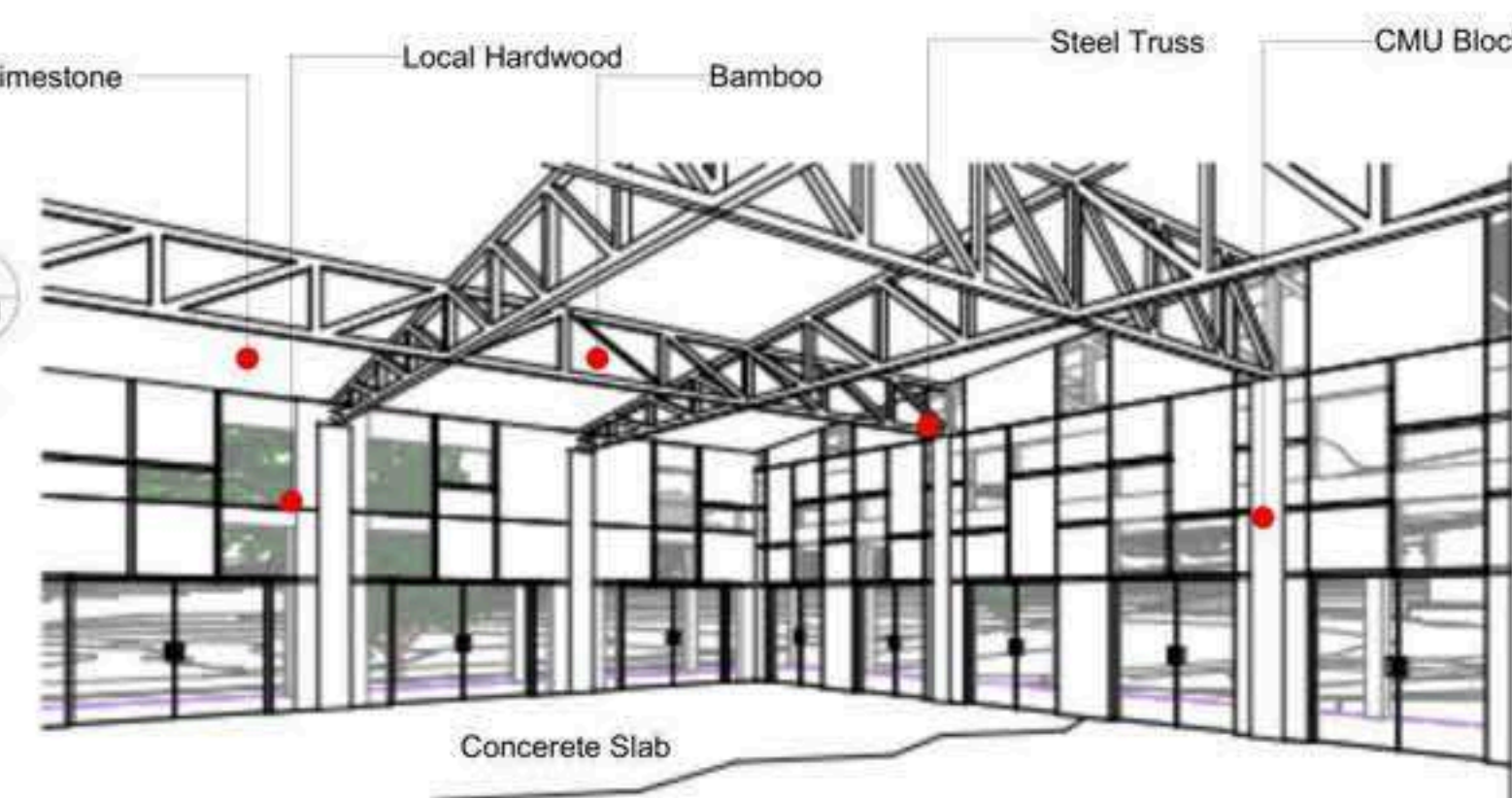




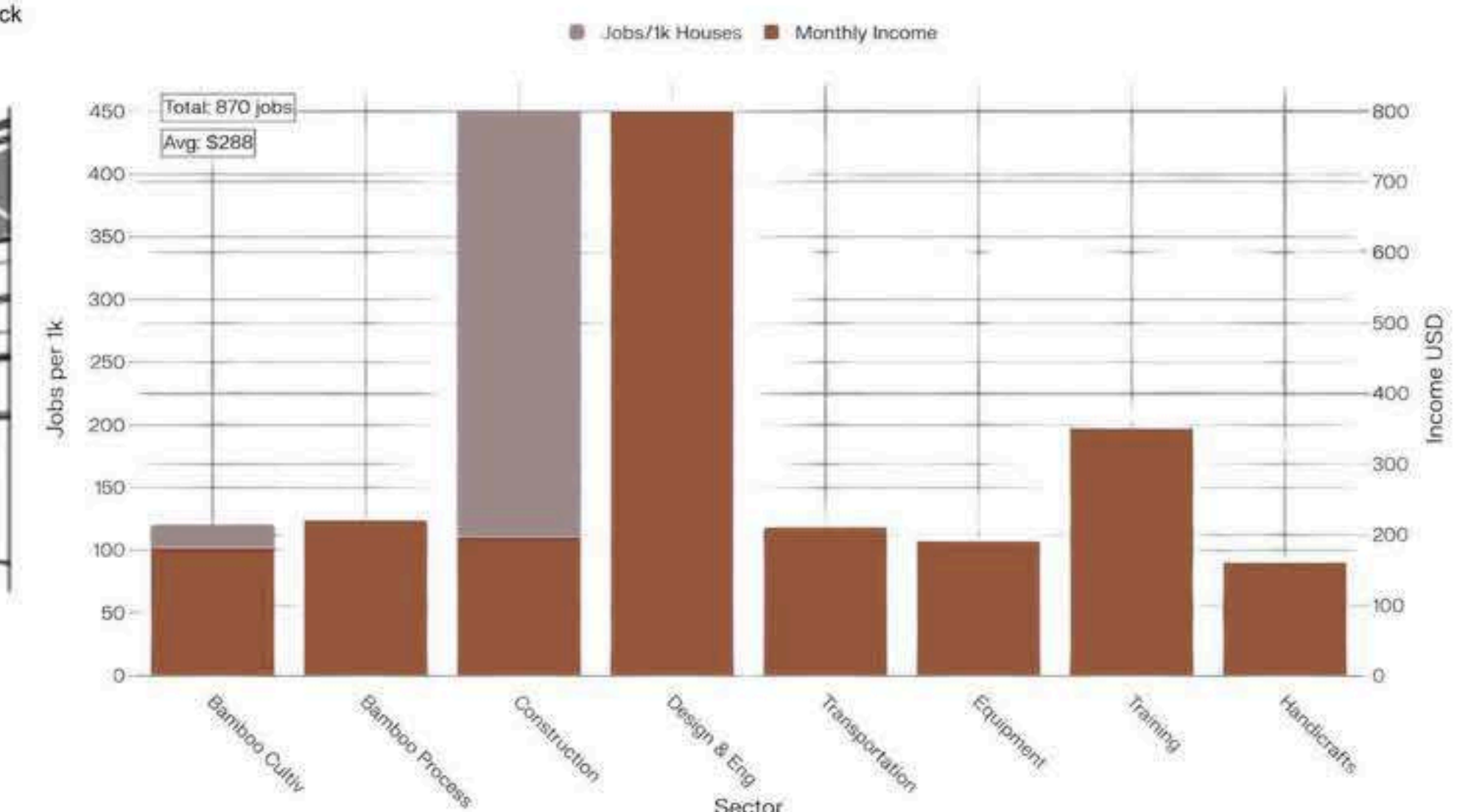
CIVIC HALL & OFFICE FLOOR PLAN

Material	Source (quarry, plantation, recycler)	Typical form	Compressive or tensile strength	Unit cost (USD/t or m²)	Carbon intensity (kg CO ₂ e/unit)	Availability rating (High/Med/Low)
Stone (Limestone)	Local quarries in hills	Cut stone blocks	30-80 MPa	\$15-25/m³	5-8/m³	High
Stone (Basalt)	Volcanic outcrops	Crushed aggregate	80-120 MPa	\$20-30/m³	3-6/m³	High
Sand (River)	River beds/deposits	Washed sand	N/A	\$8-12/m³	2-4/m³	Medium
Sand (Marine)	Coastal areas	Natural sand	N/A (may contain salt)	\$6-10/m³	1-3/m³	Medium
Clay/Earth	Local soil deposits	Adobe blocks/mortar	2-5 MPa	\$5-8/m³	1-2/m³	High
Timber (Local hardwood)	Remnant forests	Sawn lumber	25-45 MPa	\$150-300/m³	15-25/m³	Low
Bamboo	Community plantations	Poles/woven panels	40-80 MPa tensile	\$2-5/pole	0.5-1/pole	Medium
Concrete blocks (CMU)	Local block producers	Standard CMU blocks	8.2 MPa (typical local)	\$0.80-1.20/block	25-35/block	High
Crushed stone aggregate	Stone quarries	Graded aggregate	N/A	\$12-18/m³	4-7/m³	High
Lime (from limestone)	Limestone kilns	Hydrated lime powder	N/A	\$120-180/t	850-950/t	Medium
Pozzolan (volcanic ash)	Volcanic deposits	Fine powder	N/A	\$10-15/t	5-10/t	Medium
Thatch/Palm fronds	Community harvesting	Dried fronds	N/A	\$1-3/m²	0.1-0.5/m²	High

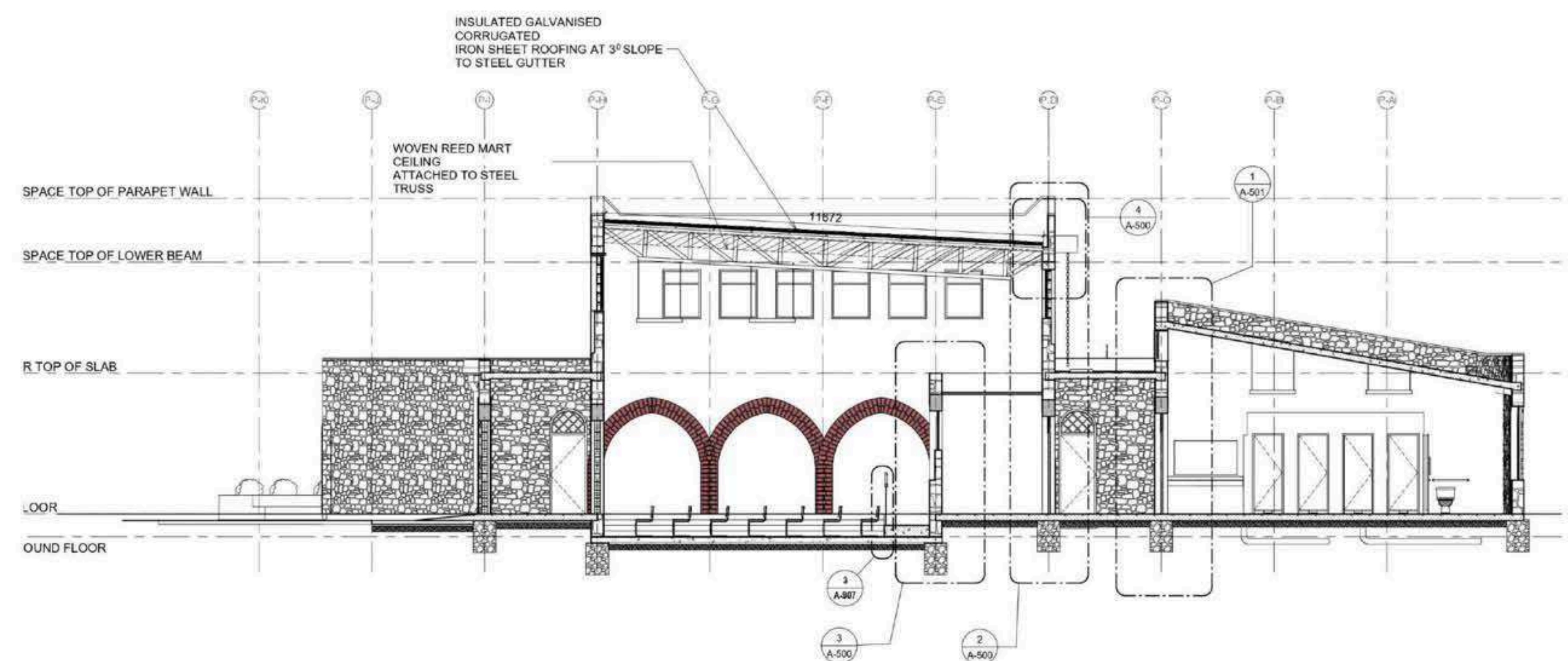
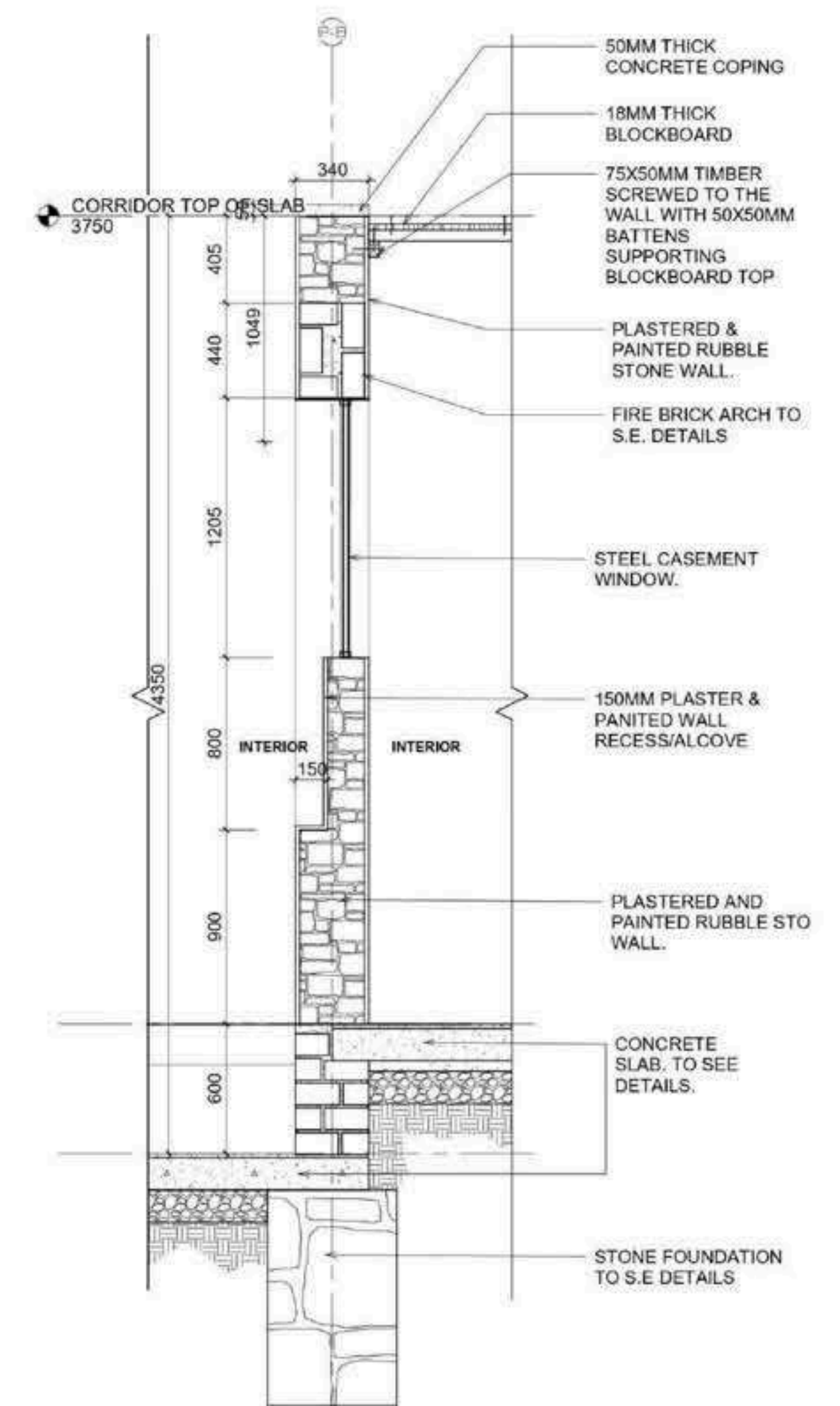
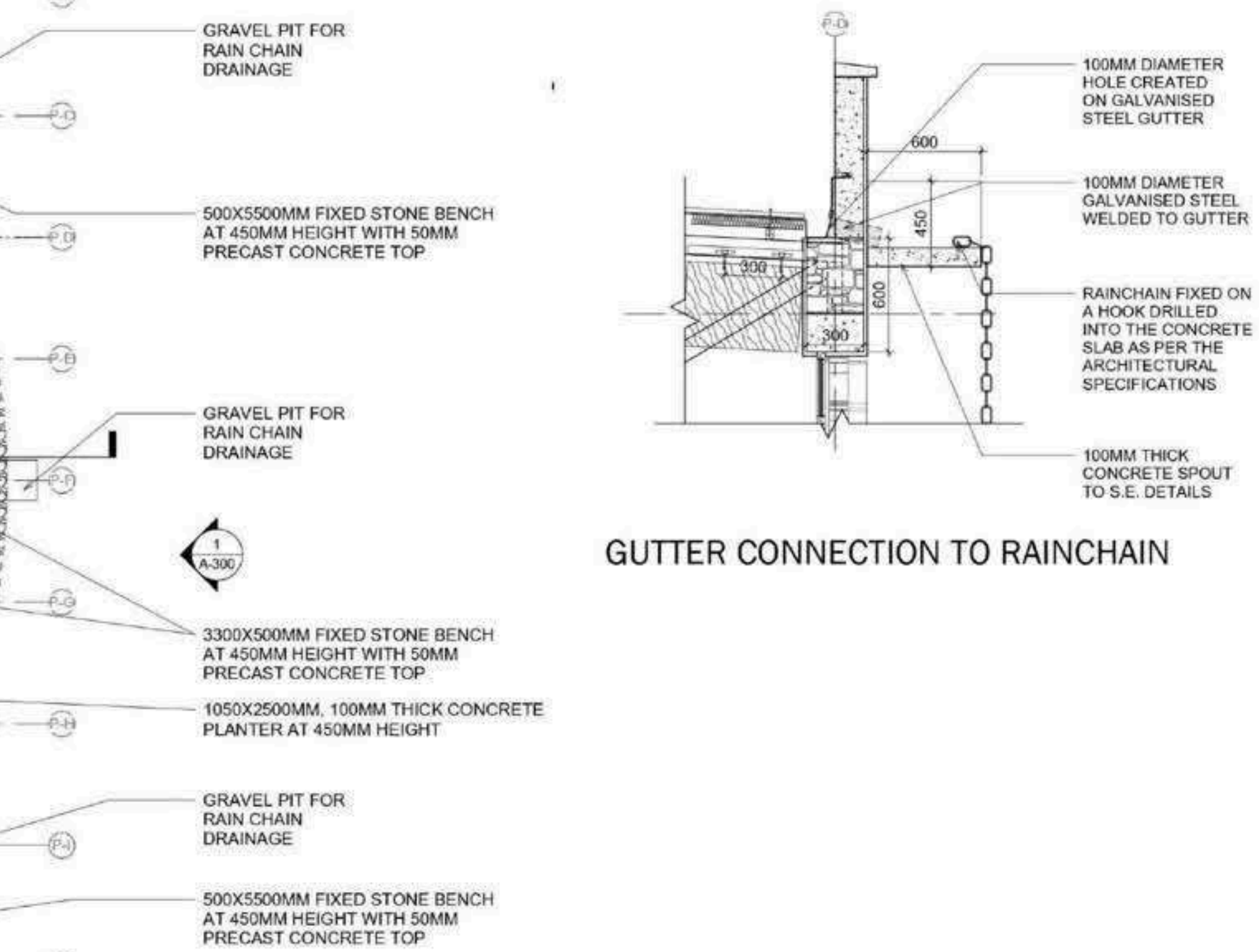
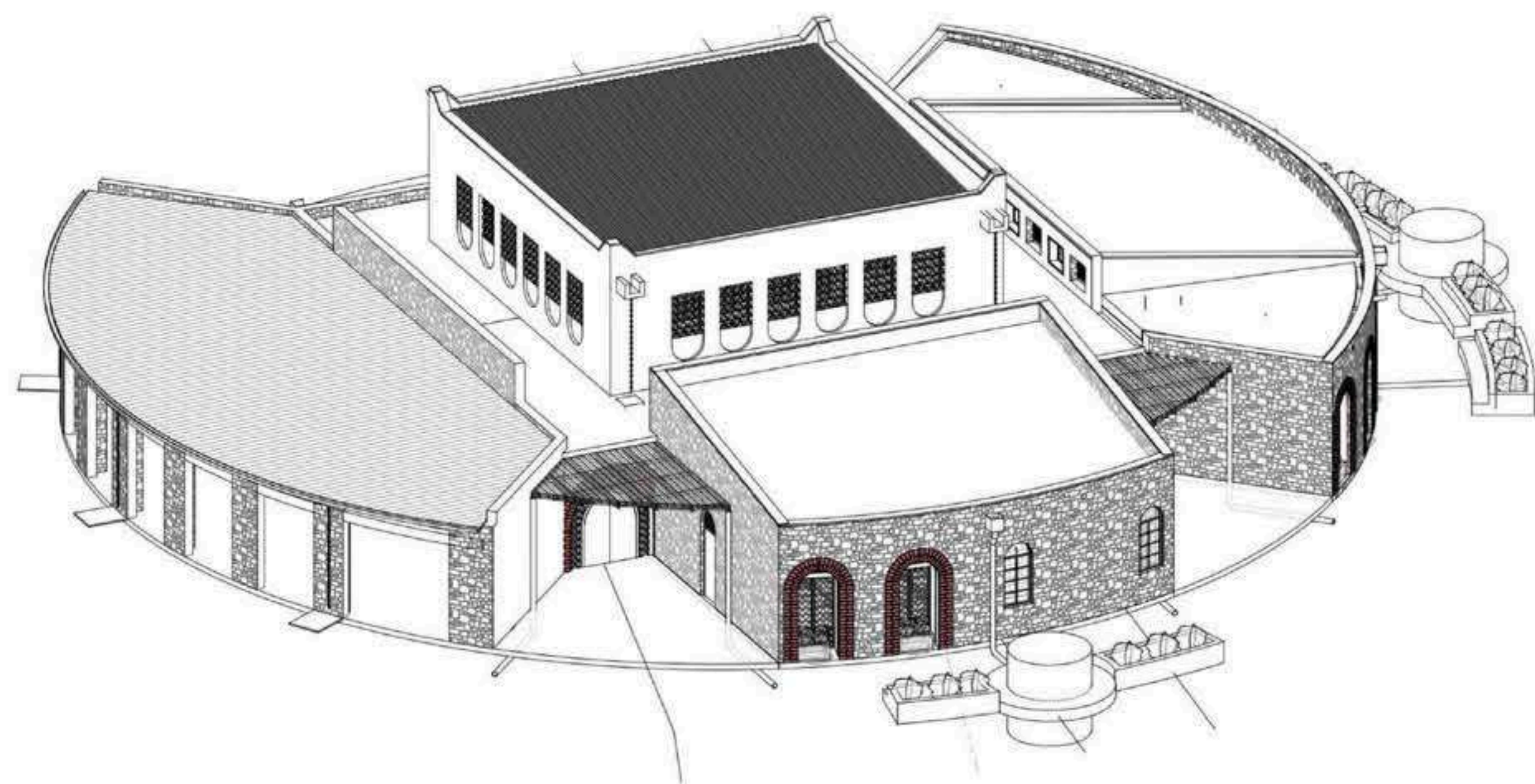
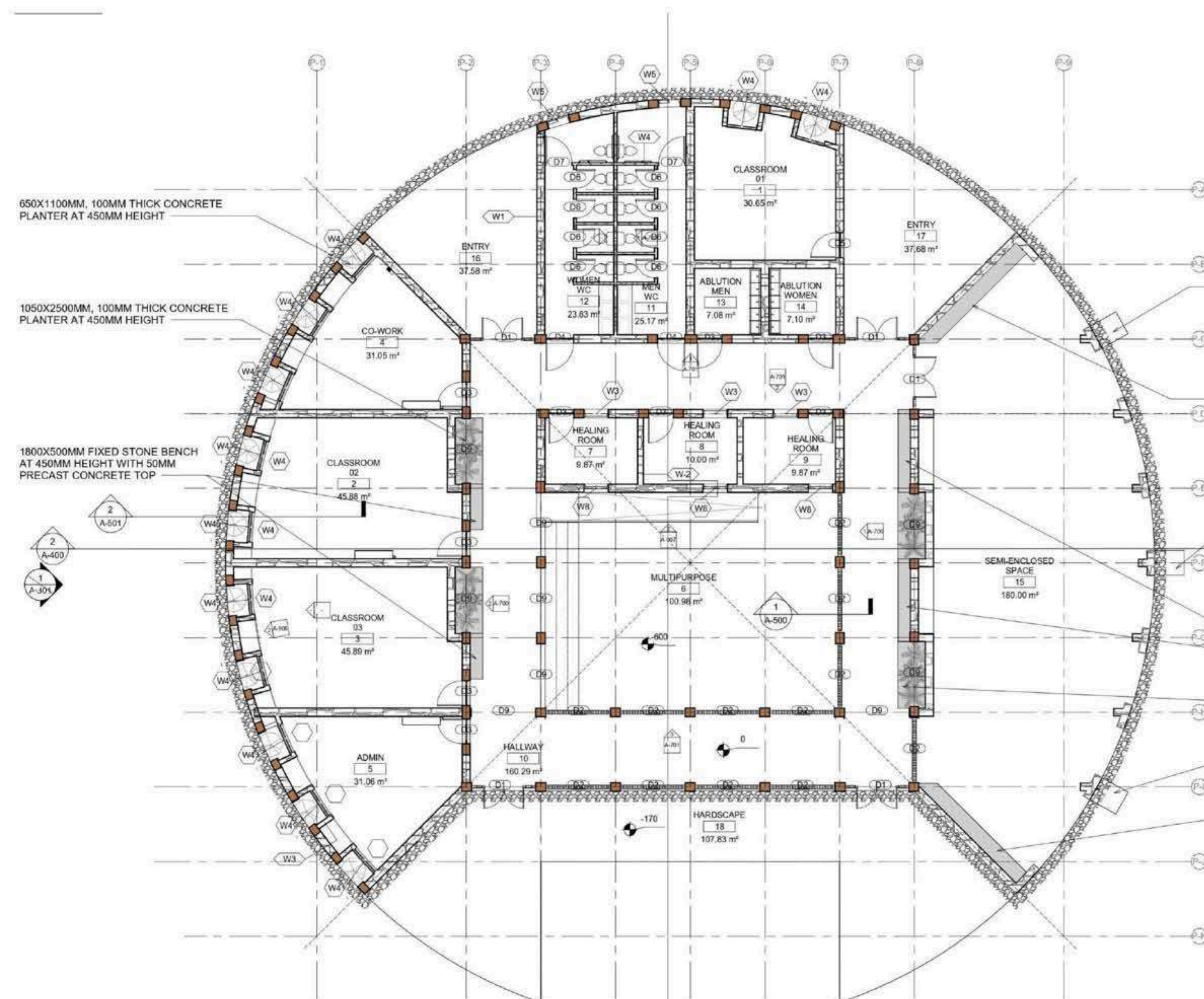
Material logic reflects what the land offers and the climate demands. Stone and earth from local quarries and soil deposits form the structural backbone — affordable, abundant, and strong enough for seismic performance when detailed correctly. Marine sand, while accessible, demands treatment; timber, increasingly rare, pushes the case for bamboo or earth-based systems. Low-carbon materials like thatch, pozzolans, and lime mortar offer scalable, culturally familiar solutions, with most key components sourced within 100 miles. The design grows from the ground it stands on — materially efficient, contextually intelligent, and socially embedded.

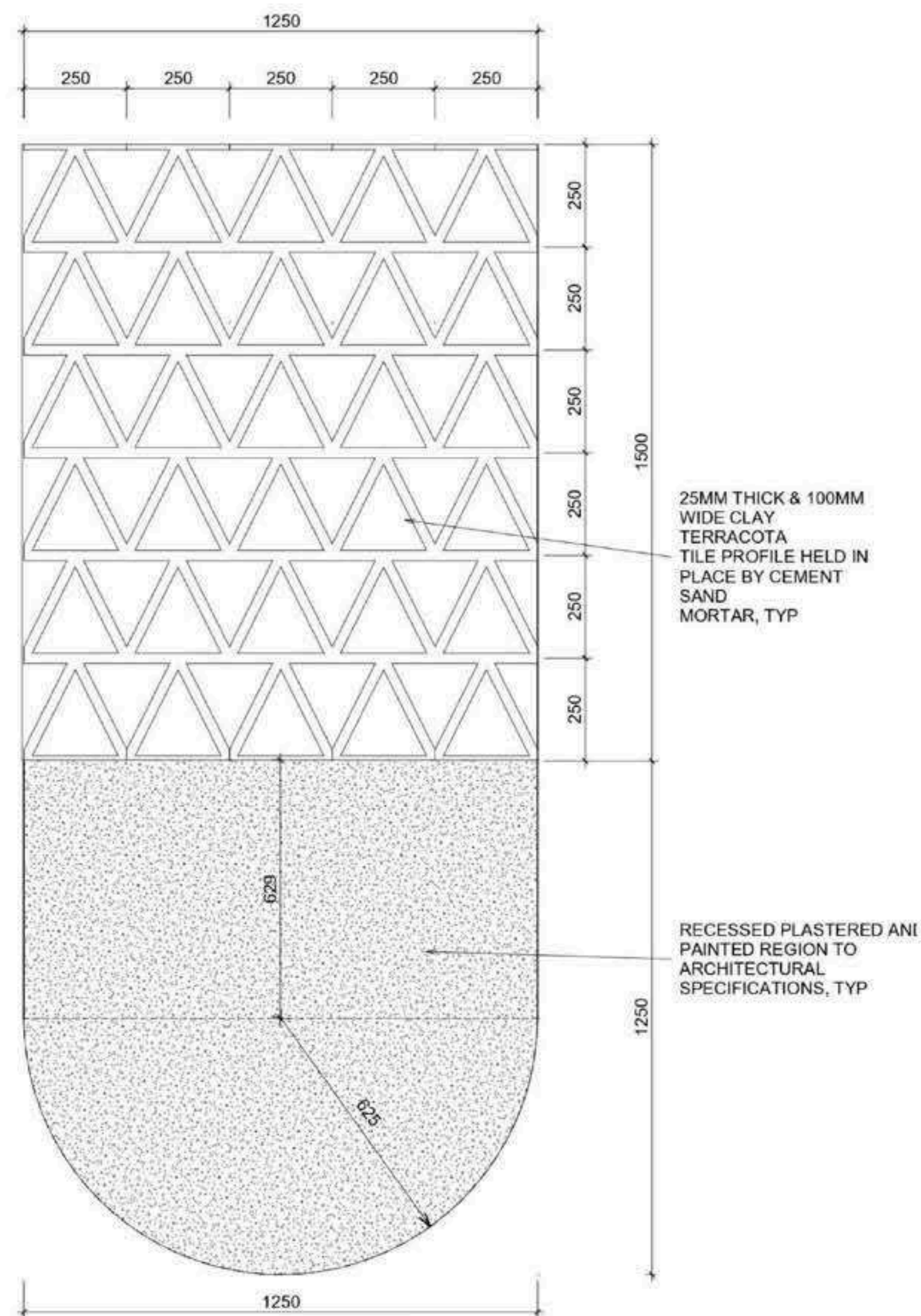


Haiti Bamboo Job Creation by Sector

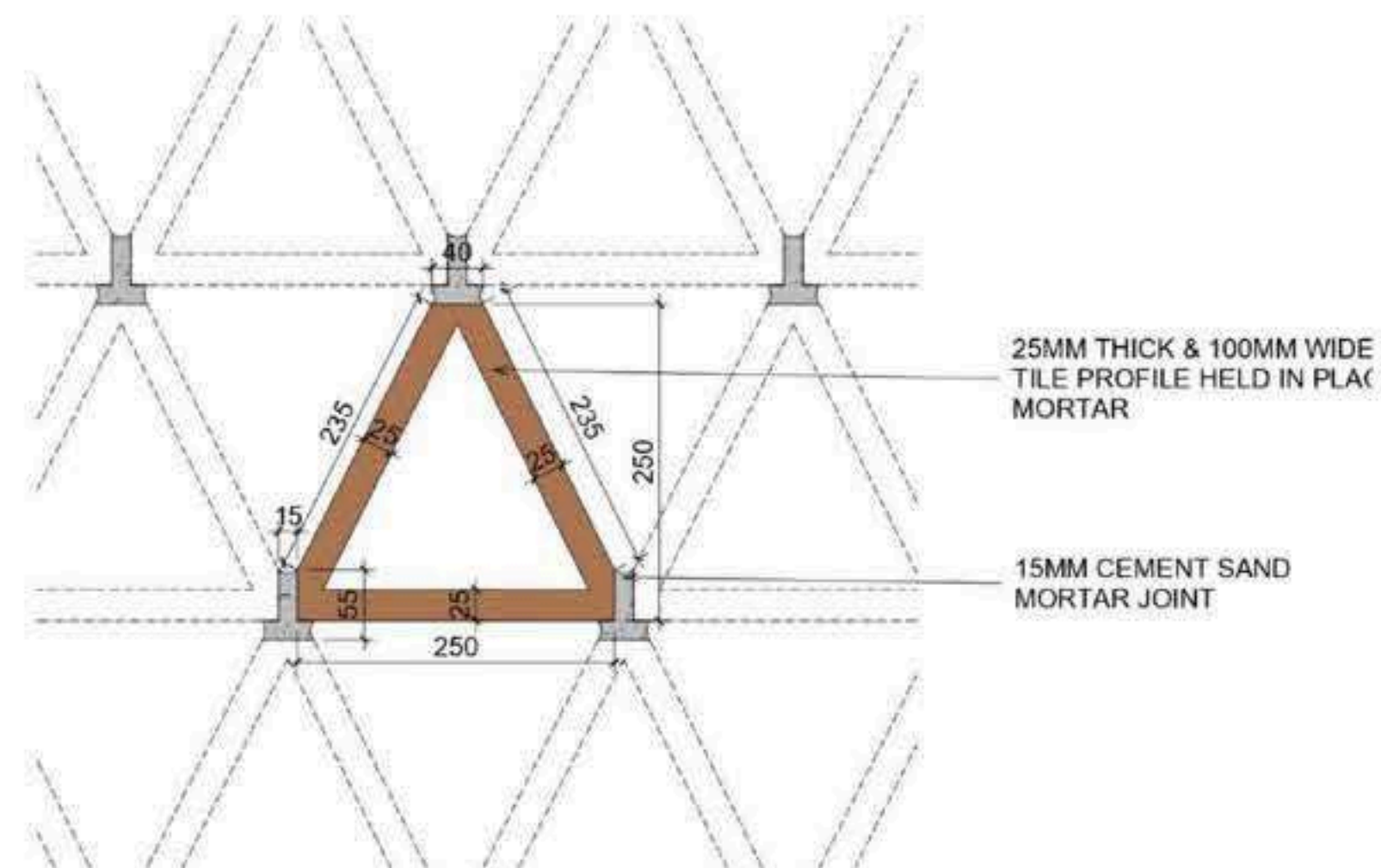


Bamboo construction presents Haiti with a transformative economic opportunity that addresses multiple development challenges simultaneously. The 67% cost reduction compared to traditional construction methods makes housing more affordable while the 870 jobs created per 1000 houses provide substantial employment opportunities across skill levels.

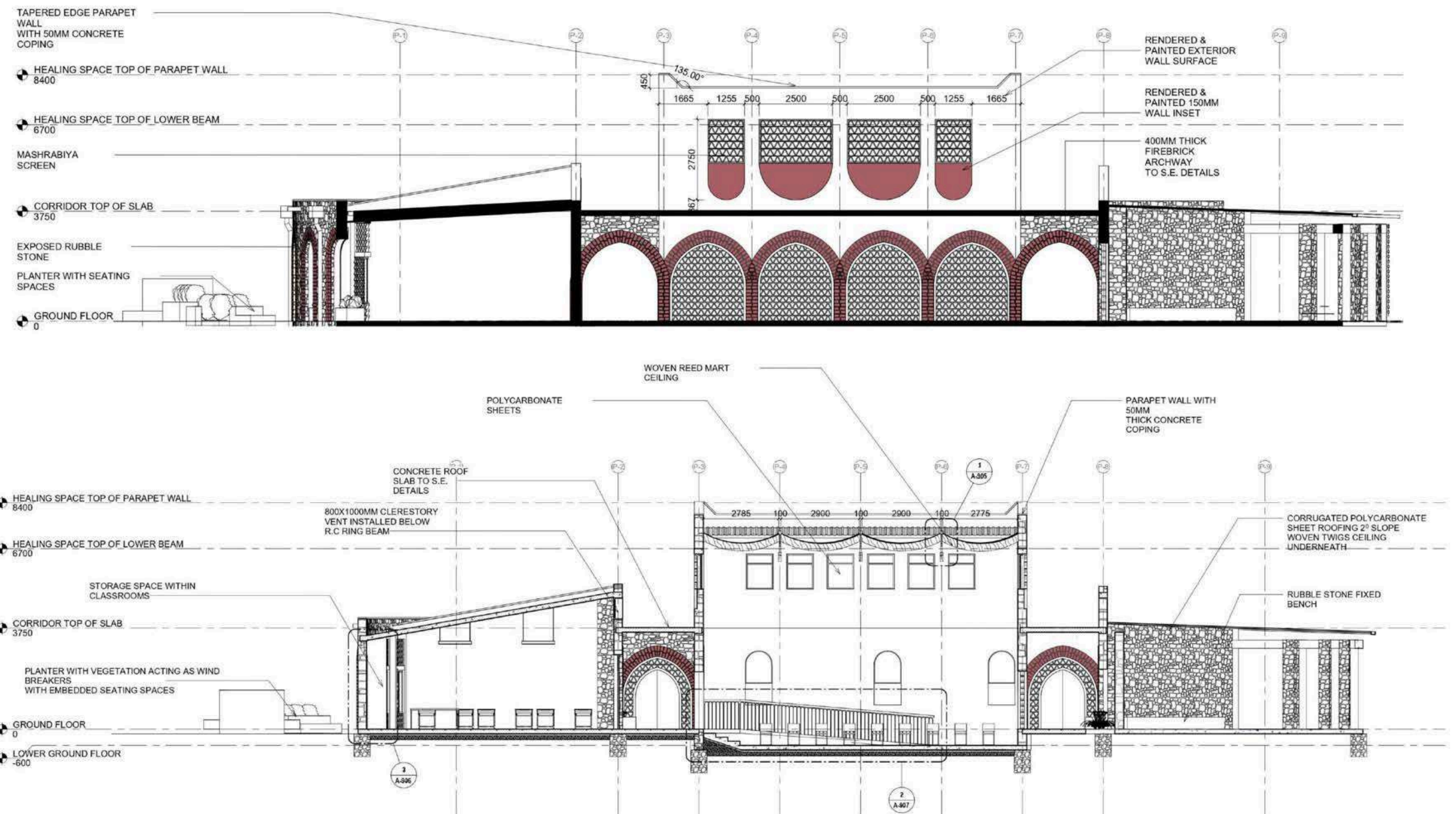




CLAY TILE MASHRABIYA (WINDOW A)



CLAY TILE ELEVATION



FDB CIVIC CENTER

FOND DES BLANC, HAITI, 2024

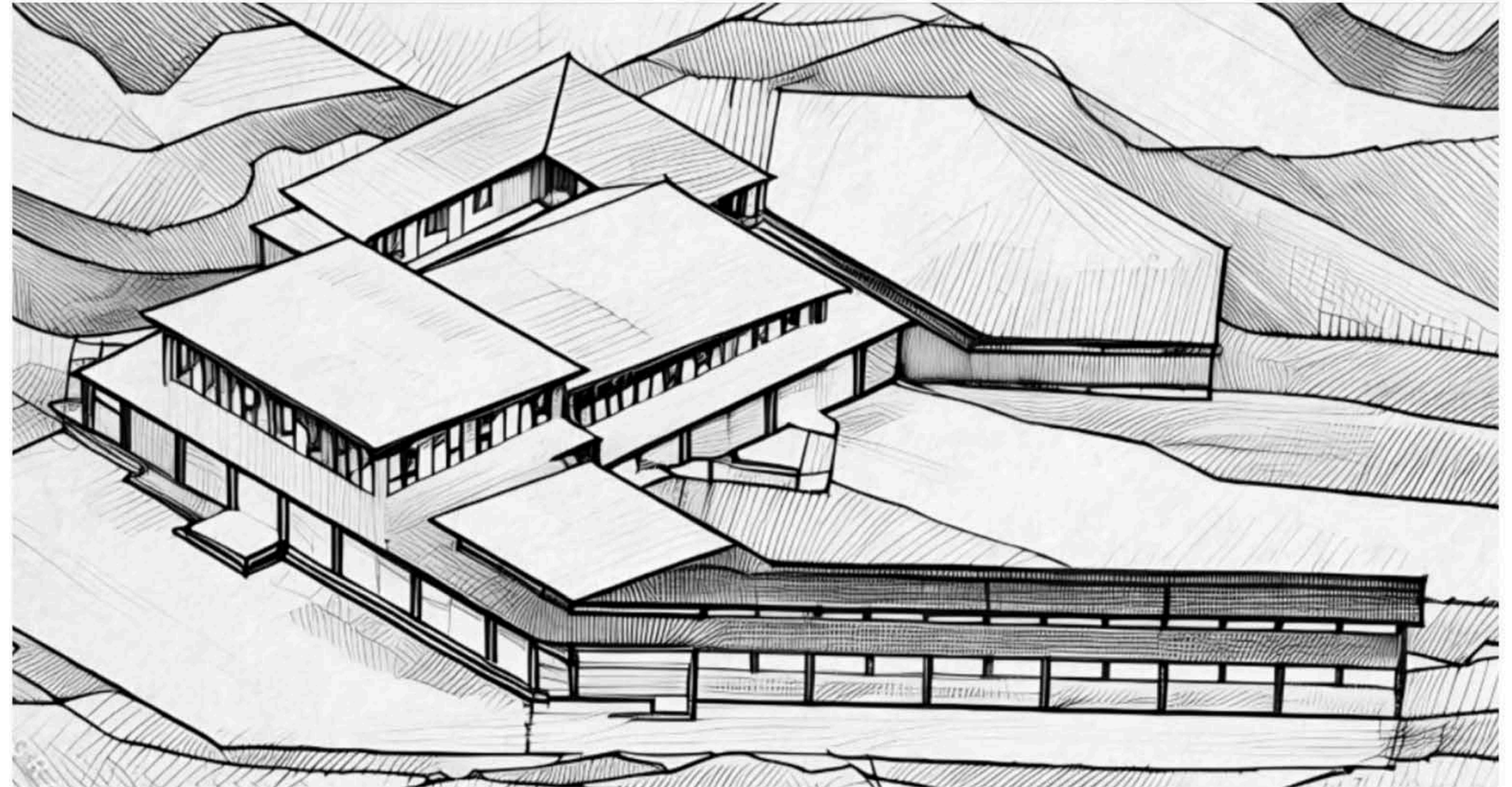
Architecture: Adaptiv
Lead Principal: Rob Freni
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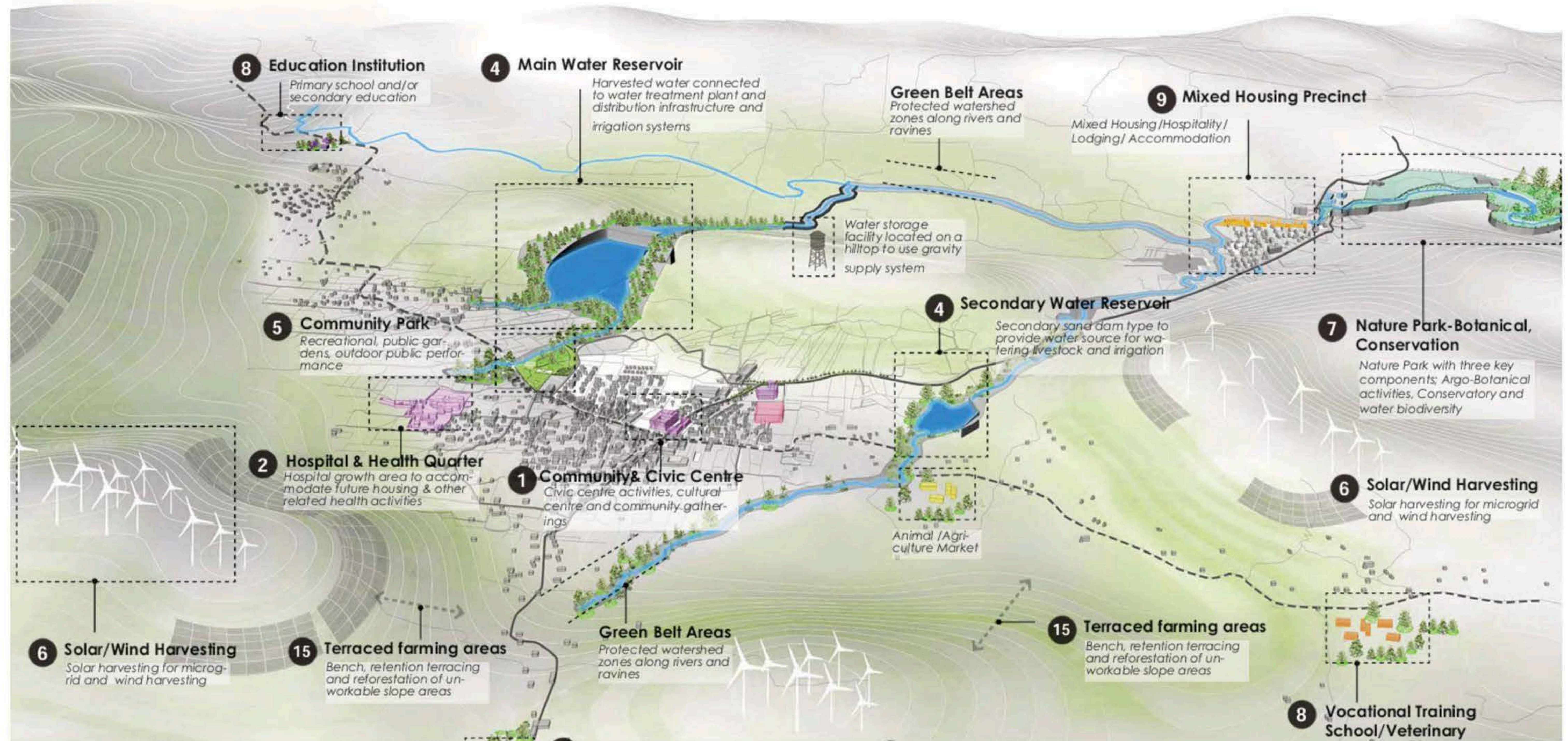
This Civic Center was imagined not just as a building, but as a flexible civic backbone — a place for gathering, governance, education, and care. In a time of deep turbulence in Haiti, the core design concept was adaptability: spaces that could shift in function, expand or contract in use, and remain relevant amid uncertainty.

The process began with early dialogue and the formation of a Community Planning Advisory Committee (CPAC), ensuring that every step was shaped by local leadership. Public launches, workshops, and household surveys grounded the design in lived realities and collective priorities.

The architecture reflects a layered response to need: robust enough for public services, open enough for informal gatherings, and dignified enough to restore trust in shared space. Locally sourced materials were prioritized to support resilience, affordability, and community identity.

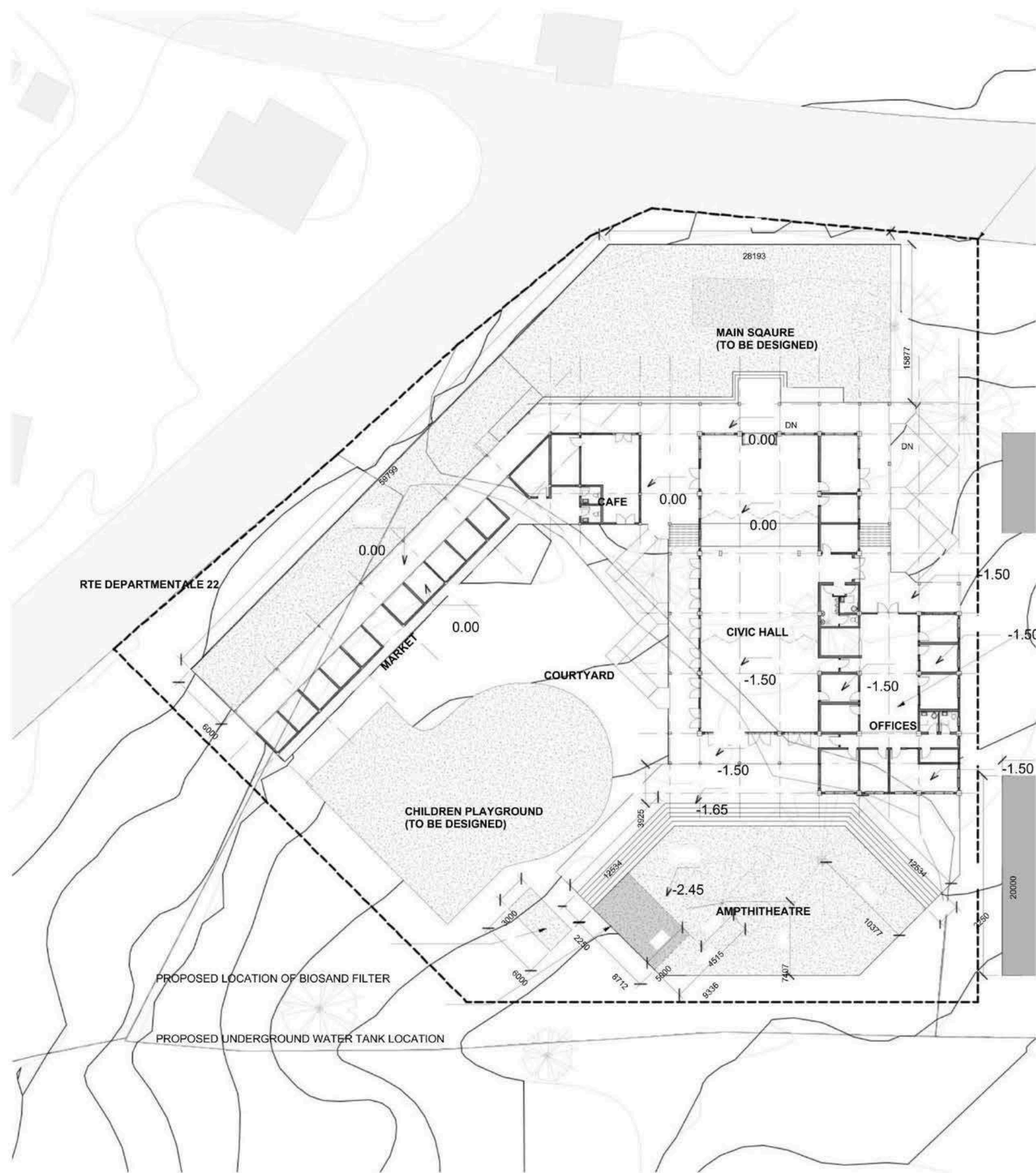
Aligned with SDG 11 (Sustainable Cities and Communities) and SDG 16 (Peace, Justice and Strong Institutions), the center is both a tool and a symbol—co-designed, place-based, and ready to evolve as the community does.





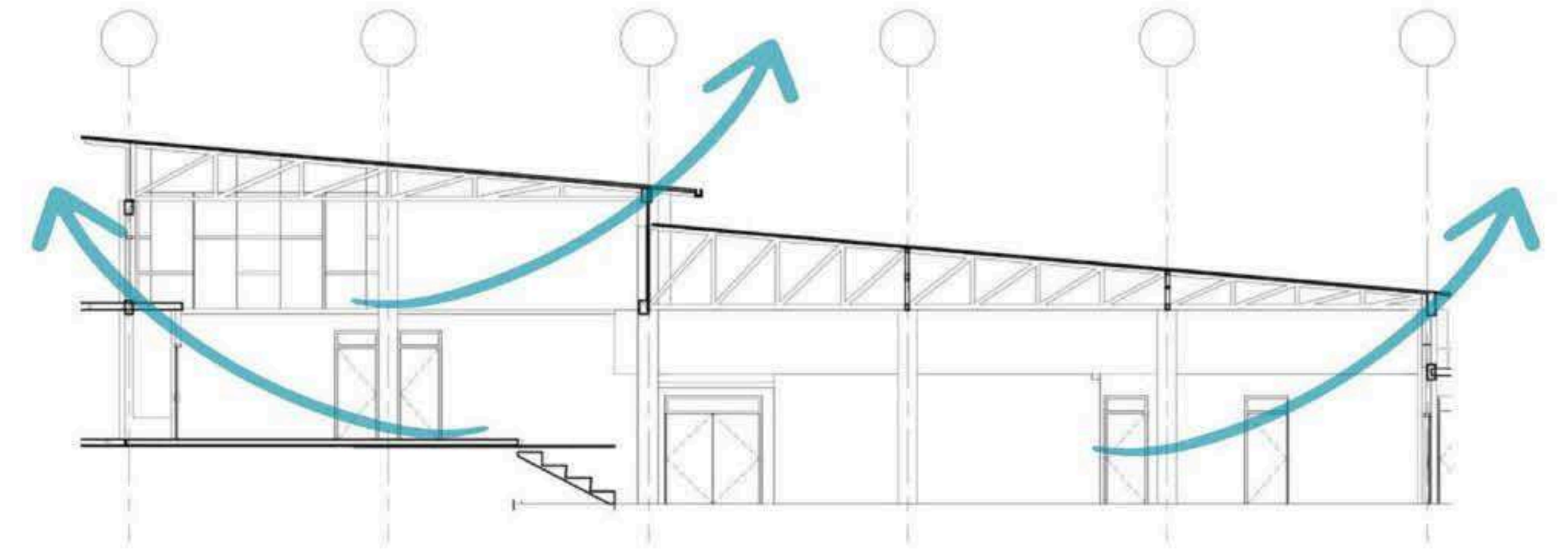
A Civic Core for Resilient Public Life

This Civic Center sits at the heart of a masterplan designed to strengthen public life through spatial, social, and environmental resilience. Strategically located at the junction of the main road, market, and village threshold, the site was chosen to amplify existing patterns of movement and social exchange. Designed for adaptability amid Haiti's ongoing turbulence, the center serves as a flexible civic core—capable of hosting governance functions, public services, and community refuge. The project draws heavily on local materials and vernacular methods, reducing embodied carbon while supporting the regional economy. Over 60% of resources are locally sourced, and the design integrates passive strategies for water and energy harvesting, enabling long-term self-sufficiency. Its planning process was deeply participatory, led by a Community Planning Advisory Committee (CPAC) made up of elders, youth, traditional leaders, and faith-based actors, ensuring cultural and contextual relevance. More than a building, the Civic Center anchors a system of green infrastructure, agricultural zones, and protected heritage areas, all aligned with SDGs 11 and 16. It is a civic space co-authored by its community—rooted in place, resilient by design, and scalable by logic.



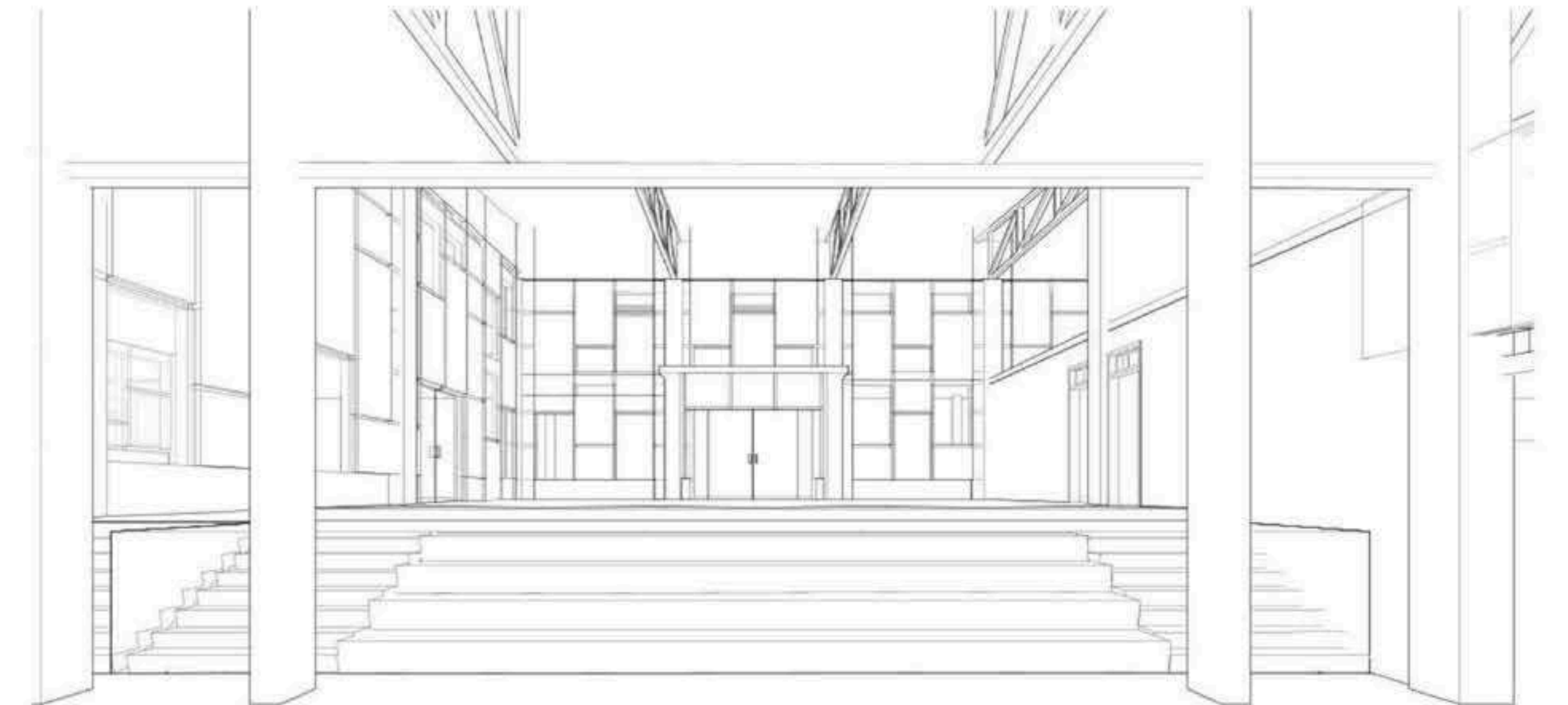
SITE LAYOUT

GEORGINA KINUTHIA • ADAPTIV • FDB CIVIC CENTRE

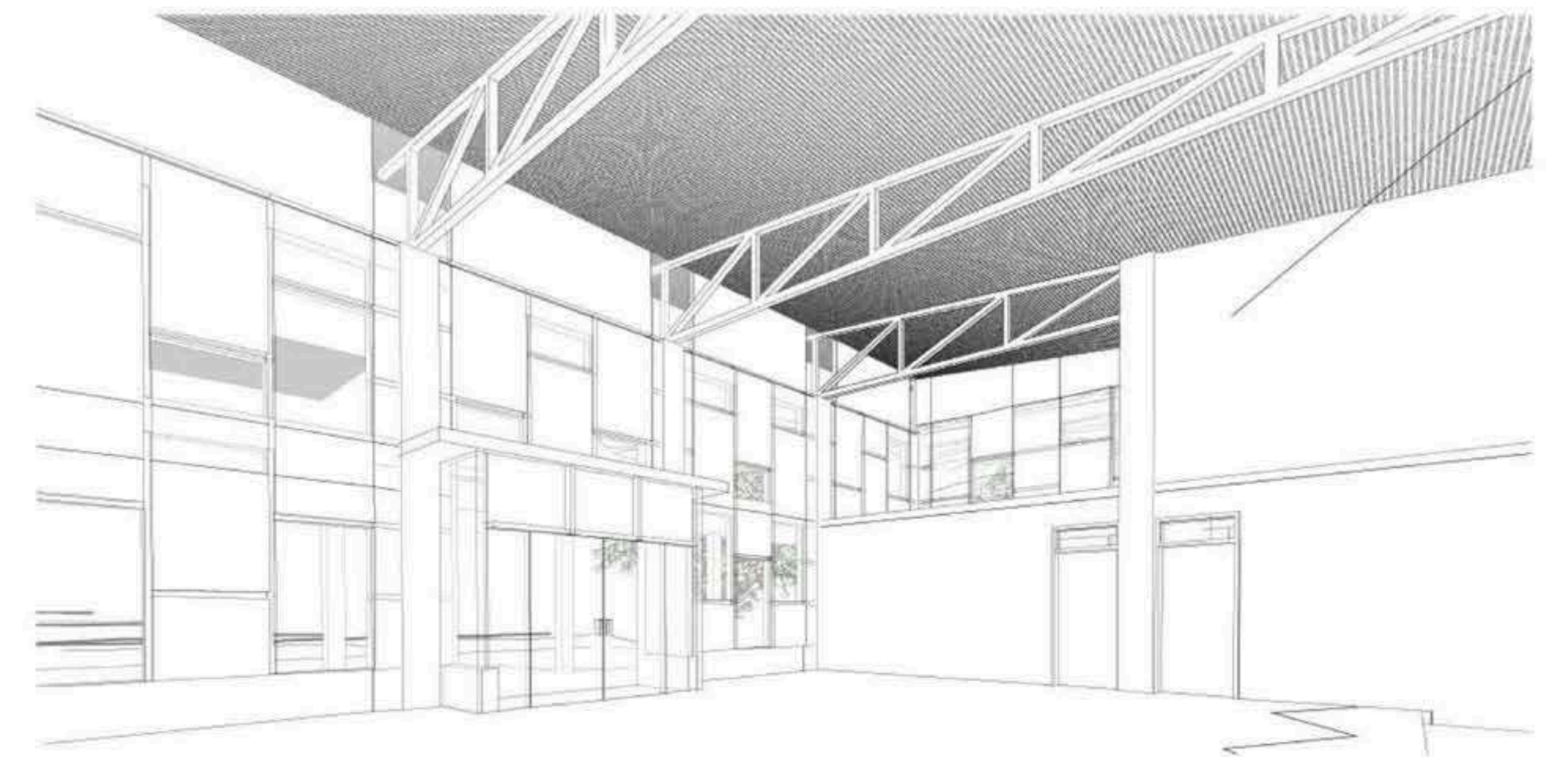


BUILDING SECTION

The carbon-conscious approach extended beyond the use of local materials to include climate-responsive strategies: stepped roofing to allow hot air to escape, maximized operability of doors and windows for natural airflow, strategic building orientation to minimize exposure to the harsh western sun, cross ventilation to enhance thermal comfort, and internal courtyards that create cooler microclimates.



INTERIOR PERSPECTIVE



SUMMITS EDUCATION

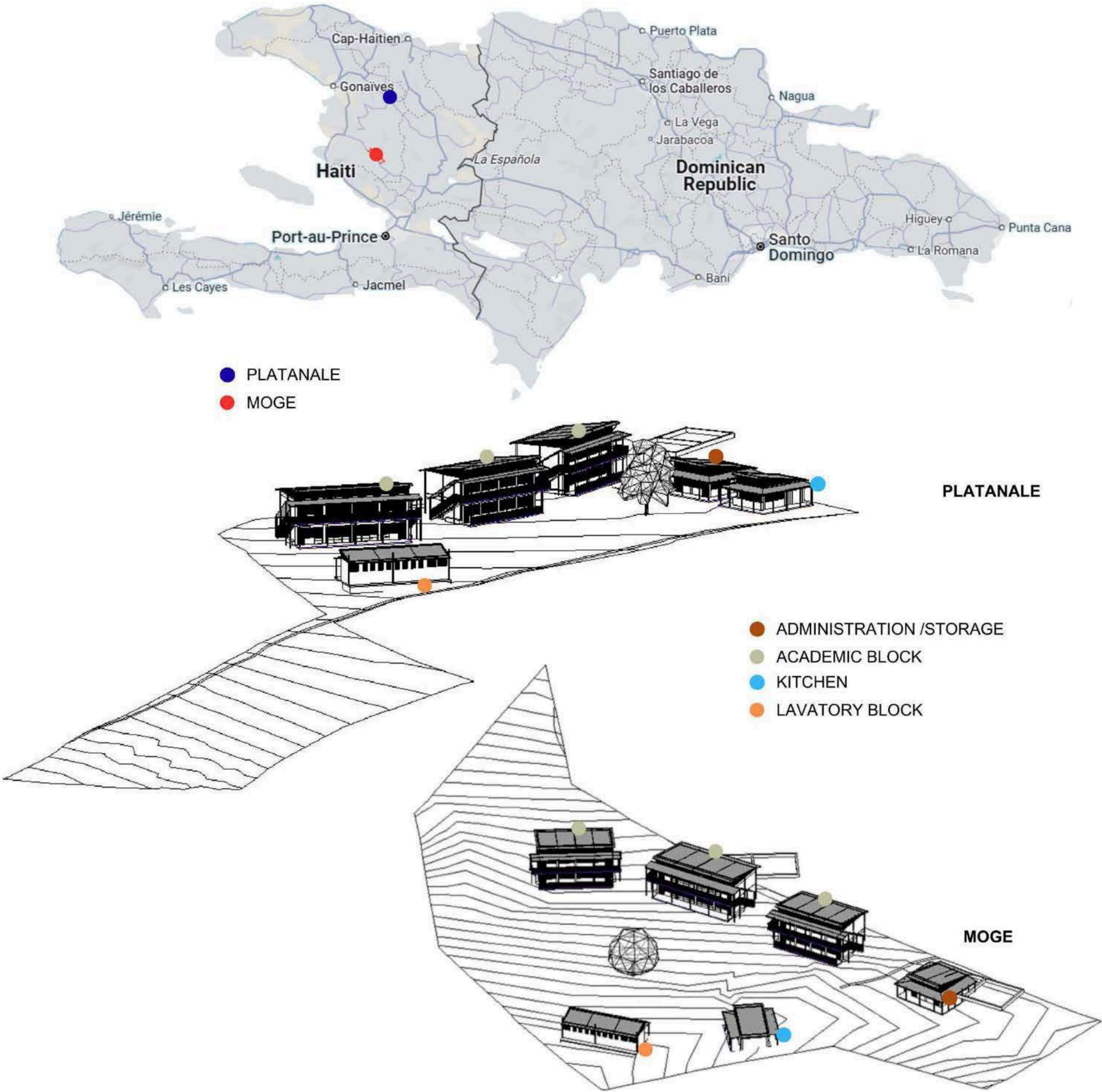
FACILITY PROTOTYPE

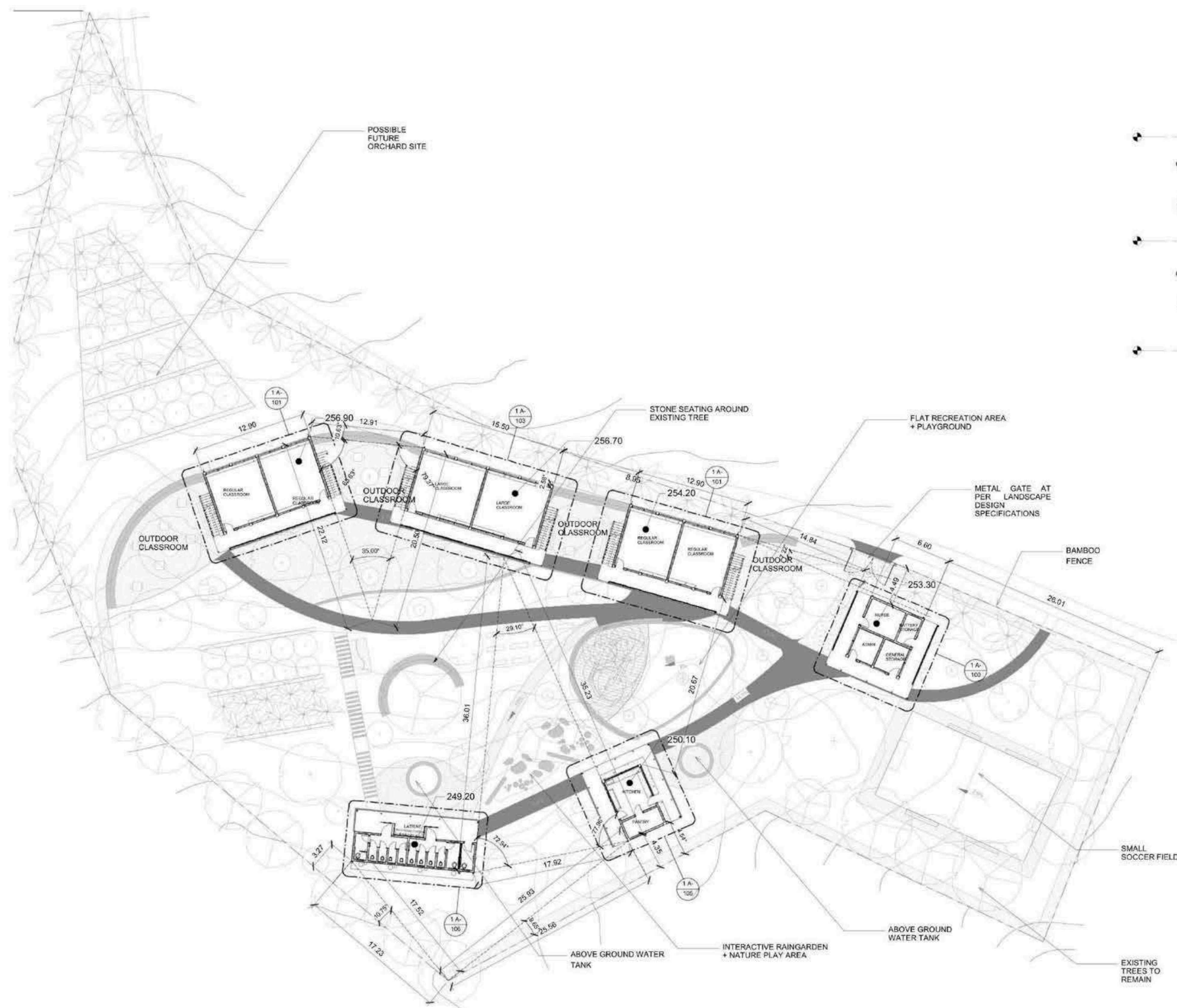
MOGE & PLATANALE, HAITI, 2024
Architecture: Adaptiv
Lead Principal: Rob Freni
Architect: Georgina Kinuthia

Education, in its most transformative form, begins with space: space that is safe, dignified, and designed for growth. In Haiti, where over 85% of primary schools are managed by non-state actors and physical infrastructure is often the weakest link, Summits Education set out to reimagine what an equitable learning environment could look like across its network of 40 schools. Following a comprehensive assessment in 2019 by Schools for the Children of the World (SCW), stark infrastructural disparities were revealed — some schools operating in well-structured facilities, others in conditions unfit for learning.

In response, Summits Education partnered with Adaptiv to develop a replicable, adaptable school prototype — rooted in climate-conscious construction, community labor systems, and local material logic. The result is a design approach not only informed by pedagogy and place, but also capable of being deployed across remote terrain. Anchored in site-specificity and systemic intent, this initiative brings spatial justice into the conversation on education — ensuring that every student, regardless of geography, learns in a space that reflects their worth.

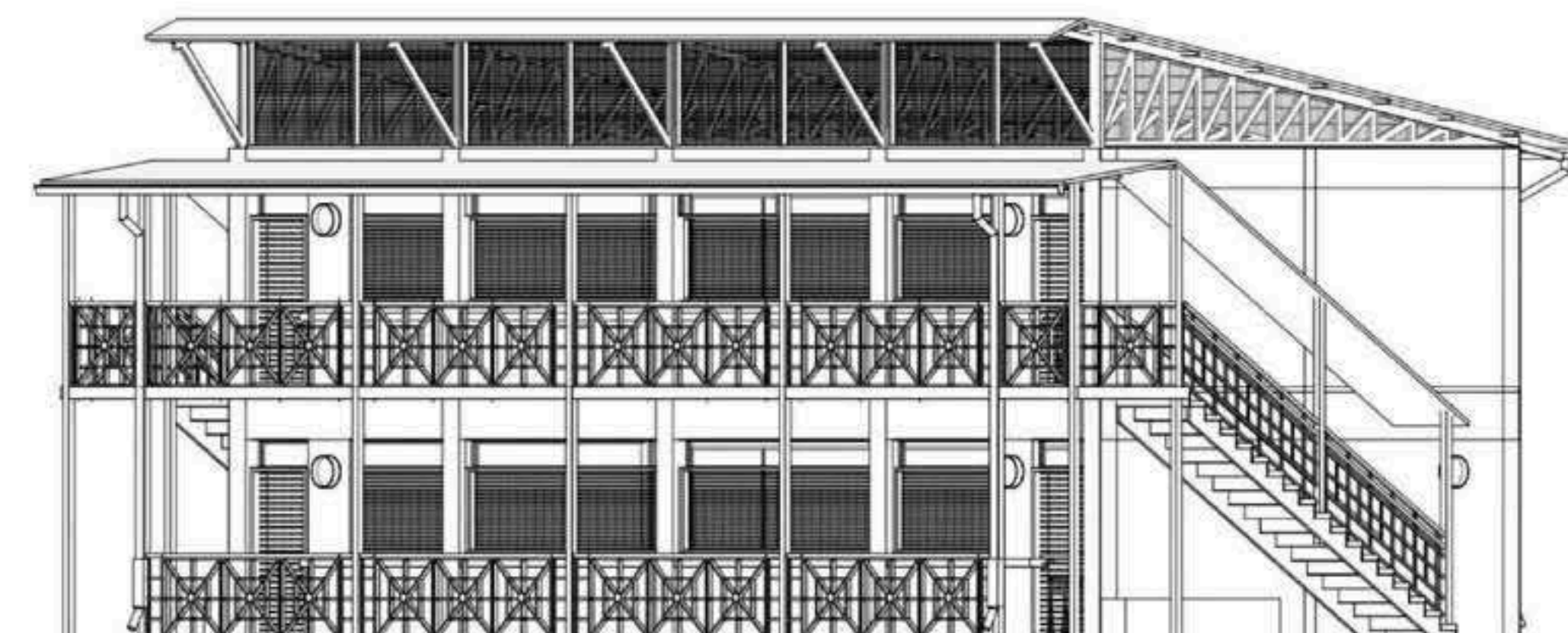
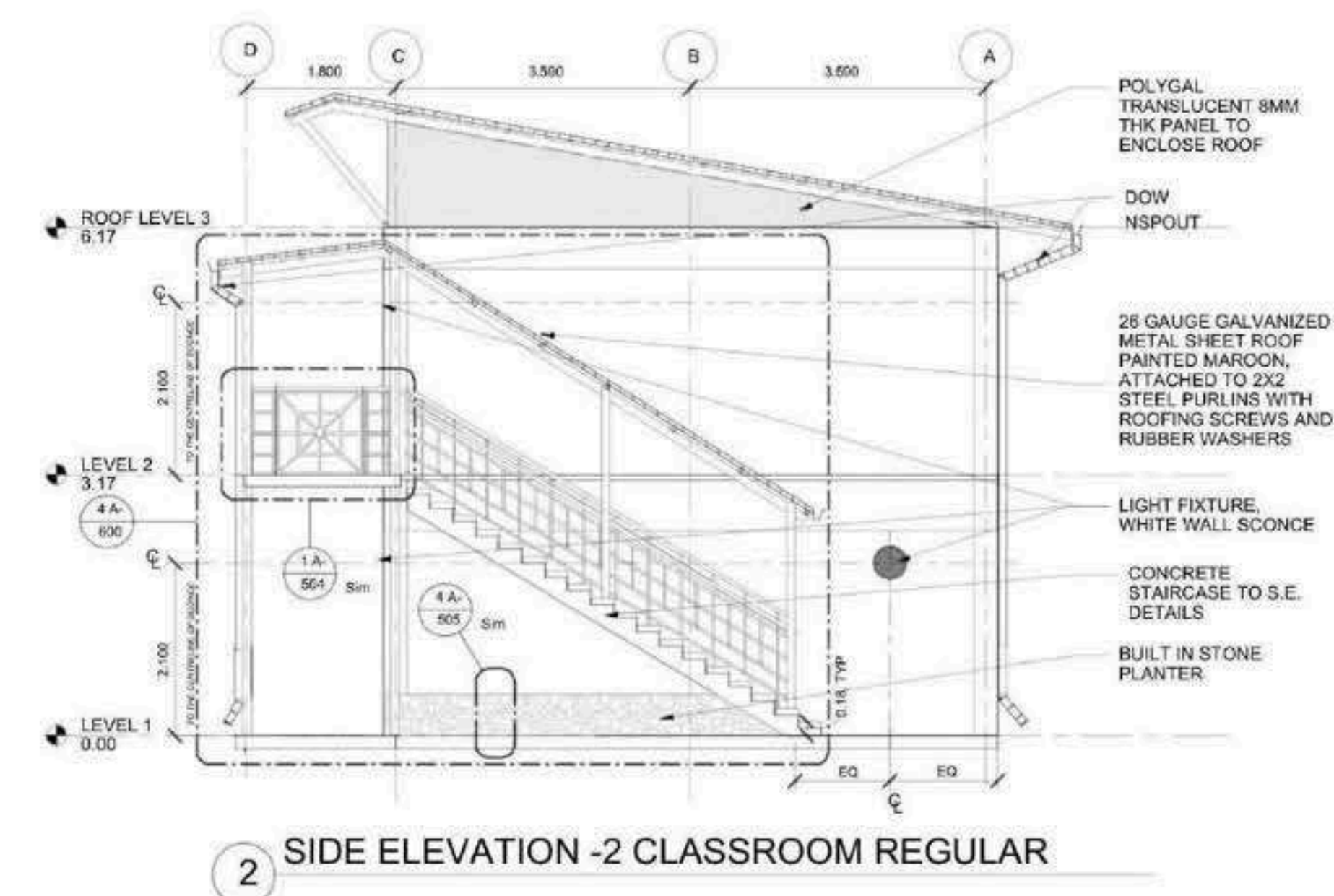
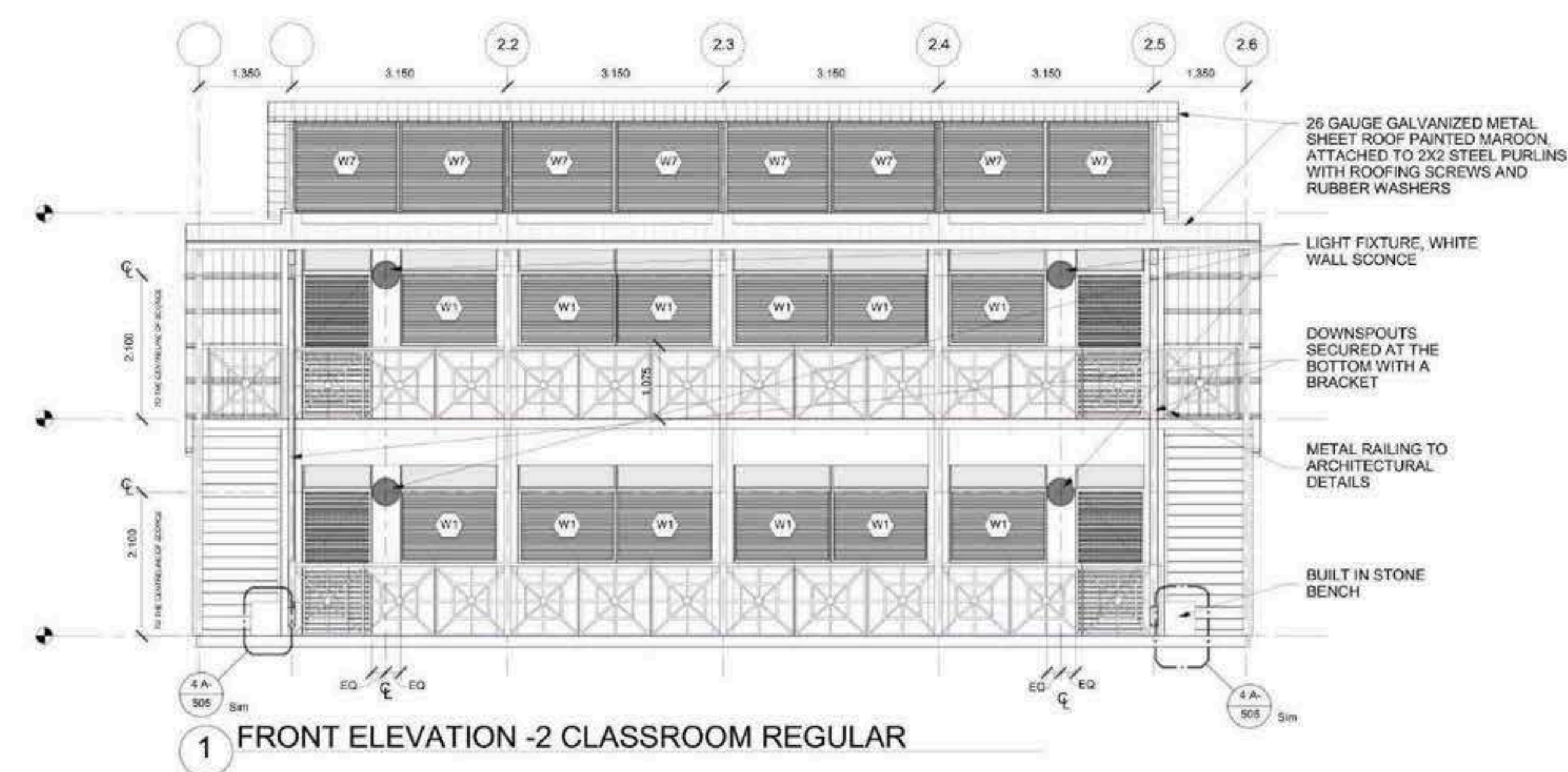
In 2024, Moge and Platanalé were selected as the first two pilot sites—real-world testbeds for this new model of spatial equity in education. Anchored in site-specificity and systemic intent, the initiative brings spatial justice into the conversation on schooling—ensuring that every student, regardless of where they live, learns in a space that reflects their worth.

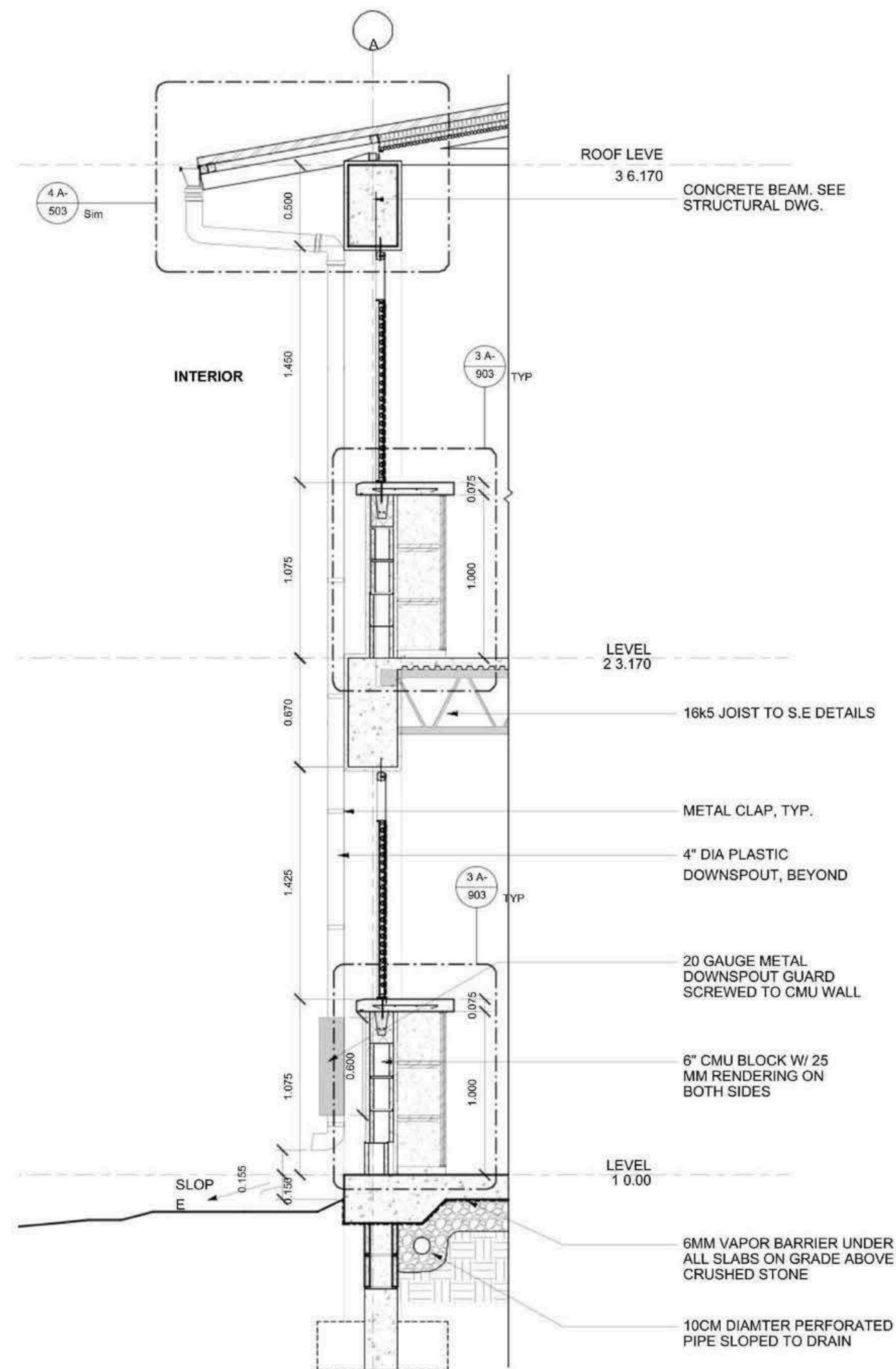




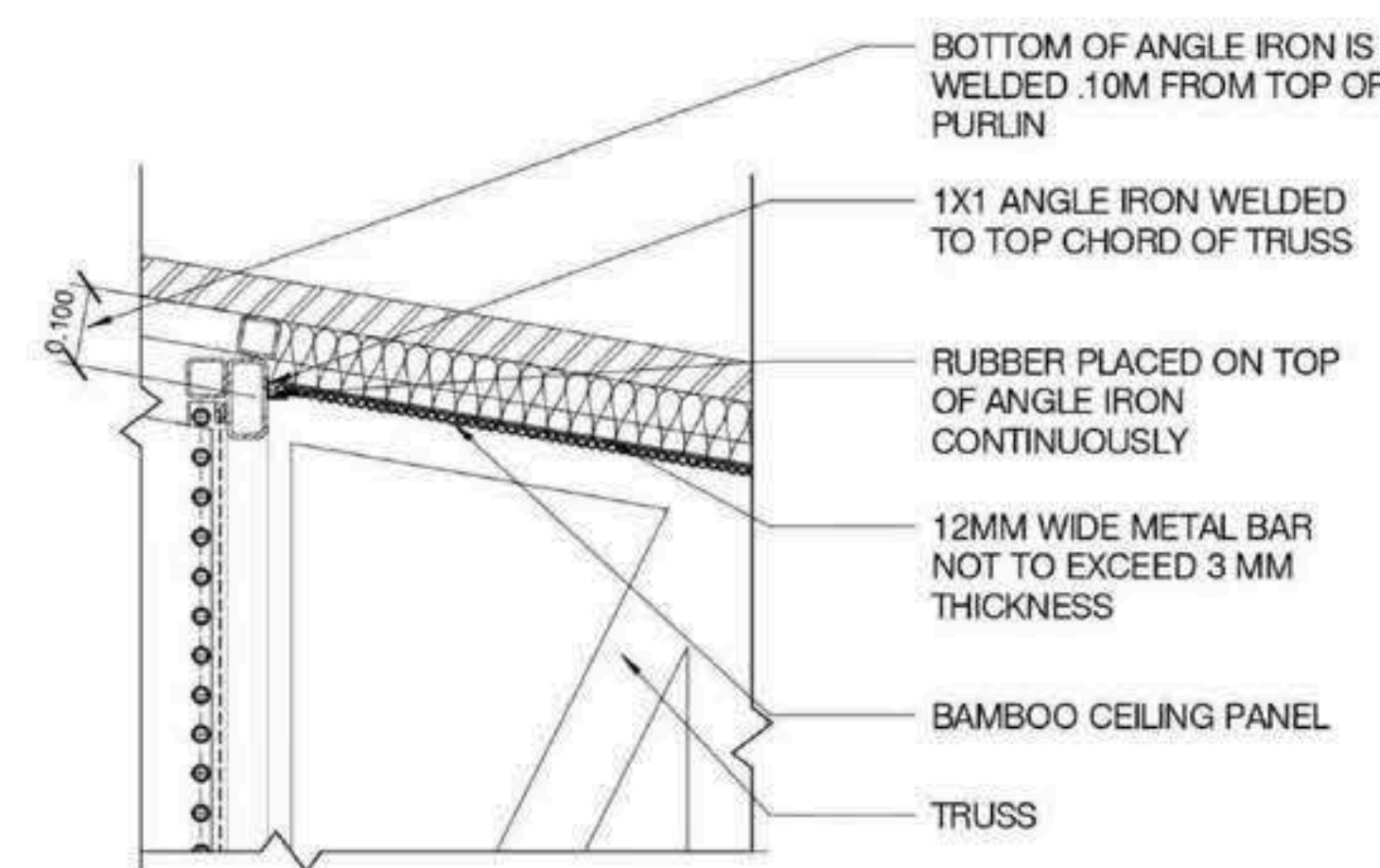
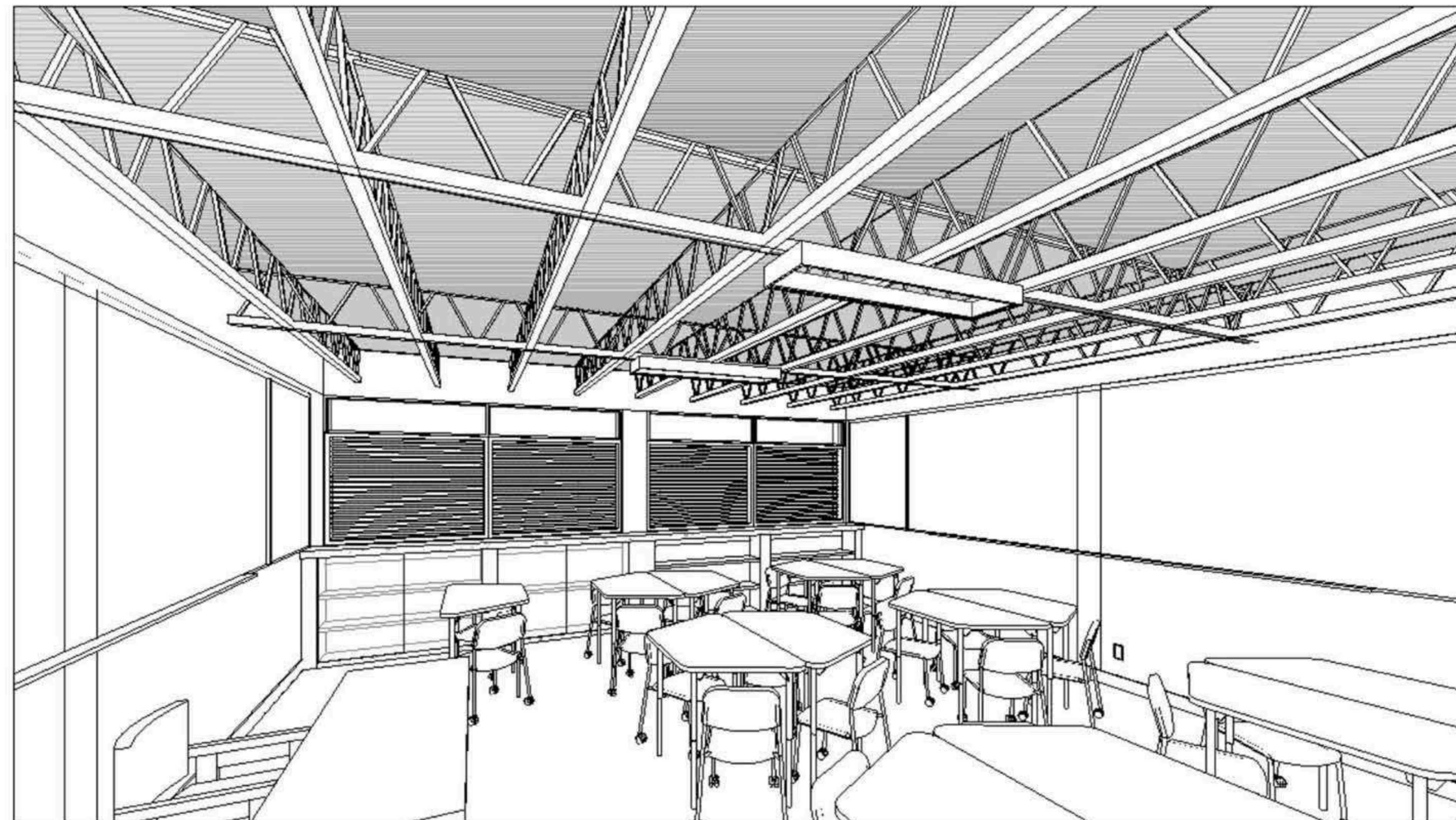
SITE PLAN _ MOGE

This site plan reflects a context-sensitive layout designed for a remote, rural setting. Buildings are arranged to optimize natural light, ventilation, and ease of access—critical in areas with limited infrastructure. Orientation follows local wind and sun paths, while open courtyards create shared spaces for learning, gathering, and play. The site's connection to footpaths and its position near the village core ensure accessibility despite Mogé's remoteness. Integrated landscaping with native vegetation supports soil stability, shade, and water management. The layout advances Summits Education's mission by creating a resilient, inclusive environment where education and community life can thrive.





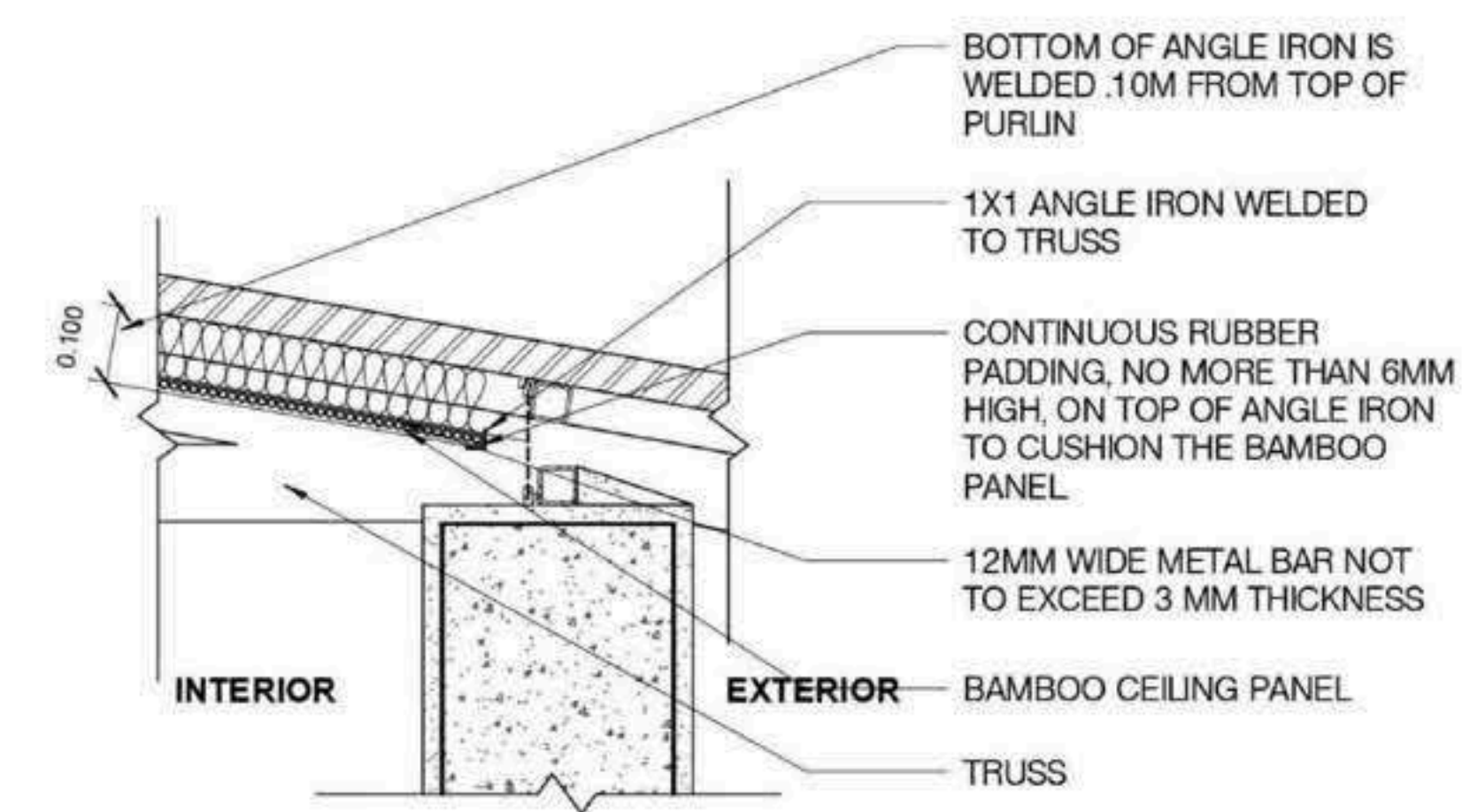
EXTERNAL WALL SECTION -ACADEMIC BLOCK



12MM WIDE METAL BAR NOT TO
EXCEED 3 MM THICKNESS,
RUNNING PERPENDICULAR TO
THE BAMBOO PIECES

1/2" - 3/8" DIAMETER BAMBOO

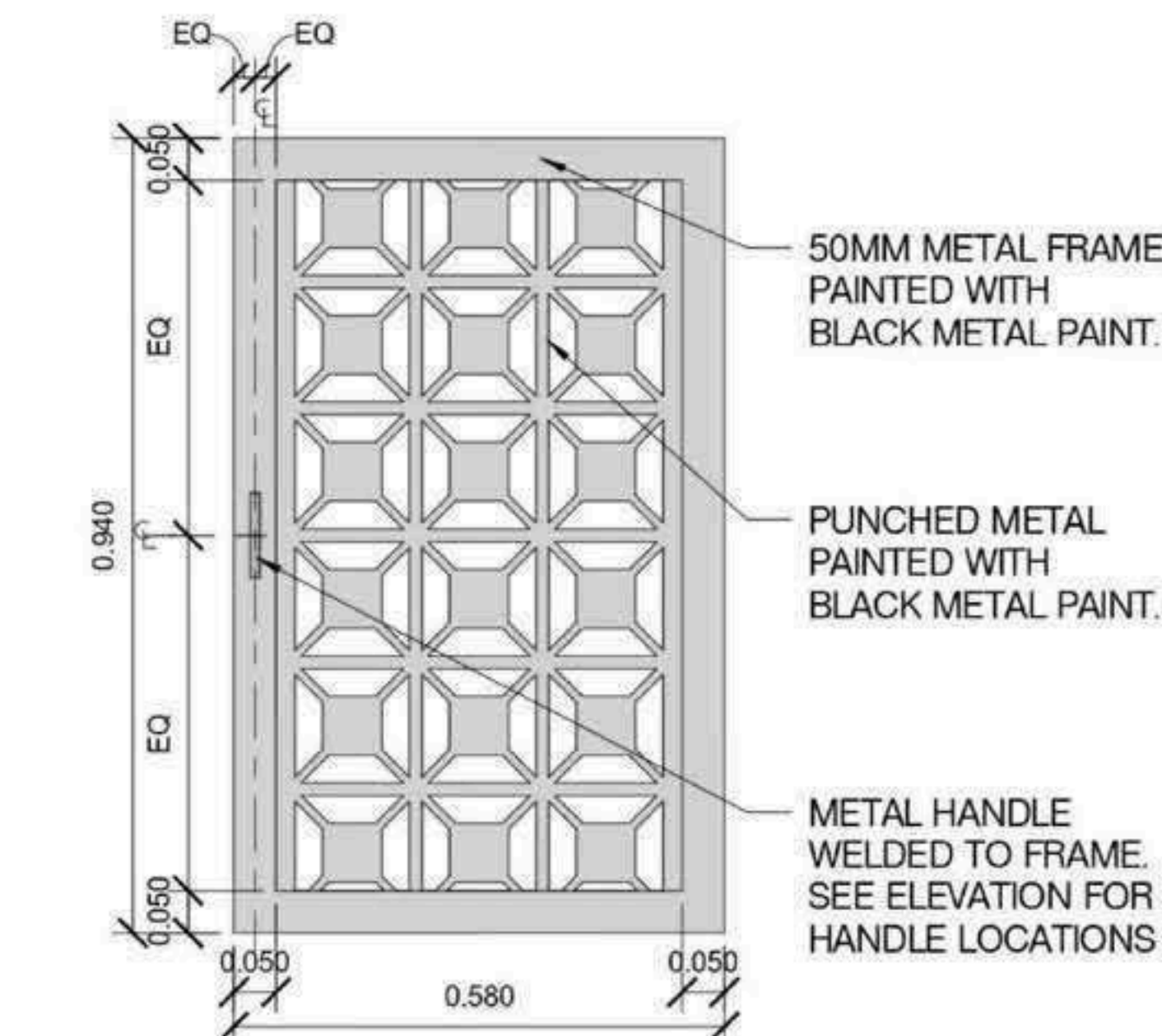
STRING WOVEN BETWEEN
BAMBOO AND AROUND FLAT
METAL ROD TO TIGHTLY
SECURE PANEL



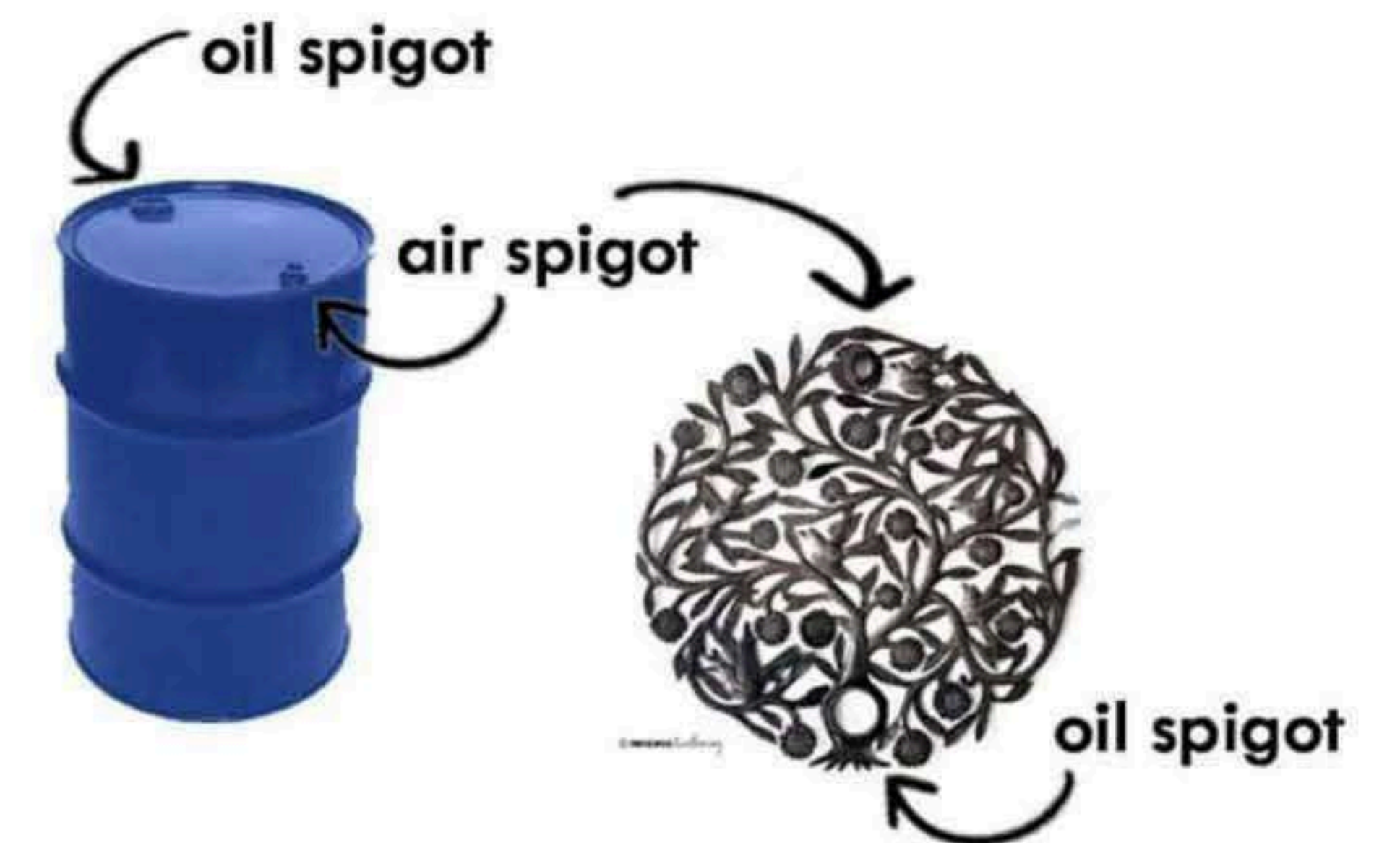
WEAVE TO EXTEND UP
AND OVER EVERY PIECE
OF BAMBOO AS SHOWN
IN THIS REFERENCE
IMAGE



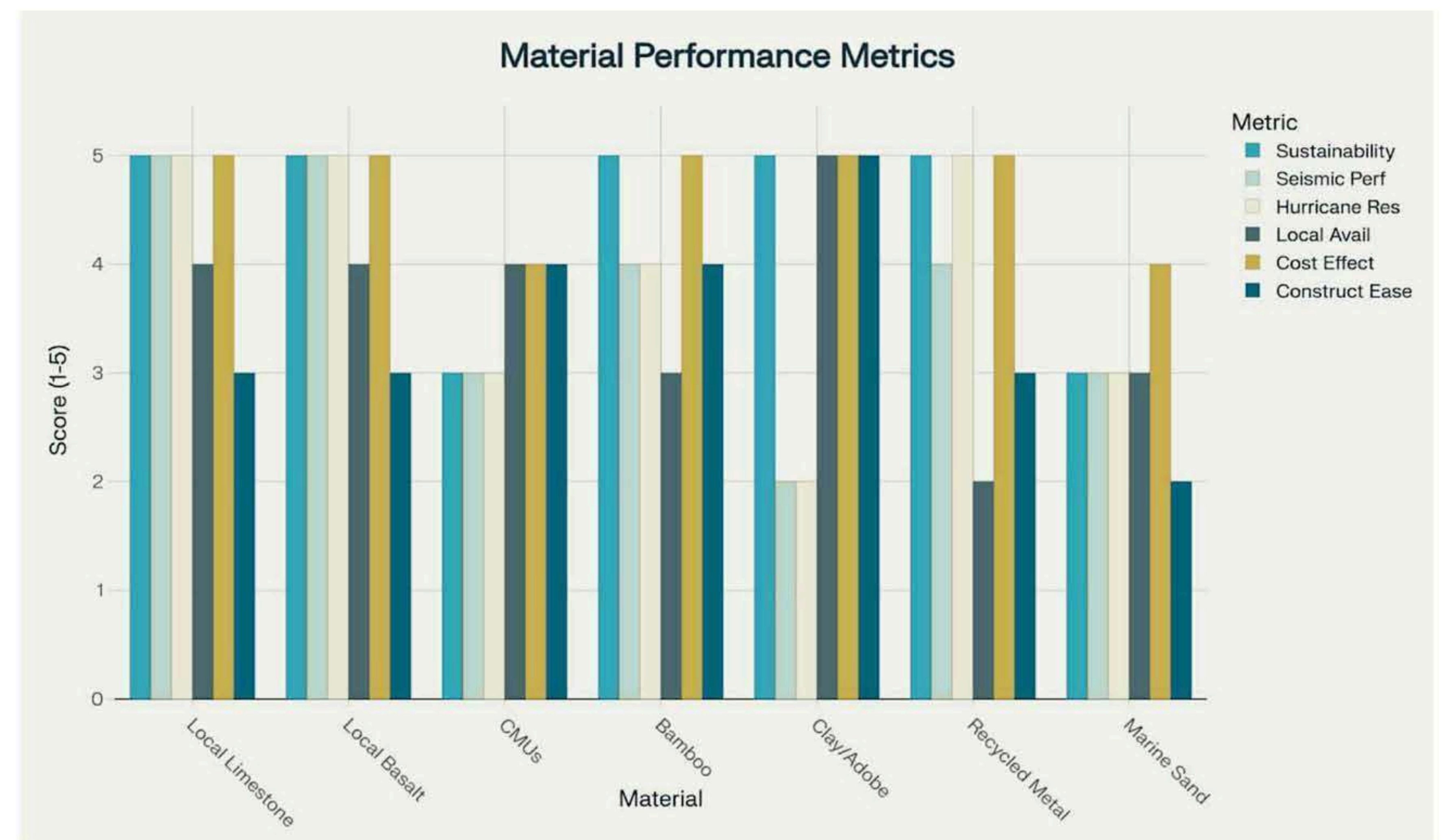
BAMBOO CEILING DETAILS



CASEWORK METAL DOORS LARGE CLASS



OIL DRUM RECYCLING



The material palette for the Mogé school design was selected to optimize sustainability, structural resilience, and local viability. Local limestone and ballast offer strong, low-embodied-energy foundations with high seismic and hurricane resistance. CMU blocks, when properly cured, deliver improved compressive strength while maintaining affordability and ease of construction. Bamboo serves as a renewable, lightweight option for shading and ventilation, with excellent tensile strength and local cultural familiarity. Clay adobe supports thermal regulation and low-carbon construction in dry zones, though it's limited to non-structural use in seismic zones. Recycled oil drums are repurposed for screens and cladding, reducing waste while adding visual texture. Where used, marine sand is washed to prevent corrosion in reinforced concrete. Altogether, this hybrid approach balances cost-effectiveness, durability, local sourcing, and ease of assembly in remote contexts.

DESIGN FEATURES INCLUDED:



Safety

Buildings and fencing are situated to define a clear point of entry to the site. The latrines are located in a highly visible location to promote safety.



Courtyard

Buildings are oriented around a courtyard to create a zone for play and promote security to the site.



Sustainable Design Strategies

Building design and siting promotes airflow to provide a comfortable space, and promote a day-lit environment. Roofs are designed to support solar panels and storage is provided for solar battery storage.



Quality Classrooms

Appropriately sized spaces with integrated storage, and acoustical capabilities.



Flexible Spaces

Indoor and outdoor gathering spaces for learning, teacher development, and community needs.



Covered Walkways

Roof extensions on the buildings provide cover around the perimeter of the central courtyard.



Promotion of Health

Access to latrines, potable water, hand sanitation and food service.



Gardens & Plantings

The inclusion of vegetation to be used for recreational and educational purposes.



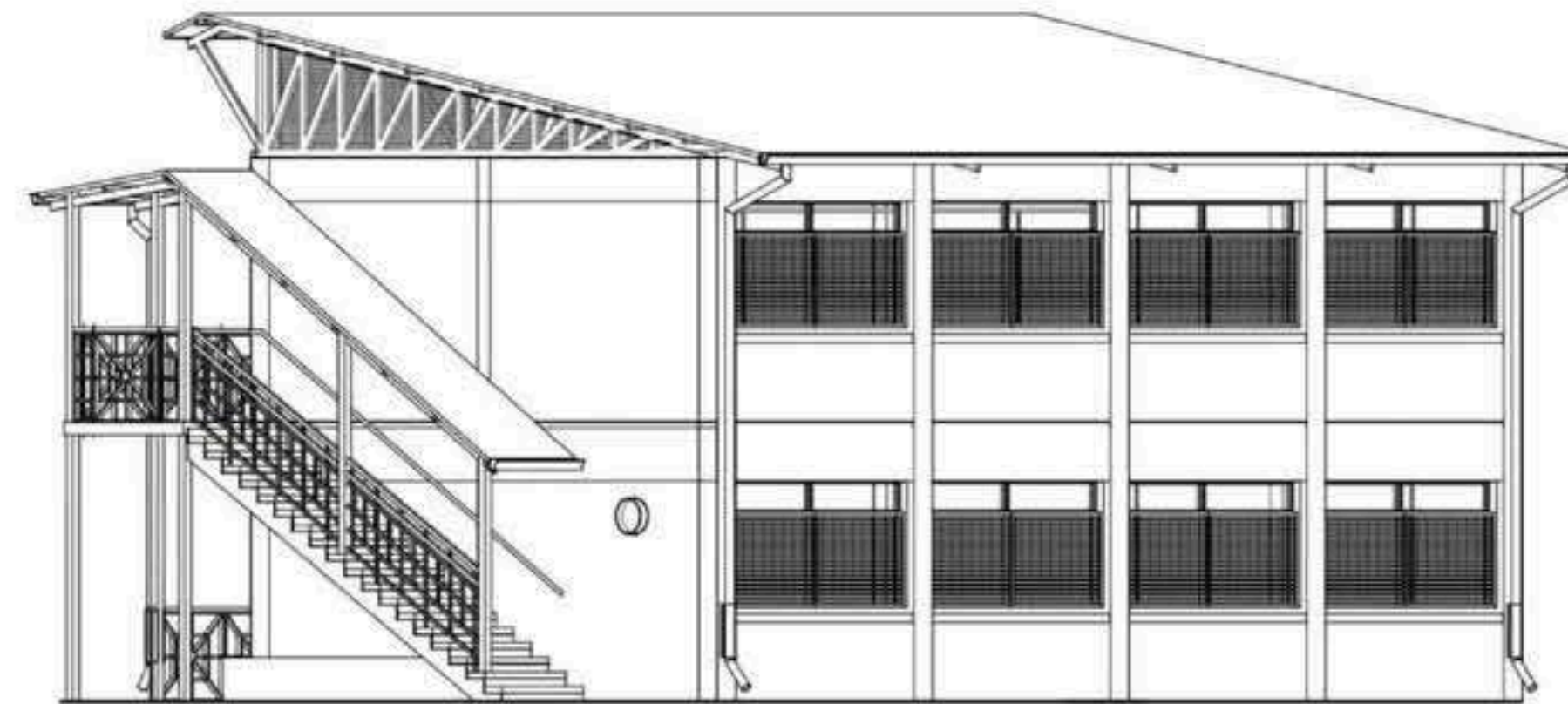
Access to Electricity

Solar panels and solar lights will provide electricity on site. Battery storage allows for electricity at night.

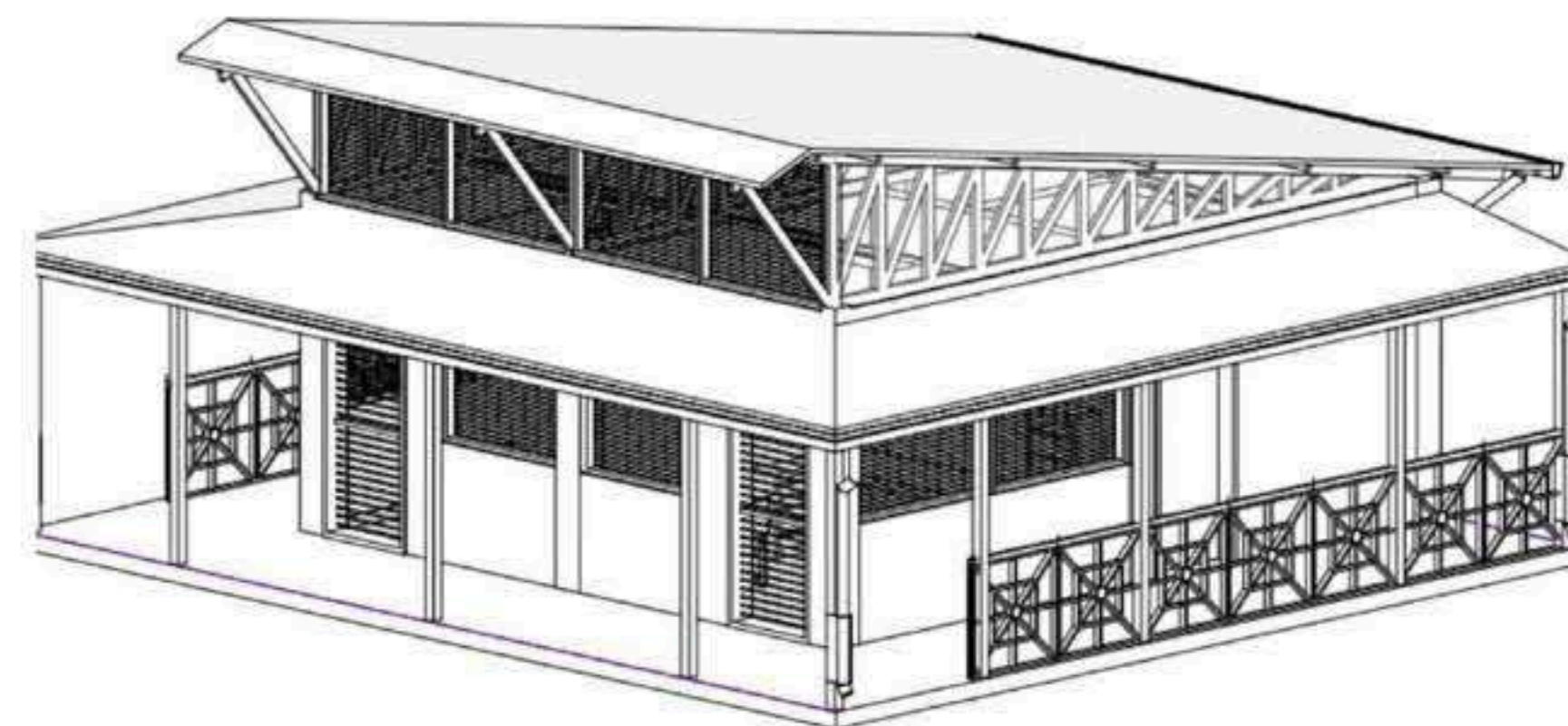


Locally Sourced

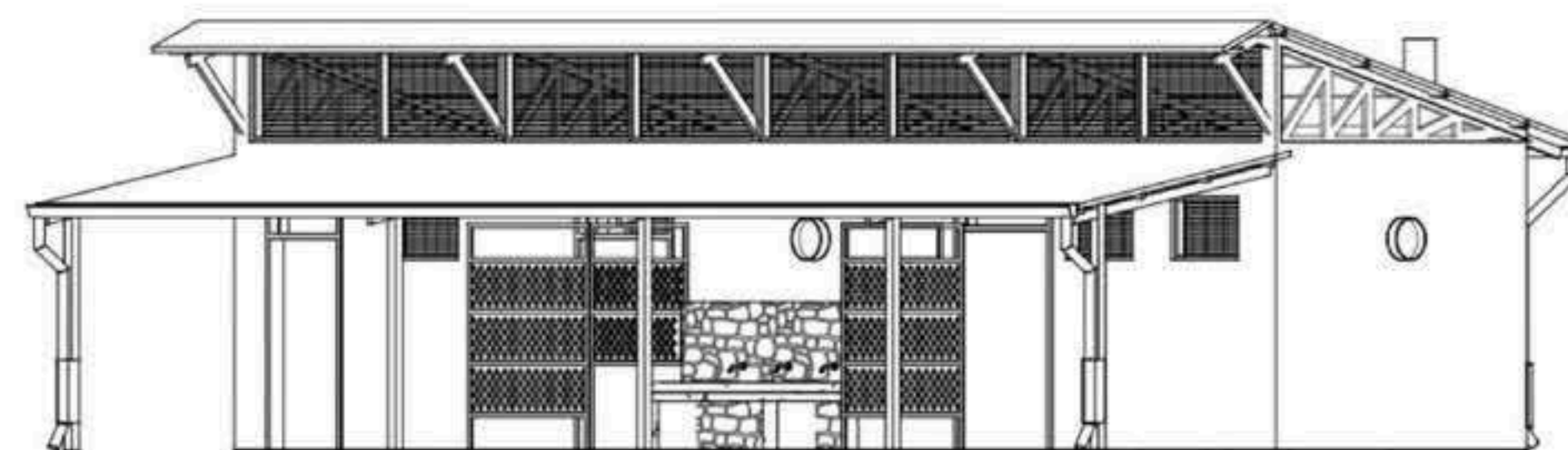
Relying on local labor and materials will provide investment in the local economy, promote sustainability, and provide job creation.



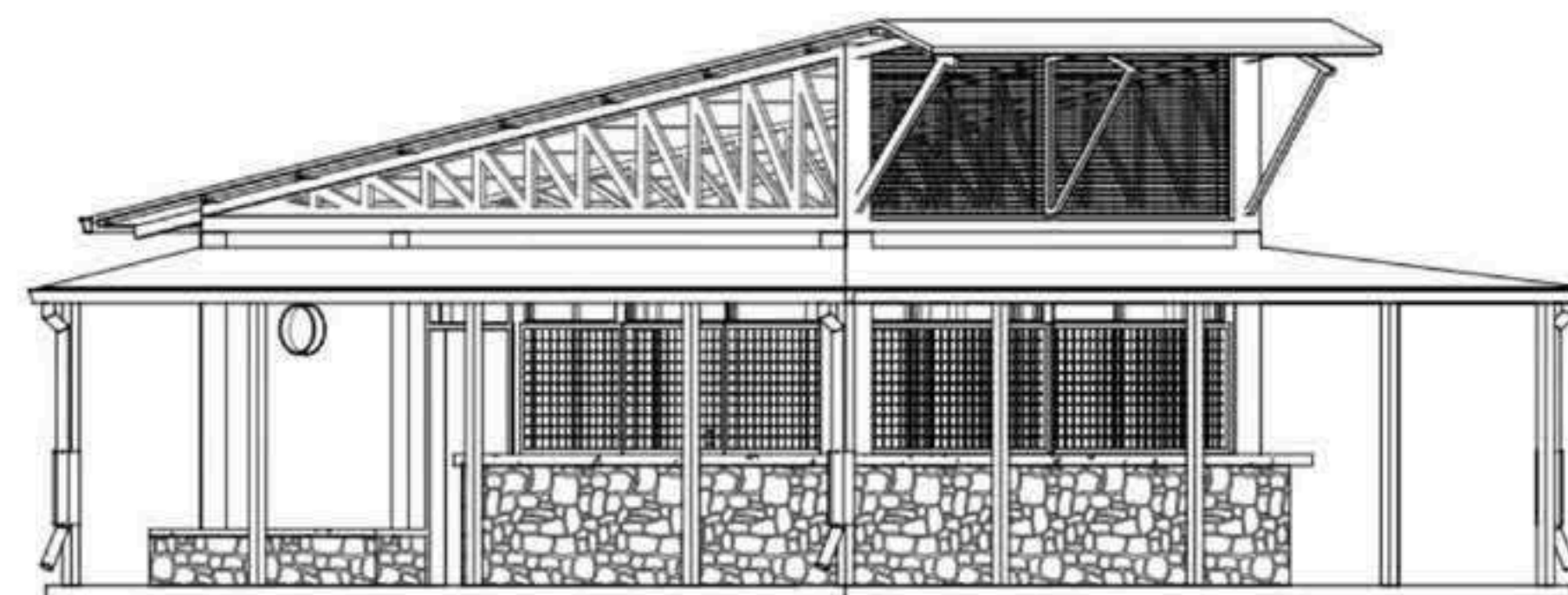
ACADEMIC BLOCK



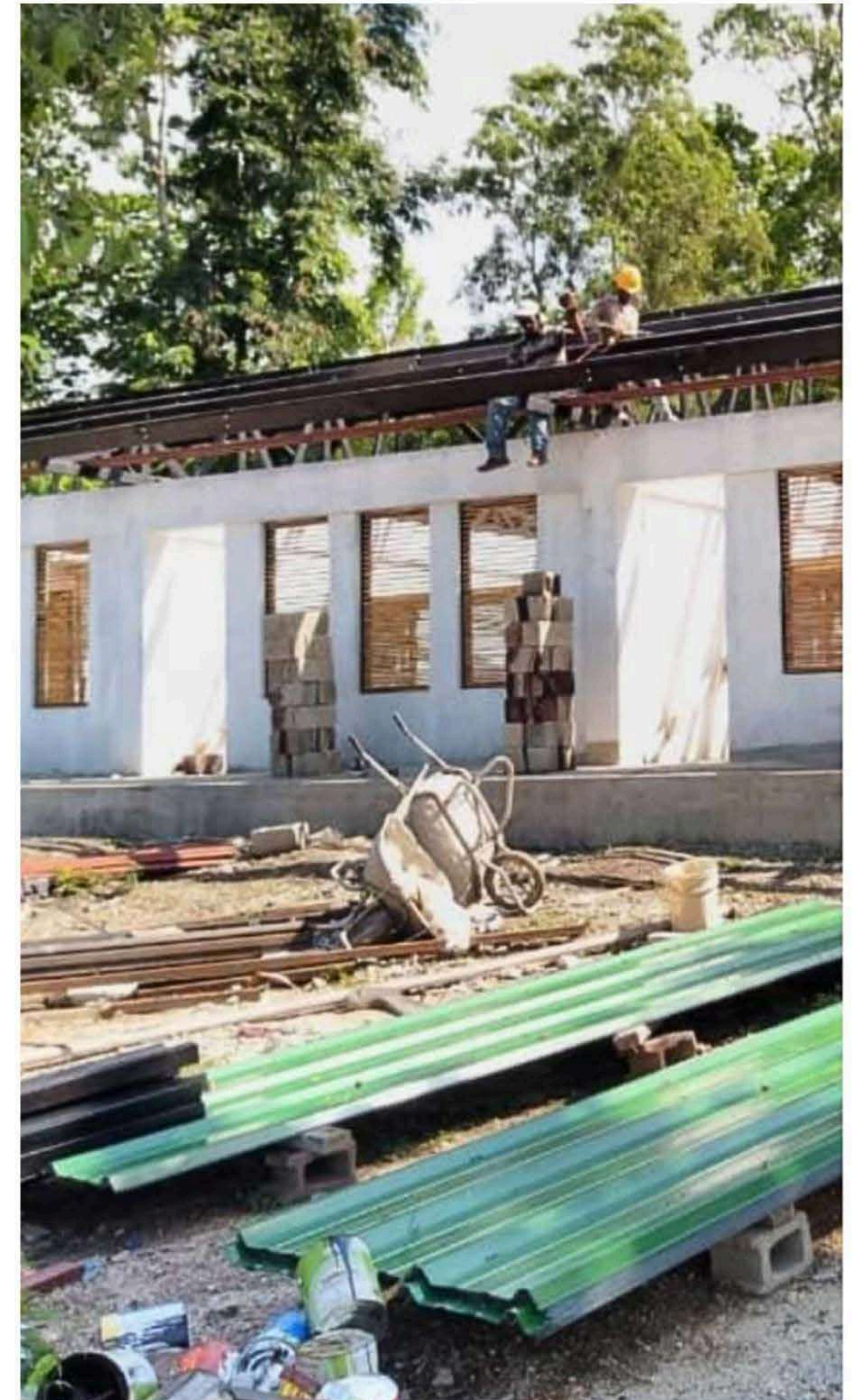
ADMINISTRATION



LAVATORY



KITCHEN



SAFE HOMES FOR GBV SURVIVORS

ALL 47 COUNTIES, KENYA
Architecture: UDesign
Lead Principal: Paul Mutahi
Architect: Georgina Kinuthia

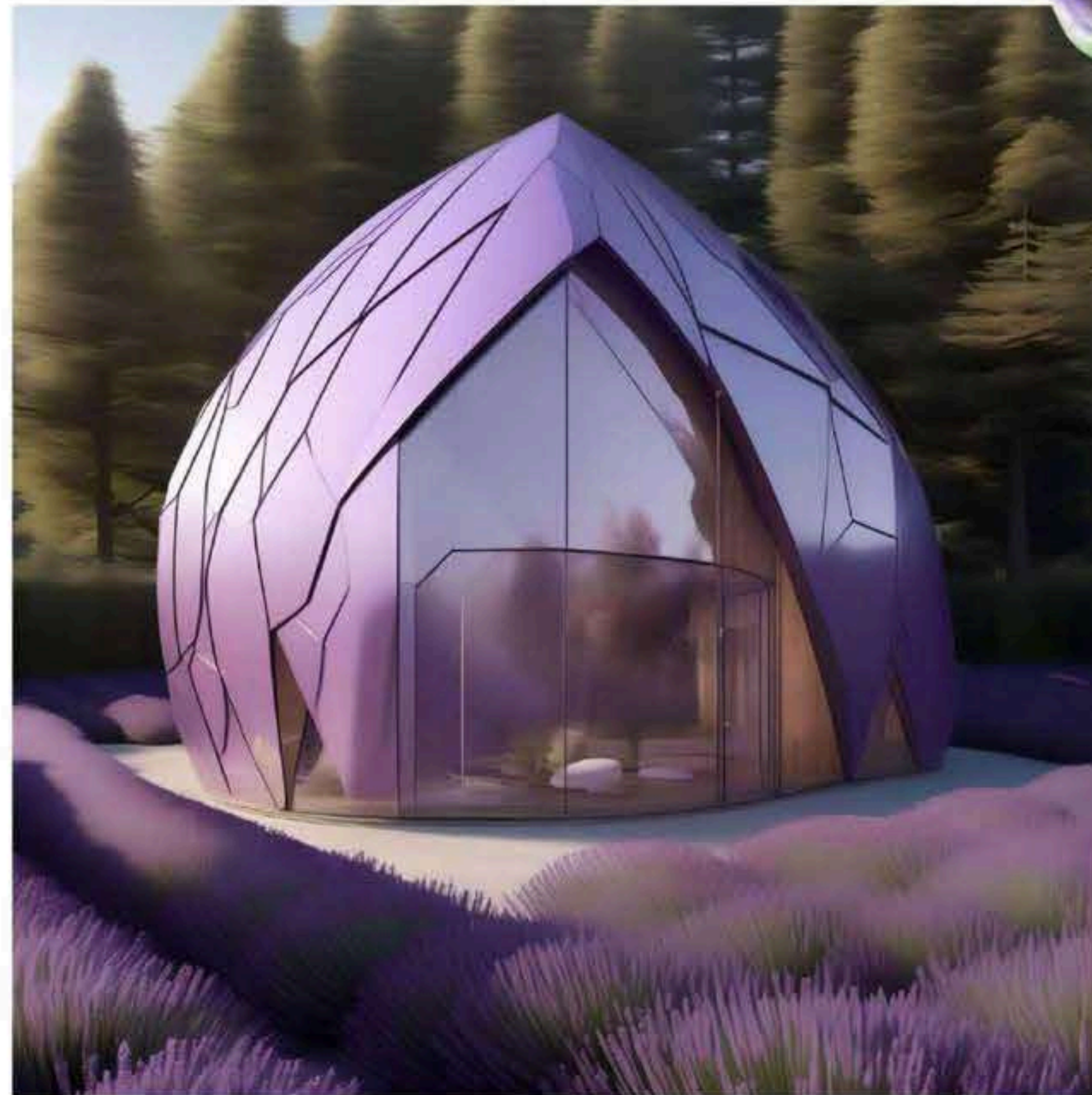
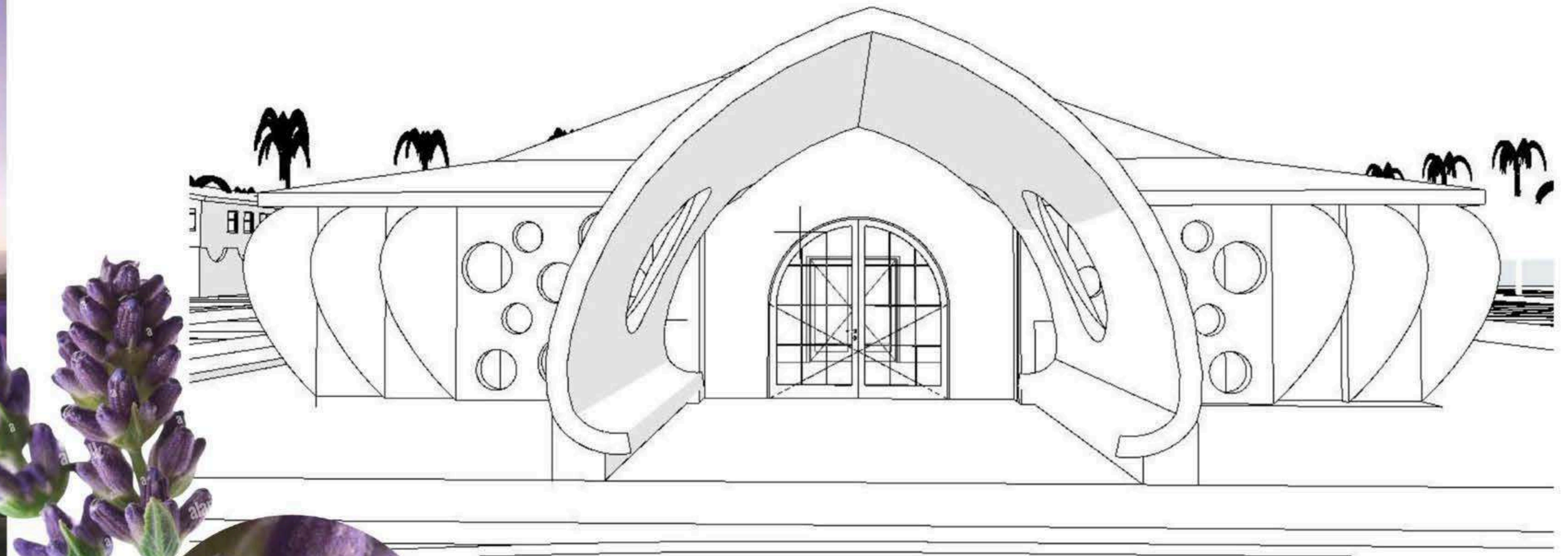
This Government of Kenya aims to establish safe homes for GBV survivors, featuring a modular design to be replicated across all 47 counties.

The design focuses on robust security, discreet locations, and trauma-informed spaces, prioritizing flexibility, privacy, and communal support areas. Therapeutic design, accessibility features, and sustainability practices contribute to survivor well-being.

Collaboration with NGOs and experts ensures the architecture aligns with specific survivor needs, fostering empowerment through education and skills training in a compassionate residential setting.

The GBV accommodation center is meticulously designed to facilitate healing by prioritizing privacy. Private rooms offer a personal retreat, while 24/7 counseling services are available through the nearby social workers' office. Additionally, dedicated spaces for meditation, prayer, and communal dining enhance the holistic healing environment. The center features a communal courtyard, fostering a sense of community while respecting the need for personal space. To ensure individualized care, the accommodation is organized to separate families, single adults, children, and young adults under careful consideration and agreement. This thoughtful arrangement aims to create a supportive and secure space for survivors on their journey to recovery.





Inspired by the calming form of a lavender flower, the architectural concept centers on creating a tranquil, restorative environment for healing women and children affected by gender-based violence. The design embraces organic shapes, soft color palettes, and nurturing spatial arrangements to foster safety, dignity, and emotional well-being. Privacy and protection are prioritized, while every element of the space is crafted to promote resilience and empowerment. Sustainability is embedded through the use of locally sourced, low-impact materials and passive design strategies that reduce energy use and enhance thermal comfort. Walkways, gardens, and gathering areas integrate natural materials and native planting to support both ecological health and sensory healing. Crucially, the project is grounded in community inclusivity—ensuring the participation of local women in both the design and long-term stewardship of the space. This co-creative process ensures that the architecture not only shelters, but uplifts—honoring cultural context, promoting collective care, and growing strength from within.

SHOFCO GIRLS' SCHOOL

NAIROBI, KENYA
Architecture: UDesign
Lead Principal: Paul Mutahi
Architect: Georgina Kinuthia

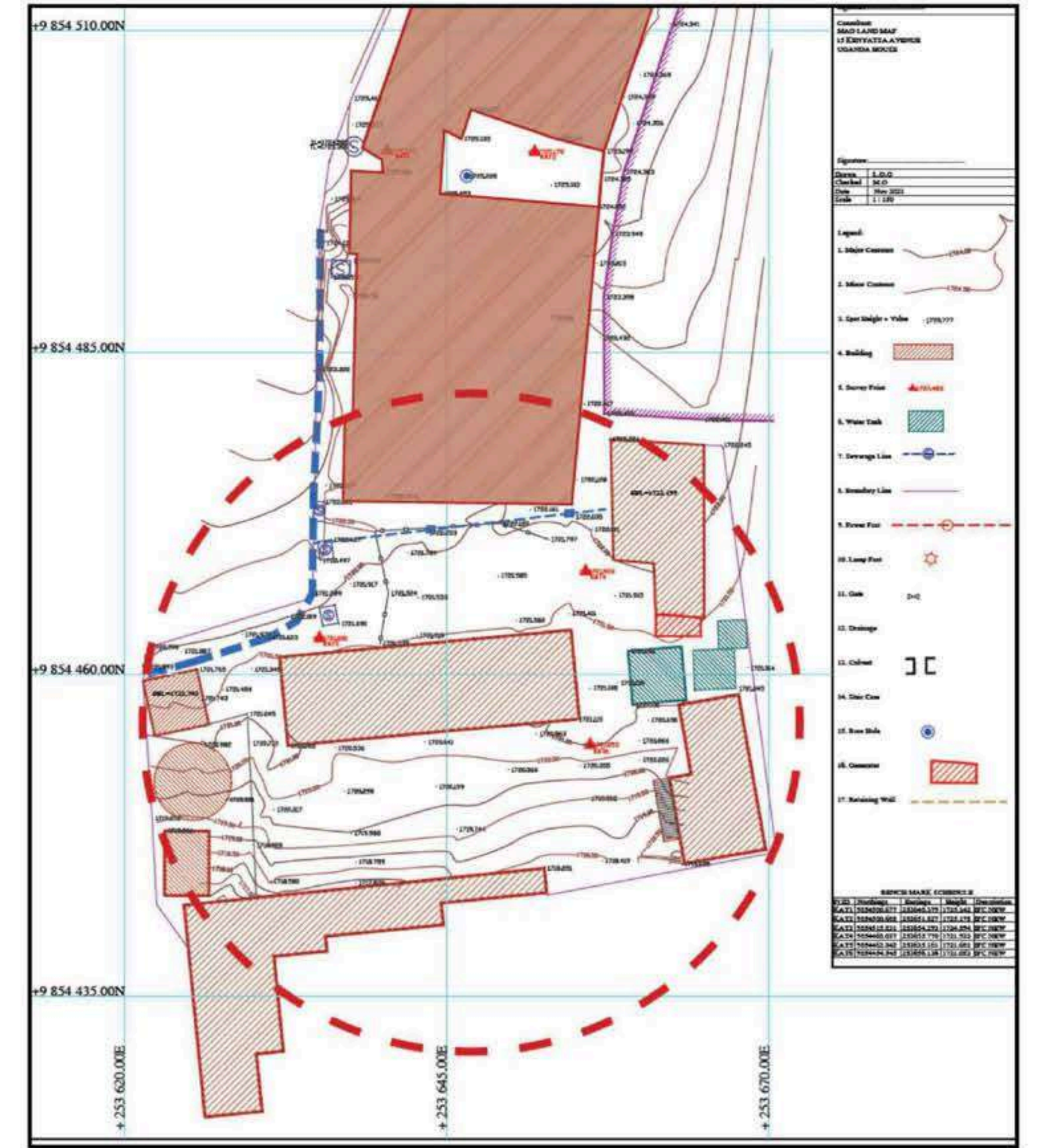
Kibera, like many informal settlements across the Global South, is often viewed through the lens of deficit. Yet for Kennedy Odede, who grew up within its dynamic and densely networked fabric, it was also a place of deep social energy, adaptability, and hope. The absence of formal infrastructure revealed not only systemic neglect, but the opportunity to radically rethink how cities could serve their most marginalized residents.

Kennedy's response—shaped by his lived experience and inspired by the spatial agency of everyday communities—was to establish SHOFCO. Together with Jessica Posner, they co-developed a model of development where spatial design and social systems work in concert. SHOFCO's hubs embed healthcare, education, safe public space, water access, and digital connectivity directly into the heart of underserved communities—treating architecture not as an object, but as a network of services and relationships.

The built interventions are intentionally human-scaled, flexible, and contextually responsive—using locally available materials, participatory design processes, and minimal ecological footprints. These spaces are not neutral; they are political. Each one signals that beauty, safety, and opportunity are not luxuries, but rights—even in the most resource-constrained environments.

SHOFCO's work invites us to shift the architectural gaze—from form to function, from skyline to ground plane, from centrality to periphery—and to see the informal city not as a problem to solve, but as a partner in building the future.





The proposed site measures approximately 1,391 m² (0.3 acres) and is currently occupied by informal iron-sheet structures. It sits within a dense urban fabric and presents both challenges and opportunities for sustainable, context-sensitive development. The site takes a trapezoidal form, oriented along an East–West axis, allowing for optimal consideration of solar paths, natural lighting, and passive cooling strategies in the building design. With a 9-meter elevation drop, the terrain slopes steeply toward the south. This condition invites a stepped or terraced architectural approach, allowing the building to follow the contours of the land while minimizing costly earthworks and improving accessibility and stormwater management. Located in Kibera, one of Nairobi’s most densely populated informal settlements, the site is bounded on all sides by residential structures. An existing school building occupies the northern edge of the plot. The proposed SHOFCO Girls' School is designed to integrate seamlessly into its surroundings while promoting economic empowerment, social inclusion, and environmental resilience.

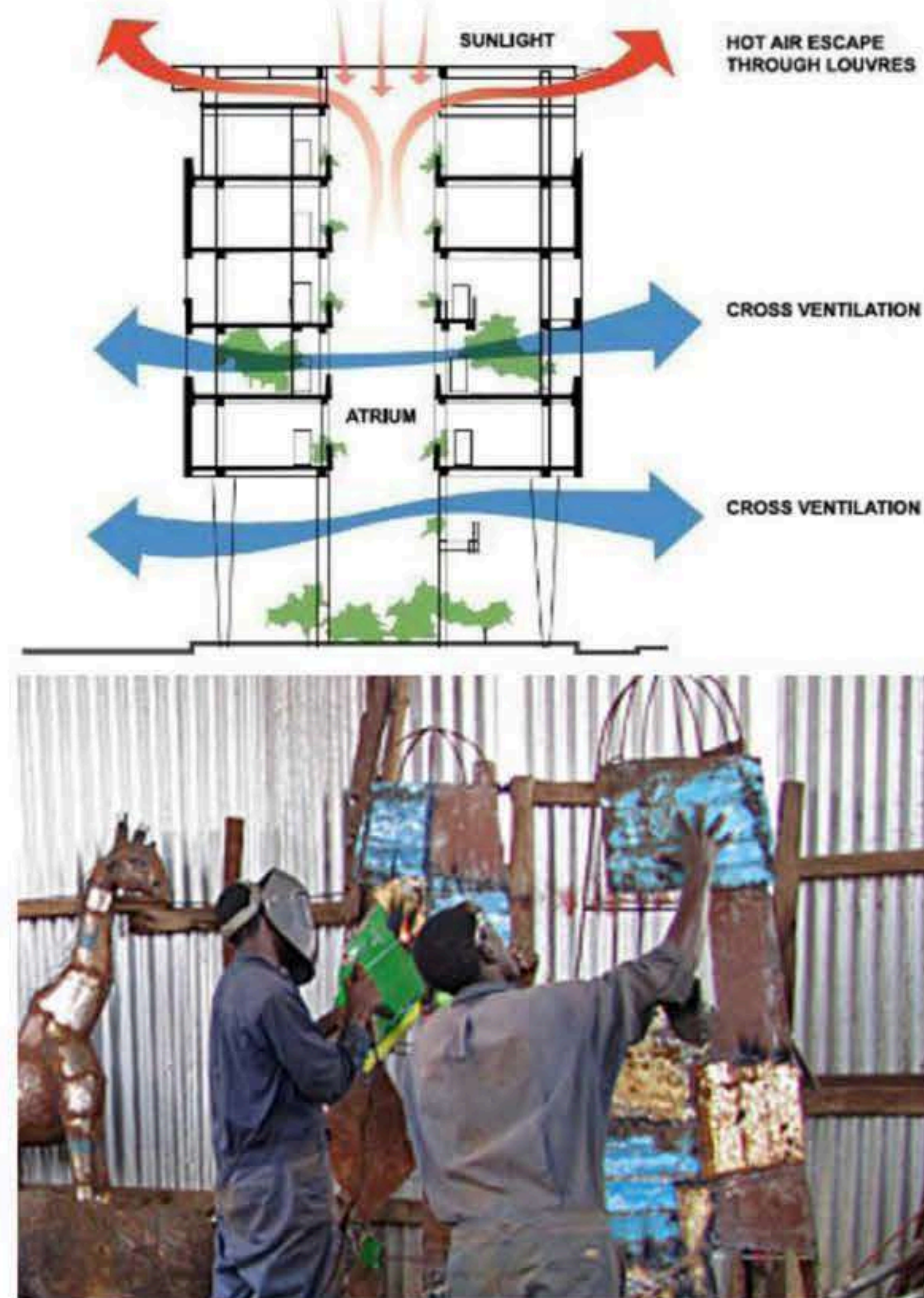
Sustainable Development Considerations

- Aligned with the UN Sustainable Development Goals (SDGs), the project seeks to incorporate the following planning and design priorities:
- Green architecture and eco-conscious construction methods
- Use of locally sourced and bio-degradable building materials
- Engagement of local labor and artisanal skills
- Deployment of renewable water systems (e.g., rainwater harvesting)
- Integration of solar and potential wind energy solutions
- Protection of existing natural elements and micro-ecosystems
- Replenishment plans for any extracted or consumed natural resources

This integrated planning approach ensures that the school not only meets functional and spatial needs, but also becomes a model of community-rooted, climate-resilient, and socially inclusive development.



Crucially, the layout was shaped by pedestrian traffic patterns, ambient noise, and the spatial pressures of the dense informal settlement—ensuring that circulation, quiet zones, and key functions respond directly to the realities of the slum environment.



The primary construction material used was compressed earth blocks, a modern evolution of traditional adobe. These uniform, solid blocks are made by compressing clay-rich soil—often sourced directly from the site—without firing, resulting in very low embodied carbon. Their moisture-regulating and thermal mass properties make them ideal for Kitui's climate, offering both comfort and sustainability at a low construction cost. Suitable for both load-bearing and non-load-bearing walls, CEBs offer durability while supporting local production and skills development.

Passive Cooling and Ventilation

The design harnesses natural ventilation through stack effect principles, aided by:

- A central atrium, allowing warm air to rise and exit while drawing in cooler air below
- Vent block louvers above openings to release hot air
- Operable windows for cross-ventilation

These strategies reduce reliance on mechanical cooling and enhance indoor comfort throughout the day.

Water, Energy, and Resource Efficiency

- Rainwater harvesting systems collect and store water for reuse
- Water-saving dual flush cisterns reduce water consumption
- Solar panels power efficient LED lighting
- Indoor motion sensor lights minimize energy use
- Use of recycled materials wherever possible, further reducing environmental impact
- Vertical landscaping adds insulation, improves air quality, and supports urban greening

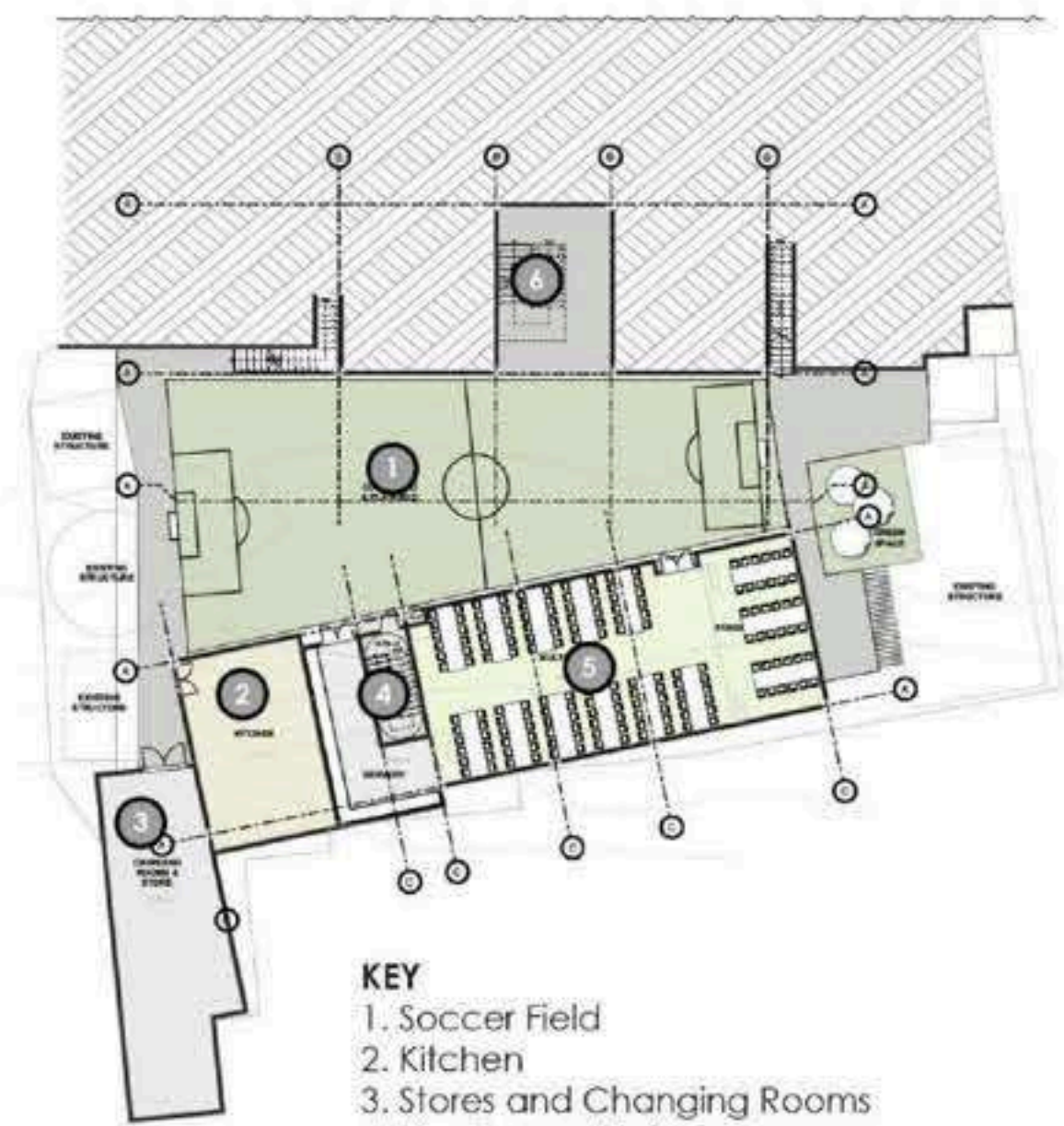
Local Labor and Artisanal Integration

The project engages the surrounding community as co-creators:

- Skilled local artisans fabricate metalwork for windows, doors, beds, and desks
- Community artists from Kibera contribute unique decorative elements to spaces like libraries and boardrooms
- Construction crews are trained in low-impact, non-polluting building techniques using local materials

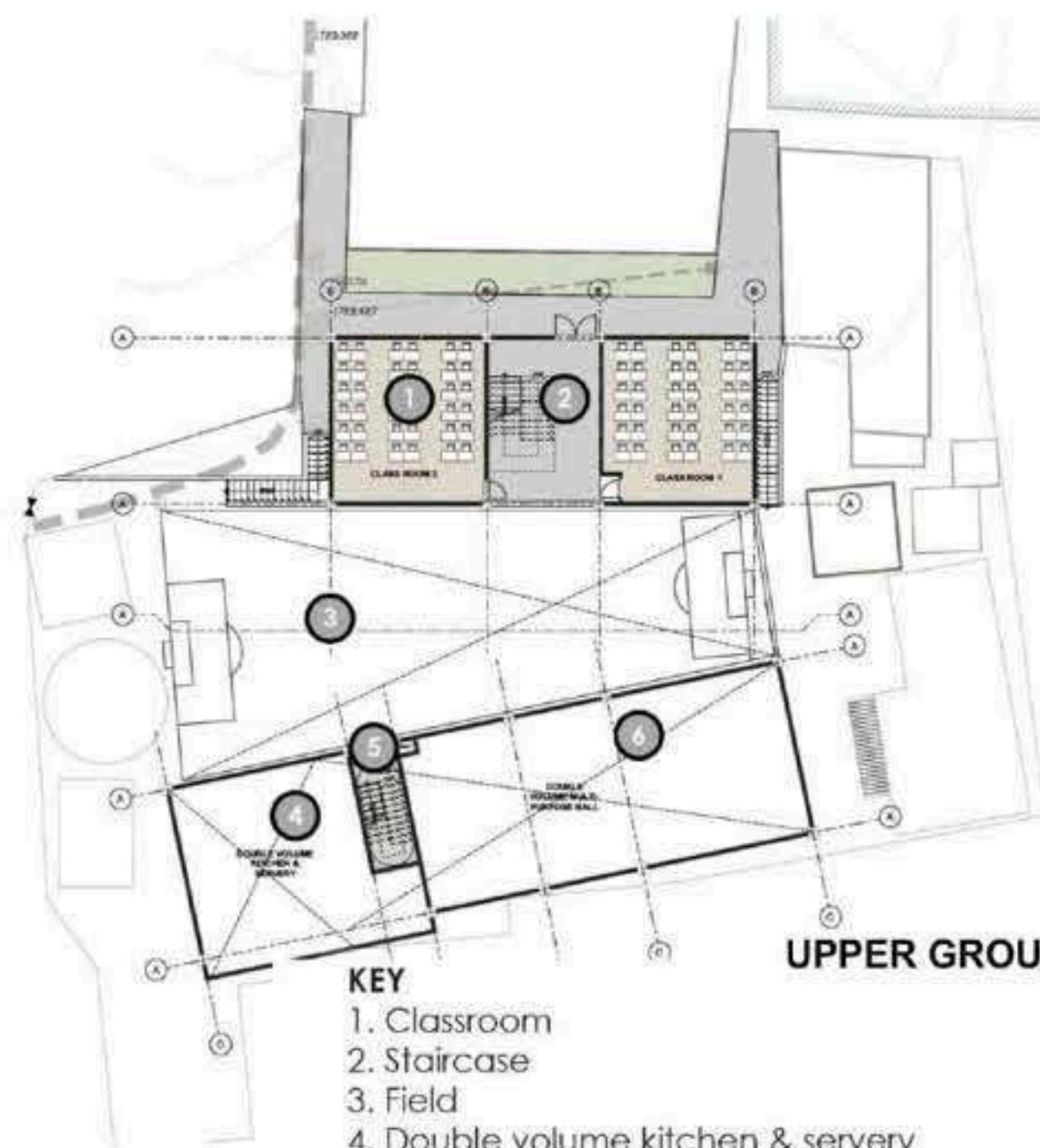
Together, these integrated strategies form a holistic, low-carbon, and community-rooted design—one that uplifts local economies while demonstrating sustainable, climate-adaptive architecture.





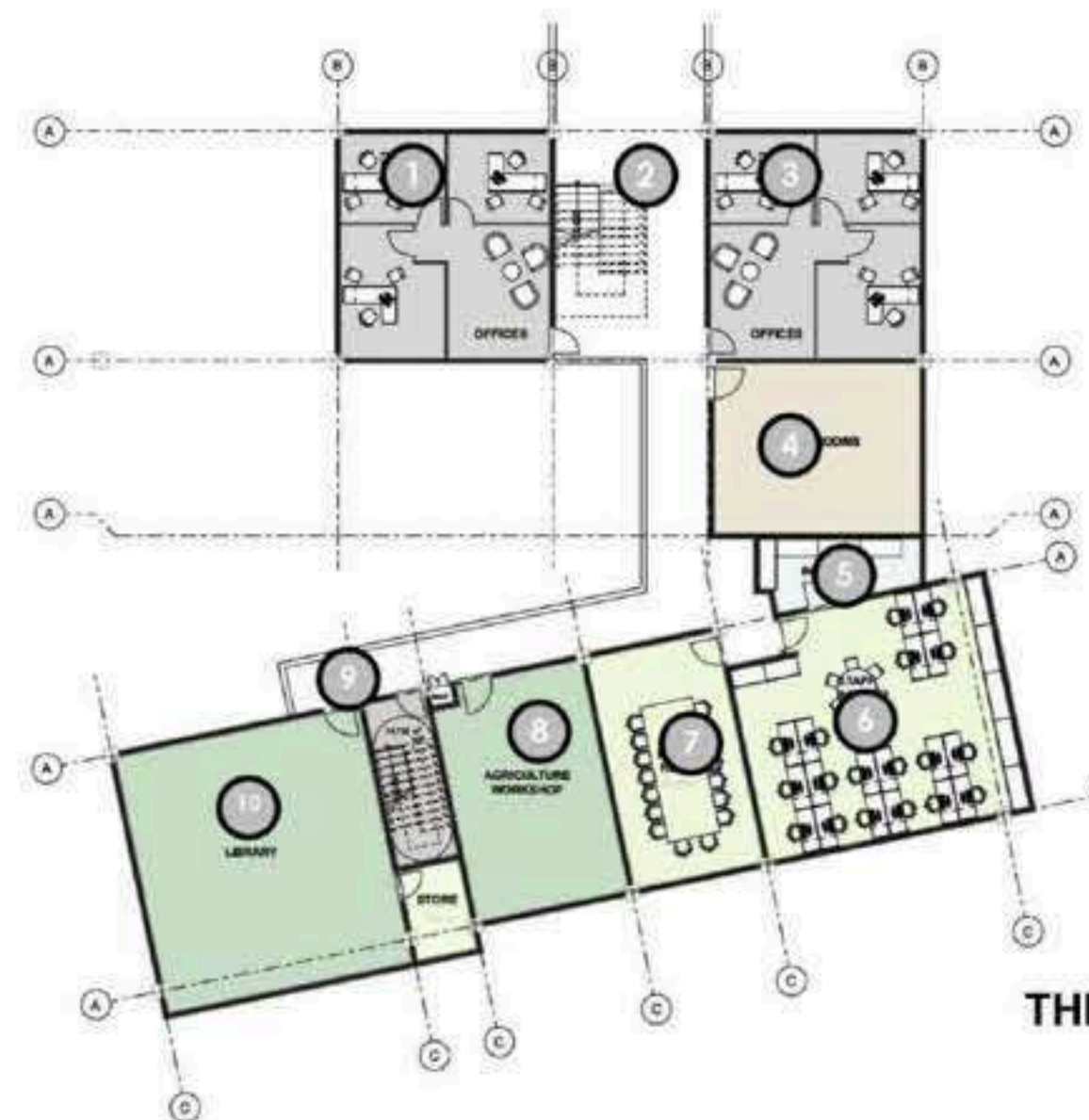
- KEY**
1. Soccer Field
 2. Kitchen
 3. Stores and Changing Rooms
 4. Fire-Escape Stairs
 5. Multipurpose Hall
 6. Staircase

LOWER GROUND FLOOR



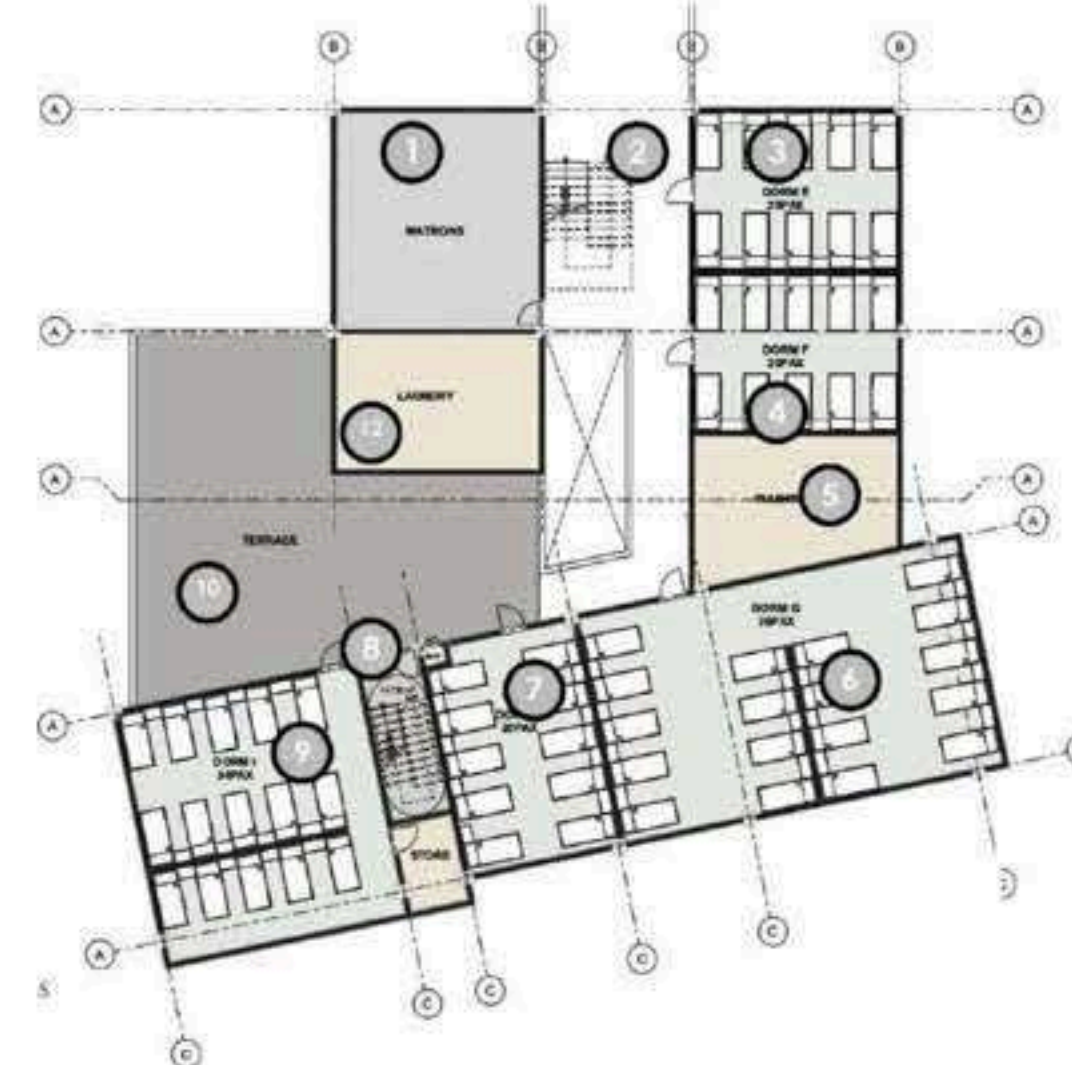
- KEY**
1. Classroom
 2. Staircase
 3. Field
 4. Double volume kitchen & servery
 5. Fire escape stairs
 6. Double volume multi-purpose hall

UPPER GROUND FLOOR



- KEY**
1. Office
 2. Stairs
 3. Office
 4. Washrooms
 5. Inventory/Store
 6. Staffroom
 7. Boardroom
 8. Agriculture Workshop
 9. Walkway
 10. Library

THIRD FLOOR LAYOUT



- KEY**
1. Matron
 2. Stairs
 3. Dorm E
 4. Dorm F
 5. Washroom
 6. Dorm C
 7. Dorm H
 8. Fire escape Stairs
 9. Dorm I
 10. Terrace

FOURTH FLOOR LAYOUT



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